Positive completeness of the finitary refinement type system for Scott-open properties

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Allowed standard prerequisites (editorial).

- Classical Scott-domain, spectral-space and Hofmann-Mislove results. Author bibliography and full retained domain/spectral appendix explicitly cite standard results; no recursive reproof required.

1 Introduction

We are interested in input-output specifications of functions that handle infinite data, such as streams or non-wellfounded trees. Consider the function $\mathsf{count} : \mathsf{Stream\,Bool} \to \mathsf{Stream\,Nat}$ defined as $\mathsf{count} = \mathsf{countrec}\ 0$, where

$$\begin{array}{rl} \mathsf{countrec}: & \mathsf{Nat} \longrightarrow \mathsf{Stream\,Bool} \longrightarrow \mathsf{Stream\,Nat} \\ \mathsf{countrec}\ n\ (b :: x) = & \mathsf{if}\ b\ \mathsf{then}\ (\mathsf{countrec}\ (1{+}n)\ x)\ \mathsf{else}\ n :: (\mathsf{countrec}\ n\ x) \end{array}$$

Given a stream of Booleans x, the stream of natural numbers $(\mathsf{count}\ x)$ retains, for each occurrence of ff in x, the number of tt's seen so far in x. It is clear that if there are infinitely many tt's and infinitely many ff's in x, then $(\mathsf{count}\ x)$ contains arbitrary large numbers.

Working with a simply-typed λ-calculus with sums and recursive types (a.k.a. FPC) [31], we are interested in specifications interpreted in a denotational semantics: since a stream (as opposed to e.g. an integer) is an inherently infinite object, a specification for (e.g.) count should hold for any stream whatsoever, and not only for those definable in a given syntax.

This leads us to consider properties on infinite datatypes in Scott domains. Logics on top of domains are known since quite a long time. We resort to Abramsky's paradigm of "Domain Theory in Logical Form" [1, 2], which allows one to systematically generate a logic from a domain representing a type. These logics are obtained by Stone duality, the core of a rich interplay between domain theory, logic and topology (cf e.g. [20, 39, 42, 4, 10, 40, 13, 12]).

In a previous work [32] (based on [7]), we considered a refinement type system which is sound and complete for saturated (i.e. upward-closed) properties on Scott domains. This strictly extends [1, 2]. Indeed, saturated properties cover least and greatest fixpoints of negation-free formulae in modal μ-calculi on the corresponding types. On streams, this includes usual modalities of Linear Temporal Logic (LTL), such as $\Diamond$ (*eventually*, i.e. "there is a position such that") and $\Box$ (*always*, i.e. "for all positions") [33]. Logics like LTL, CTL (Computation Tree Logic) or modal μ-calculi are widely used to formulate properties on infinite objects (see e.g. [5, 14, 8]).

On the other hand, only compact-open properties are available in [1, 2]. But for each compact-open set K of streams, there is some $n \in \mathbb{N}$ such that membership to K is completely determined by prefixes of length n. This excludes proper uses of $\Diamond$ and $\Box$. However, the systems in [7, 32] are *infinitary*, in the sense that some rules have infinitely many premises.

In this paper, we devise a *finitary* refinement type system which is sound and complete for a family of Scott-open properties allowing for specifications of infinitary behaviours. This

extends [1, 2]. Types are refined by a (negation-free) fixpoint-like logic which relies on the well-known fact that the Scott domains interpreting recursive types are *spectral spaces*.

In a spectral space (and more generally in a *stably compact space* [24]), there is a symmetry between the open sets and the compact-saturated ones. We reflect this symmetry in the logic: we have *positive* formulae which are closed under least fixpoints and define Scott-open sets, and *negative* formulae which are closed under greatest fixpoints and define compact-saturated sets. Input-output specifications of functions are written using a realizability implication $\Vdash\!\!\mapsto$ with the expected (contra)variance w.r.t. polarities.

On streams, positive formulae are closed under $\Diamond$, while negative formulae are closed under $\Box$. Given a positive formula φ and a negative formula ψ, we have a positive formula $\Box\psi \Vdash\!\!\mapsto \Diamond\varphi$ which selects those functions that take a stream whose elements all satisfy ψ to a stream with at least one element satisfying φ. Consider now the composite

$$\mathsf{count} \circ \mathsf{bft} : \ \mathsf{Tree}\,\mathsf{Bool} \longrightarrow \mathsf{Stream}\,\mathsf{Nat}$$

where $\mathsf{bft} \colon \mathsf{Tree}\,\sigma \to \mathsf{Stream}\,\sigma$ is a breadth-first tree traversal. Negative formulae on trees are closed under the usual CTL-like modalities $\forall\Box$ (*invariantly*, i.e. "for all nodes") and $\exists\Box$ (*potentially always*, i.e. "there is an infinite path on which for all nodes"). Thanks to the realizability implication, for each $n \in \mathbb{N}$ there is a positive formula which expresses the following specification for $\mathsf{count} \circ \mathsf{bft}$: Given a totally defined tree of Booleans, if the children of the root have an infinite path with only tt's and an infinite path with only ff's, then $\mathsf{count} \circ \mathsf{bft}$ produces a stream which contains a totally defined number $\geq n$.

We prove a *Positive Completeness* result: our type system derives all sound positive specifications of λ-terms (thus including the above positive specifications for $\mathsf{count} \circ \mathsf{bft}$).

Positive completeness implies that checking positive formulae on λ-terms is semi-decidable. This is in line with the intuition that membership to a Scott-open set is akin to a reachability problem: evaluate the (semantics of the) λ-term until knowing that the open set is met. Our refinement type system essentially amounts to a syntactic approach to this reachability problem. We also note that checking positive formulae on λ-terms is *not* decidable.

Besides, an adaptation of Higher-Order Model Checking to input-output specifications for a specific format of higher-order tree transducers (weaker than our FPC) has been shown to be undecidable in [22].

At the technical level, our logic allows for prenex normal forms. To this end, least and greatest fixpoints are decomposed as finite iterations indexed by variables which are existentially and universally quantified, respectively. This builds on ideas in [19] (which is based on guarded recursion and states no completeness result).

Organisation of the paper. Our simply-typed λ-calculus is discussed in §2. The logic is introduced in §3, while refinement types are presented in §4. Section 5 covers denotational semantics, soundness and positive completeness, as well as some necessary material on spectral spaces. Finally, the conclusion (§6) discusses some further works and compare with more related works. Proofs are available in the Appendices.

2 The Pure System

The *pure types* (notation $\tau, \sigma, \dots$) are the closed types over the grammar

$$\tau \ ::= \ \mathsf{Unit} \ \mid \ \tau \times \tau \ \mid \ \tau \to \tau \ \mid \ \alpha \ \mid \ \mathsf{rec}\,\alpha.\tau \ \mid \ \tau + \tau$$

where α ranges over an infinite supply of *type variables*, and where $\mathsf{rec}\,\alpha.\tau$ binds α in τ.

$$\frac{}{\mathcal{E} \vdash \langle\rangle : \mathsf{Unit}} \qquad \frac{\mathcal{E}, x : \tau \vdash M : \tau}{\mathcal{E} \vdash \mathsf{fix}\, x.M : \tau} \qquad \frac{\mathcal{E} \vdash M : \tau[\mathsf{rec}\,\alpha.\tau/\alpha]}{\mathcal{E} \vdash \mathsf{fold}(M) : \mathsf{rec}\,\alpha.\tau} \qquad \frac{\mathcal{E} \vdash M : \mathsf{rec}\,\alpha.\tau}{\mathcal{E} \vdash \mathsf{unfold}(M) : \tau[\mathsf{rec}\,\alpha.\tau/\alpha]}$$

$$\frac{\mathcal{E} \vdash M : \tau_i}{\mathcal{E} \vdash \mathsf{in}_i(M) : \tau_1 + \tau_2} \qquad \frac{\mathcal{E} \vdash M : \tau_1 + \tau_2 \qquad \mathcal{E}, x_1 : \tau_1 \vdash N_1 : \sigma \qquad \mathcal{E}, x_2 : \tau_2 \vdash N_2 : \sigma}{\mathcal{E} \vdash \mathsf{case}\ M\ [x_1 \mapsto N_1 \mid x_2 \mapsto N_2] : \sigma}$$

Figure 1 Typing rules for pure types, where $i \in \{1, 2\}$.

Table 1 Functions on Streams and Trees.

countrec :	$\mathsf{Nat} \longrightarrow \mathsf{Stream\,Bool} \longrightarrow \mathsf{Stream\,Nat}$
$:=$	$\mathsf{fix}\, g.\lambda n.\lambda x.\ \mathsf{if}\ (\mathsf{hd}\, x)\ \mathsf{then}\ (g\ (1{+}n)\ (\mathsf{tl}\, x))\ \mathsf{else}\ (n :: (g\ n\ (\mathsf{tl}\, x)))$
count $:=$	$(\mathsf{countrec}\ 0) :\ \mathsf{Stream\,Bool} \longrightarrow \mathsf{Stream\,Nat}$
extract :	$\mathsf{Cont}\,\sigma \longrightarrow \sigma$
$:=$	$\mathsf{fix}\, e.\lambda c.\, \mathsf{unfold}\ c\ e$
Over :	$\mathsf{Cont}\,\sigma$
$:=$	$\mathsf{fix}\, c.\, \mathsf{fold}(\lambda k.\ k\ c)$
bftrec :	$\mathsf{Tree}\,\sigma \longrightarrow \mathsf{Cont}(\mathsf{Stream}\,\sigma) \longrightarrow \mathsf{Cont}(\mathsf{Stream}\,\sigma)$
$:=$	$\mathsf{fix}\, g.\lambda t.\lambda c.\, \mathsf{fold}\,\big(\lambda k.\ (\mathsf{label}\, t) :: \big(\mathsf{unfold}\ c\ \big(k \circ (g(\mathsf{left}\, t)) \circ (g(\mathsf{right}\, t))\big)\big)\big)$
bft $:=$	$\lambda t.\ \mathsf{extract}(\mathsf{bftrec}\ t\ \mathsf{Over}) :\ \mathsf{Tree}\,\sigma \longrightarrow \mathsf{Stream}\,\sigma$

We consider λ-terms from the grammar

$$M, N \ ::= \ \langle\rangle \mid \langle M, N\rangle \mid \pi_1(M) \mid \pi_2(M) \mid x \mid \lambda x.M \mid MN \mid \mathsf{fix}\, x.M \\ \mid \mathsf{fold}(M) \mid \mathsf{unfold}(M) \mid \mathsf{in}_1(M) \mid \mathsf{in}_2(M) \mid \mathsf{case}\ M\ [x_1 \mapsto N_1 \mid x_2 \mapsto N_2]$$

The term formers $\mathsf{fold}, \mathsf{unfold}, \pi_1, \pi_2, \mathsf{in}_1, \mathsf{in}_2$ are often written in curried form, so that e.g. $(\mathsf{fold}\, M)$ stands for $\mathsf{fold}(M)$. We write $M \circ N$ for $\lambda x.M(N\, x)$.

Terms are typed with judgements of the form $\mathcal{E} \vdash M : \tau$, where the *typing context* $\mathcal{E}$ is a list $x_1 : \sigma_1, \dots, x_n : \sigma_n$ with $x_i \neq x_j$ if $i \neq j$. The typing rules are those in Figure 1, together with the *basic rules* in Figure 7a *restricted to pure types* (i.e. with $\mathcal{E}$ as above and T, U pure types). Of course, each type τ is inhabited by the "diverging" term $\Omega_\tau := \mathsf{fix}\, x.x : \tau$.

▶ **Example 2.1.** The type $\mathsf{Bool} := \mathsf{Unit} + \mathsf{Unit}$ represents Booleans, with $\mathsf{tt} := \mathsf{in}_1(\langle\rangle)$ and $\mathsf{ff} := \mathsf{in}_2(\langle\rangle)$. The type of natural numbers is $\mathsf{Nat} := \mathsf{rec}\,\alpha.\,\mathsf{Unit} + \alpha$. For each $n \in \mathbb{N}$, we have a term $\mathsf{n} : \mathsf{Nat}$ defined as $0 := \mathsf{fold}(\mathsf{in}_1\,\langle\rangle)$ and $\mathsf{n}+1 := (1{+})\mathsf{n}$, where $(1{+}) : \mathsf{Nat} \to \mathsf{Nat}$ is the successor function $\lambda x.\, \mathsf{fold}(\mathsf{in}_2\, x)$. We can represent "$+\infty$" as $\mathsf{inf} := \mathsf{fix}\, x.\, 1{+}x : \mathsf{Nat}$.

The type of streams over σ is $\mathsf{Stream}\,\sigma := \mathsf{rec}\,\alpha.\,\sigma \times \alpha$. It is equipped with the constructor $\mathsf{Cons} := \lambda h.\lambda t.\, \mathsf{fold}\,\langle h, t\rangle : \sigma \to \mathsf{Stream}\,\sigma \to \mathsf{Stream}\,\sigma$. We use the infix notation $(M :: N)$ for $(\mathsf{Cons}\, M\, N)$. The usual *head* and *tail* functions are $\mathsf{hd} := \lambda s.\, \pi_1(\mathsf{unfold}\, s) : \mathsf{Stream}\,\sigma \to \sigma$ and $\mathsf{tl} := \lambda s.\, \pi_2(\mathsf{unfold}\, s) : \mathsf{Stream}\,\sigma \to \mathsf{Stream}\,\sigma$.

The type of binary trees over σ is $\mathsf{Tree}\,\sigma := \mathsf{rec}\,\alpha.\,\sigma \times (\alpha \times \alpha)$. The constructor $\mathsf{Node} : \sigma \to \mathsf{Tree}\,\sigma \to \mathsf{Tree}\,\sigma \to \mathsf{Tree}\,\sigma$ and the destructors $\mathsf{label} : \mathsf{Tree}\,\sigma \to \sigma$ and $\mathsf{left}, \mathsf{right} : \mathsf{Tree}\,\sigma \to \mathsf{Tree}\,\sigma$ are defined similarly as Cons, hd, tl on streams. ⌟

▶ **Example 2.2.** Table 1 defines functions on streams and trees mentioned in §1. The stream function count was discussed in §1. The notation $\mathsf{if}\ M\ \mathsf{then}\ N_1\ \mathsf{else}\ N_2$ stands for the term $\mathsf{case}\ M\ [_ \mapsto N_1 \mid _ \mapsto N_2]$. On trees, the function bft implements Martin

$$\frac{}{\mathsf{True} \in \mathcal{L}^s(\Sigma;\tau)} \qquad \frac{\varphi,\psi \in \mathcal{L}^s(\Sigma;\tau)}{\varphi \wedge \psi \in \mathcal{L}^s(\Sigma;\tau)} \qquad \frac{}{\mathsf{False} \in \mathcal{L}^s(\Sigma;\tau)} \qquad \frac{\varphi,\psi \in \mathcal{L}^s(\Sigma;\tau)}{\varphi \vee \psi \in \mathcal{L}^s(\Sigma;\tau)}$$

(a) Propositional connectives.

$$\frac{}{[\langle\rangle] \in \mathcal{L}^s(\Sigma;\mathsf{Unit})} \qquad \frac{\varphi \in \mathcal{L}^s(\Sigma;\tau_1)}{[\pi_1]\varphi \in \mathcal{L}^s(\Sigma;\tau_1 \times \tau_2)} \qquad \frac{\varphi \in \mathcal{L}^s(\Sigma;\tau_2)}{[\pi_2]\varphi \in \mathcal{L}^s(\Sigma;\tau_1 \times \tau_2)}$$

$$\frac{\varphi \in \mathcal{L}^s(\Sigma;\tau[\mathsf{rec}\,\alpha.\tau/\alpha])}{[\mathsf{fold}]\varphi \in \mathcal{L}^s(\Sigma;\mathsf{rec}\,\alpha.\tau)} \qquad \frac{\varphi \in \mathcal{L}^s(\Sigma;\tau_1)}{[\mathsf{in}_1]\varphi \in \mathcal{L}^s(\Sigma;\tau_1+\tau_2)} \qquad \frac{\varphi \in \mathcal{L}^s(\Sigma;\tau_2)}{[\mathsf{in}_2]\varphi \in \mathcal{L}^s(\Sigma;\tau_1+\tau_2)}$$

(b) Modalities.

$$\frac{(p\colon\tau) \in \Sigma}{p \in \mathcal{L}^s(\Sigma;\tau)} \qquad \frac{\varphi \in \mathcal{L}^s(\Sigma;\tau) \quad p \notin \mathrm{dom}(\Sigma)}{\varphi \in \mathcal{L}^s(\Sigma,p\colon\sigma;\tau)} \qquad \frac{\varphi \in \mathcal{L}^s(\Sigma,p\colon\tau;\tau)}{(\mu^t p)\varphi \in \mathcal{L}^s(\Sigma;\tau)} \qquad \frac{\varphi \in \mathcal{L}^s(\Sigma,p\colon\tau;\tau)}{(\nu^t p)\varphi \in \mathcal{L}^s(\Sigma;\tau)}$$

(c) Bounded iteration, where t is an iteration term.

$$\frac{\varphi \in \mathcal{L}^+(\Sigma;\tau) \quad k\ \mathsf{Pos}\ \varphi}{(\exists k)\varphi \in \mathcal{L}^+(\Sigma;\tau)} \qquad \frac{\varphi \in \mathcal{L}^-(\Sigma;\tau) \quad k\ \mathsf{Neg}\ \varphi}{(\forall k)\varphi \in \mathcal{L}^-(\Sigma;\tau)} \qquad \frac{\psi \in \mathcal{L}^{-s}(;\sigma) \quad \varphi \in \mathcal{L}^s(;\tau)}{\psi \Vdash \varphi \in \mathcal{L}^s(;\sigma \to \tau)}$$

(d) Quantification and realizability implication, with $-s$ defined as $-\pm = \pm$, $-+ = -$ and $-- = +$.

Figure 2 Formation rules for $\mathcal{L}^s$ (the predicates Pos and Neg are defined in Figure 3).

$$\frac{k \notin \mathrm{FV}(\varphi)}{k\ \mathsf{Pos}\ \varphi} \qquad \frac{k\ \mathsf{Pos}\ \varphi,\psi}{k\ \mathsf{Pos}\ \varphi \star \psi} \qquad \frac{k\ \mathsf{Pos}\ \varphi}{k\ \mathsf{Pos}\ [\triangle]\varphi} \qquad \frac{k\ \mathsf{Neg}\ \psi \quad k\ \mathsf{Pos}\ \varphi}{k\ \mathsf{Pos}\ \psi \Vdash \varphi} \qquad \frac{k\ \mathsf{Pos}\ \varphi}{k\ \mathsf{Pos}\ (\exists \ell)\varphi}$$

$$\frac{k \notin \mathrm{FV}(\varphi)}{k\ \mathsf{Neg}\ \varphi} \qquad \frac{k\ \mathsf{Neg}\ \varphi,\psi}{k\ \mathsf{Neg}\ \varphi \star \psi} \qquad \frac{k\ \mathsf{Neg}\ \varphi}{k\ \mathsf{Neg}\ [\triangle]\varphi} \qquad \frac{k\ \mathsf{Pos}\ \psi \quad k\ \mathsf{Neg}\ \varphi}{k\ \mathsf{Neg}\ \psi \Vdash \varphi} \qquad \frac{k\ \mathsf{Neg}\ \varphi}{k\ \mathsf{Neg}\ (\forall \ell)\varphi}$$

$$\frac{k\ \mathsf{Pos}\ \varphi}{k\ \mathsf{Pos}\ (\mu^t p)\varphi} \qquad \frac{k\ \mathsf{Pos}\ \varphi \quad k \notin \mathrm{FV}(t)}{k\ \mathsf{Pos}\ (\nu^t p)\varphi} \qquad \frac{k\ \mathsf{Neg}\ \varphi}{k\ \mathsf{Neg}\ (\nu^t p)\varphi} \qquad \frac{k\ \mathsf{Neg}\ \varphi \quad k \notin \mathrm{FV}(t)}{k\ \mathsf{Neg}\ (\mu^t p)\varphi}$$

Figure 3 The predicates Pos and Neg, where $\triangle \in \{\pi_1, \pi_2, \mathsf{in}_1, \mathsf{in}_2, \mathsf{fold}\}$, and where $\star$ is $\wedge$ or $\vee$.

Hofmann's breadth-first traversal (see e.g. [6, 19, 32, 21]). It uses the recursive type $\mathsf{Cont}\,\sigma := \mathsf{rec}\,\alpha.\,(\alpha \to \sigma) \to \sigma$. Note that the recursive type corresponding to $\mathsf{Cont}\,\sigma$ in [6, 19, 21] has an additional constructor (in place of our fixpoint $\mathsf{Over}\colon\ \mathsf{Cont}\,\sigma$). See Example 5.2 (§5.1). ⌟

3 The Logic

The main ingredient of this paper is the logic we use to annotate pure types when forming refinement types. This logic is in essence *manysorted*: for each pure type τ we have formulae φ *of type* τ, notation $\varphi \in \mathcal{L}^s(\tau)$, where s is $+$ for positive formulae and $-$ for a negative ones.

Positive and negative formulae are in essence closed under least and greatest fixpoints, respectively, and thus may contain fixpoint variables p. But least fixpoints $(\mu p)\varphi$ and greatest fixpoints $(\nu p)\psi$ are actually not directly available. They are defined as $(\exists k)(\mu^k p)\varphi$ and $(\forall \ell)(\nu^\ell p)\psi$, where $(\mu^k p)\varphi$ and $(\nu^\ell p)\psi$ represent bounded iterations $\varphi^k(\mathsf{False})$ and $\psi^\ell(\mathsf{True})$, respectively. This decomposition allows for a notion of *prenex* form (Proposition 3.9, §3.2). A fixpoint logic with iteration variables was already considered in [36].

Formulae are presented in §3.1, while deduction is discussed in §3.2.

3.1 Formulae

We assume given infinitely many *iteration variables* k, ℓ, etc. *Iteration terms*, intended to range over natural numbers, are given by the grammar

$$t \ ::= \ k \mid 0 \mid t+1$$

We also assume given *fixpoint variables* p, q, etc. *Fixpoint contexts* Σ are lists of the form $p_1 : \sigma_1, \dots, p_n : \sigma_n$ where $p_i \neq p_j$ whenever $i \neq j$. We let $\mathrm{dom}(\Sigma)$ be $\{p_1, \dots, p_n\}$. Figure 2 gathers formation rules for sets $\mathcal{L}^s(\Sigma;\tau)$ indexed by fixpoint contexts and types. We write $\mathcal{L}^s(\tau)$ or $\mathcal{L}^s(;\tau)$ for $\mathcal{L}^s(\Sigma;\tau)$ when the fixpoint context Σ is empty.

▶ **Definition 3.1** (Formulae). *The sets $\mathcal{L}^{\pm}(\Sigma;\tau)$, $\mathcal{L}^{+}(\Sigma;\tau)$ and $\mathcal{L}^{-}(\Sigma;\tau)$ of respectively* neutral *$(\pm)$*, positive *$(+)$ and* negative *$(-)$* formulae *are defined by the rules in Figure 2 with $s \in \{\pm, +, -\}$, and using the predicates* Pos *and* Neg *defined in Figure 3.*

$\mathcal{L}^{\omega}(\tau)$ is the fragment of $\mathcal{L}^{\pm}(\tau)$ without *the rules in Figure 2c (but with $\Vdash\!\!\mapsto$ from Figure 2d). $\mathcal{L}^{\wedge}(\tau)$ is the fragment of $\mathcal{L}^{\omega}(\tau)$ without binary disjunctions $\vee$ (but with* False*).*

Note that $\mathcal{L}^{\pm}$, $\mathcal{L}^{\omega}$ and $\mathcal{L}^{\wedge}$ are stable under the realizability implication $\Vdash\!\!\mapsto$. In fact, $\mathcal{L}^{\pm}(\Sigma;\tau) = \mathcal{L}^{+}(\Sigma;\tau) \cap \mathcal{L}^{-}(\Sigma;\tau)$ and $\mathcal{L}^{\omega}(\tau) \subseteq \mathcal{L}^{\pm}(\tau)$ are defined by the rules in Figure 2a–2c and Figure 2a–2b respectively, together with

$$\frac{\psi \in \mathcal{L}^{\pm}(;\sigma) \qquad \varphi \in \mathcal{L}^{\pm}(;\tau)}{\psi \Vdash\!\!\mapsto \varphi \in \mathcal{L}^{\pm}(;\sigma \to \tau)} \qquad\qquad \frac{\psi \in \mathcal{L}^{\omega}(\sigma) \qquad \varphi \in \mathcal{L}^{\omega}(\tau)}{\psi \Vdash\!\!\mapsto \varphi \in \mathcal{L}^{\omega}(\sigma \to \tau)}$$

On the other hand, $\Vdash\!\!\mapsto$ is contravariant in its first argument w.r.t. $+$ and $-$:

$$\frac{\psi \in \mathcal{L}^{-}(;\sigma) \qquad \varphi \in \mathcal{L}^{+}(;\tau)}{\psi \Vdash\!\!\mapsto \varphi \in \mathcal{L}^{+}(;\sigma \to \tau)} \qquad\qquad \frac{\psi \in \mathcal{L}^{+}(;\sigma) \qquad \varphi \in \mathcal{L}^{-}(;\tau)}{\psi \Vdash\!\!\mapsto \varphi \in \mathcal{L}^{-}(;\sigma \to \tau)} \tag{1}$$

In any case, $\varphi \Vdash\!\!\mapsto \psi$ contains no free fixpoint variables. Also, formulae $\varphi \in \mathcal{L}^{\omega}(\tau)$ have no fixpoint variables nor iteration variables at all. When possible, we write $\mathcal{L}$ for $\mathcal{L}^{s}$.

Note that in Figure 3, we could have had a rule stating that $k\ \mathsf{Pos}\ (\forall\ell)\varphi$ if $k\ \mathsf{Pos}\ \varphi$. But this would have been useless for the left rule in Figure 2d, since a positive formula has no positive occurrence of a $(\forall\ell)$. A dual remark applies to $k\ \mathsf{Neg}\ (\exists\ell)\varphi$.

The semantics of formulae is discussed in §5.3. Their intended meaning is as follows. The formula $\psi \Vdash\!\!\mapsto \varphi \in \mathcal{L}(\sigma \to \tau)$ is intended to select those $M : \sigma \to \tau$ such that φ holds on $MN : \tau$ whenever ψ holds on $N : \sigma$. The formula $[\mathsf{fold}]\varphi$ holds on M if, and only if, φ holds on $\mathsf{unfold}(M)$. For $i = 1, 2$, the formula $[\pi_i]\varphi$ selects those $M : \tau_1 \times \tau_2$ such that φ holds on $\pi_i(M)$, while $[\mathsf{in}_i]\varphi$ selects those $\mathsf{in}_j(M) : \tau_1 + \tau_2$ such that $j = i$ and φ holds on M. Hence $[\mathsf{in}_1]\mathsf{True} \wedge [\mathsf{in}_2]\mathsf{True}$ is a contradiction and $[\mathsf{in}_1]\mathsf{True} \vee [\mathsf{in}_2]\mathsf{True}$ fails on $\Omega_{\tau_1+\tau_2}$.

▶ **Example 3.2.** We derive some composite modalities in $\mathcal{L}^{\omega}$ on the datatypes of Example 2.1. On Bool, we have formulae $[\mathsf{tt}] \coloneqq [\mathsf{in}_1][\langle\rangle]$ and $[\mathsf{ff}] \coloneqq [\mathsf{in}_2][\langle\rangle]$ which hold on $\mathsf{tt}, \mathsf{ff} : \mathsf{Bool}$, respectively. The formula $[\mathsf{tt}] \vee [\mathsf{ff}]$, which expresses totality at type Bool, fails on Ω_{Bool}.

On Nat, we have $[0] \coloneqq [\mathsf{fold}][\mathsf{in}_1][\langle\rangle]$ and a composite modality $[1+]\varphi \coloneqq [\mathsf{fold}][\mathsf{in}_2]\varphi$. By taking $[\mathsf{n}+1] \coloneqq [1+][\mathsf{n}]$, we have for each $n \in \mathbb{N}$ a formula $[\mathsf{n}] \in \mathcal{L}(\mathsf{Nat})$ which holds on $\mathsf{n} : \mathsf{Nat}$. We also have a composite modality $[\geq \mathsf{n}]\varphi$, defined as $[\geq 0]\varphi \coloneqq \varphi$ and $[\geq \mathsf{n}+1] \coloneqq [1+][\geq \mathsf{n}]\varphi$, such that $[\geq \mathsf{n}]\mathsf{True}$ holds on $\mathsf{inf} : \mathsf{Nat}$ and on $\mathsf{m} : \mathsf{Nat}$ for all $m \geq n$.

On streams $\mathsf{Stream}\,\sigma$, the composite modalities $[\mathsf{hd}]$ and $[\mathsf{tl}]$ are defined as $[\mathsf{hd}]\psi \coloneqq [\mathsf{fold}][\pi_1]\psi$ and $[\mathsf{tl}]\varphi \coloneqq [\mathsf{fold}][\pi_2]\varphi$. Given $\psi \in \mathcal{L}(\sigma)$ and $\varphi \in \mathcal{L}(\mathsf{Stream}\,\sigma)$, the formulae $[\mathsf{hd}]\psi \in \mathcal{L}(\mathsf{Stream}\,\sigma)$ and $[\mathsf{tl}]\varphi \in \mathcal{L}(\mathsf{Stream}\,\sigma)$ select those streams M such that ψ holds on $(\mathsf{hd}\,M)$ and such that φ holds on $(\mathsf{tl}\,M)$, respectively. We write $\bigcirc\varphi$ for $[\mathsf{tl}]\varphi$.

Similarly, on trees $\mathsf{Tree}\,\sigma$ one can define $[\mathsf{label}]$, $[\mathsf{left}]$ and $[\mathsf{right}]$ such that $[\mathsf{label}]\psi \in \mathcal{L}(\mathsf{Tree}\,\sigma)$ given $\psi \in \mathcal{L}(\sigma)$, and $[\mathsf{left}]\varphi, [\mathsf{right}]\varphi \in \mathcal{L}(\mathsf{Tree}\,\sigma)$ given $\varphi \in \mathcal{L}(\mathsf{Tree}\,\sigma)$. We also have composite modalities $\exists\bigcirc\varphi \coloneqq [\mathsf{left}]\varphi \vee [\mathsf{right}]\varphi$ and $\forall\bigcirc\varphi \coloneqq [\mathsf{left}]\varphi \wedge [\mathsf{right}]\varphi$ which allow for taking, respectively, disjunctions and conjunctions over the children of a node. ⌟

We now discuss some uses of fixpoint and iteration variables in $\mathcal{L}^{\pm}$, $\mathcal{L}^{+}$ and $\mathcal{L}^{-}$. Let $\varphi \in \mathcal{L}(\Sigma, p\colon \tau; \tau)$. Given $\psi \in \mathcal{L}(\Sigma;\tau)$ and $n \in \mathbb{N}$, we define $(p.\varphi)^n(\psi) \in \mathcal{L}(\Sigma;\tau)$ as

$$(p.\varphi)^0(\psi) \quad\coloneqq\quad \psi \qquad \text{and} \qquad (p.\varphi)^{n+1}(\psi) \quad\coloneqq\quad \varphi[(p.\varphi)^n(\psi)/p] \tag{2}$$

If t is a closed iteration term representing $n \in \mathbb{N}$ (i.e. t is the nth successor of 0), then the formulae $(\mu^t p)\varphi$ and $(\nu^t p)\varphi$ should be read as $(p.\varphi)^n(\mathsf{False})$ and $(p.\varphi)^n(\mathsf{True})$, respectively (cf Equation (5) in §3.2 below). But in general, the iteration term t may contain free iteration variables. In fact, we shall see in Example 5.11 (§5.3) that the least fixpoint of $\varphi \in \mathcal{L}^{+}(p\colon \tau;\tau)$ and the greatest fixpoint of $\psi \in \mathcal{L}^{-}(p\colon \tau;\tau)$ can be represented respectively as

$$(\exists k)(\mu^k p)\varphi \in \mathcal{L}^{+}(\tau) \qquad \text{and} \qquad (\forall \ell)(\nu^\ell p)\psi \in \mathcal{L}^{-}(\tau) \tag{3}$$

▶ **Example 3.3.** The representation of fixpoints in Equation (3) allows for formulae expressing possibly unbounded behaviours. For instance, totality at type Nat is expressed by the *positive* formula $[\mathsf{tot}] \coloneqq (\exists k)(\mu^k p)([\mathsf{0}] \vee [\mathsf{1}+]p)$, intended to hold on all $\mathsf{n}\colon \mathsf{Nat}$ but not on $\mathsf{inf}\colon \mathsf{Nat}$.

On streams $\mathsf{Stream}\,\sigma$, the usual LTL-like *eventually* modality $\Diamond$ is available on *positive* formulae $\varphi \in \mathcal{L}^{+}(\mathsf{Stream}\,\sigma)$, namely $\Diamond\varphi \coloneqq (\exists k)(\mu^k p)(\varphi \vee \bigcirc p)$. The formula $\Diamond\varphi$ is intended to hold on those $M\colon \mathsf{Stream}\,\sigma$ such that φ holds on $(\mathsf{tl}^n\, M)$ for some $n \in \mathbb{N}$.

Regarding greatest fixpoints, the CTL-like modalities $\forall\Box$ and $\exists\Box$ on $\mathsf{Tree}\,\sigma$ are available for *negative* formulae: given $\psi \in \mathcal{L}^{-}(\mathsf{Tree}\,\sigma)$, we let $\forall\Box\psi \coloneqq (\forall\ell)(\nu^\ell p)(\psi \wedge \forall\bigcirc p)$ and $\exists\Box\psi \coloneqq (\forall\ell)(\nu^\ell p)(\psi \wedge \exists\bigcirc p)$. The intended meaning of $\forall\Box[\mathsf{label}]\psi \in \mathcal{L}^{-}(\mathsf{Tree}\,\sigma)$ is to select those trees of σ's whose node labels all satisfy $\psi \in \mathcal{L}^{-}(\sigma)$, while $\exists\Box[\mathsf{label}]\psi \in \mathcal{L}^{-}(\mathsf{Tree}\,\sigma)$ asks ψ to hold on all labels in some infinite path.

We thus obtain for each $n \in \mathbb{N}$ the following positive formula

$$\forall\Box[\mathsf{label}]([\mathsf{tt}] \vee [\mathsf{ff}]) \;\wedge\; \exists\bigcirc\exists\Box[\mathsf{label}][\mathsf{tt}] \;\wedge\; \exists\bigcirc\exists\Box[\mathsf{label}][\mathsf{ff}] \quad \Vdash\!\!\to \quad \Diamond[\mathsf{hd}][\geq \mathsf{n}][\mathsf{tot}] \tag{4}$$

in $\mathcal{L}^{+}(\mathsf{Tree}\,\mathsf{Bool} \to \mathsf{Stream}\,\mathsf{Nat})$, which expresses the specification for $\mathsf{count} \circ \mathsf{bft}$ devised in §1: Given a totally defined tree of Booleans, if the children of the root have an infinite path with only tt's and an infinite path with only ff's, then $\mathsf{count} \circ \mathsf{bft}$ produces a stream which contains a totally defined number $\geq n$. Note that fixpoints of the same polarity can be nested, e.g. $\Diamond[\mathsf{hd}][\geq \mathsf{n}][\mathsf{tot}] \in \mathcal{L}^{+}(\mathsf{Stream}\,\mathsf{Nat})$ with $[\mathsf{tot}] \in \mathcal{L}^{+}(\mathsf{Nat})$. ⌟

▶ Remark 3.4 (Non-Examples). Given a *negative* $\psi \in \mathcal{L}^{-}(\mathsf{Stream}\,\sigma)$, we have the usual *always* modality $\Box\psi \coloneqq (\forall\ell)(\nu^\ell p)(\psi \wedge \bigcirc p) \in \mathcal{L}^{-}(\mathsf{Stream}\,\sigma)$ which holds on M if ψ holds on $(\mathsf{tl}^n\, M)$ for all $n \in \mathbb{N}$. A composite $\Box\Diamond[\mathsf{hd}]\varphi$ would require a stream to have infinitely many elements satisfying φ. But $\Box\Diamond[\mathsf{hd}]\varphi$ is not a formula since $\Diamond[\mathsf{hd}]\varphi$ is not negative.

Also, in order to universally quantify over $n \in \mathbb{N}$ in the r.-h.s. of (4), one could think of $(\forall\ell)\Diamond[\mathsf{hd}](\mu^\ell p)([\mathsf{tot}] \vee [\mathsf{1}+]p)$. This is not a formula since ψ has to be negative in $(\forall\ell)\psi$. ⌟

3.2 Deduction

Deduction will enter the refinement type system via subtyping (Figure 6 in §4).

$$\frac{}{\varphi \vdash \varphi} \qquad \frac{\psi \vdash \theta \qquad \theta \vdash \varphi}{\psi \vdash \varphi} \qquad (\mathsf{D})\frac{}{\psi \wedge (\varphi \vee \theta) \vdash (\psi \wedge \varphi) \vee (\psi \wedge \theta)}$$

$$\frac{\psi \vdash \varphi_1 \qquad \psi \vdash \varphi_2}{\psi \vdash \varphi_1 \wedge \varphi_2} \qquad \frac{\psi_1 \vdash \varphi}{\psi_1 \wedge \psi_2 \vdash \varphi} \qquad \frac{\psi_1 \vdash \varphi}{\psi_1 \wedge \psi_2 \vdash \varphi} \qquad \frac{}{\psi \vdash \mathsf{True}}$$

$$\frac{\psi \vdash \varphi_1}{\psi \vdash \varphi_1 \vee \varphi_2} \qquad \frac{\psi \vdash \varphi_2}{\psi \vdash \varphi_1 \vee \varphi_2} \qquad \frac{\psi_1 \vdash \varphi \qquad \psi_2 \vdash \varphi}{\psi_1 \vee \psi_2 \vdash \varphi} \qquad \frac{}{\mathsf{False} \vdash \varphi}$$

(a) Propositional rules.

$$(\mathsf{I})\frac{}{[\mathsf{in}_1]\varphi \wedge [\mathsf{in}_2]\psi \vdash \mathsf{False}} \qquad \frac{\psi \vdash \varphi}{[\triangle]\psi \vdash [\triangle]\varphi} \qquad \frac{\psi' \vdash \psi \qquad \varphi \vdash \varphi'}{\psi \Vdash \varphi \vdash \psi' \Vdash \varphi'}$$

$$\frac{}{[\triangle]\varphi_1 \wedge [\triangle]\varphi_2 \vdash [\triangle](\varphi_1 \wedge \varphi_2)} \qquad \frac{\triangle \notin \{\mathsf{in}_1, \mathsf{in}_2\}}{\mathsf{True} \vdash [\triangle]\mathsf{True}} \qquad \frac{}{(\forall k)[\triangle]\varphi \vdash [\triangle](\forall k)\varphi}$$

$$\frac{}{[\triangle](\varphi_1 \vee \varphi_2) \vdash [\triangle]\varphi_1 \vee [\triangle]\varphi_2} \qquad \frac{}{[\triangle]\mathsf{False} \vdash \mathsf{False}} \qquad \frac{}{[\triangle](\exists k)\varphi \vdash (\exists k)[\triangle]\varphi}$$

(b) Modal rules, where $\triangle \in \{\pi_1, \pi_2, \mathsf{in}_1, \mathsf{in}_2, \mathsf{fold}\}$.

$$\frac{\psi \vdash \varphi[t/k]}{\psi \vdash (\exists k)\varphi} \qquad (\exists\mathsf{L})\frac{\theta \wedge \varphi \vdash \psi \qquad k \notin \mathrm{FV}(\psi, \theta)}{\theta \wedge (\exists k)\varphi \vdash \psi} \qquad (\exists\mathsf{M})\frac{}{(\exists k)(\exists \ell)\varphi \vdash (\exists k)\varphi[k/\ell]}$$

$$\frac{\varphi[t/k] \vdash \psi}{(\forall k)\varphi \vdash \psi} \qquad (\forall\mathsf{R})\frac{\psi \vdash \varphi \vee \theta \qquad k \notin \mathrm{FV}(\psi, \theta)}{\psi \vdash (\forall k)\varphi \vee \theta} \qquad (\forall\mathsf{M})\frac{}{(\forall k)\varphi[k/\ell] \vdash (\forall k)(\forall \ell)\varphi}$$

$$(\iota/\exists)\frac{k \notin \mathrm{FV}(t)}{(\iota^t p)(\exists k)\varphi \vdash (\exists k)(\iota^t p)\varphi} \qquad (\forall/\iota)\frac{k \notin \mathrm{FV}(t)}{(\forall k)(\iota^t p)\varphi \vdash (\iota^t p)(\forall k)\varphi}$$

(c) Quantifier rules, where $\iota \in \{\mu, \nu\}$.

$$\frac{}{(\mu^0 p)\varphi \vdash \mathsf{False}} \qquad \frac{(\mu^t p)\varphi \vdash \psi}{(\mu^{t+1} p)\varphi \vdash \varphi[\psi/p]} \qquad \frac{}{\varphi[(\mu^t p)\varphi/p] \vdash (\mu^{t+1} p)\varphi} \qquad \frac{\psi \vdash \varphi}{(\mu^t p)\psi \vdash (\mu^t p)\varphi}$$

$$\frac{}{\mathsf{True} \vdash (\nu^0 p)\varphi} \qquad \frac{\psi \vdash (\nu^t p)\varphi}{\varphi[\psi/p] \vdash (\nu^{t+1} p)\varphi} \qquad \frac{}{(\nu^{t+1} p)\varphi \vdash \varphi[(\nu^t p)\varphi/p]} \qquad \frac{\psi \vdash \varphi}{(\nu^t p)\psi \vdash (\nu^t p)\varphi}$$

(d) Bounded iteration.

$$(\mathsf{WF})\frac{k \notin \mathrm{FV}(\varphi)}{(\forall k)\psi \Vdash \varphi \vdash (\exists k)(\psi \Vdash \varphi)} \qquad (\mathsf{CC})\frac{\psi \in \mathcal{L}^{\pm} \qquad k \notin \mathrm{FV}(\psi)}{\psi \Vdash (\exists k)\varphi \vdash (\exists k)(\psi \Vdash \varphi)}$$

$$(\exists/\Vdash)\frac{k \notin \mathrm{FV}(\varphi)}{(\forall k)(\psi \Vdash \varphi) \vdash (\exists k)\psi \Vdash \varphi} \qquad (\Vdash/\forall)\frac{k \notin \mathrm{FV}(\psi)}{(\forall k)(\psi \Vdash \varphi) \vdash \psi \Vdash (\forall k)\varphi}$$

$$(\Vdash/\wedge)\frac{}{(\psi \Vdash \varphi_1) \wedge (\psi \Vdash \varphi_2) \vdash \psi \Vdash (\varphi_1 \wedge \varphi_2)} \qquad \frac{}{\mathsf{True} \vdash (\psi \Vdash \mathsf{True})}$$

$$(\vee/\Vdash)\frac{}{(\psi_1 \Vdash \varphi) \wedge (\psi_1 \Vdash \varphi) \vdash (\psi_1 \vee \psi_2) \Vdash \varphi} \qquad \frac{}{\mathsf{True} \vdash (\mathsf{False} \Vdash \varphi)}$$

$$(\Vdash/\vee)\frac{\delta \in \mathcal{L}^{\wedge}}{\delta \Vdash (\varphi_1 \vee \varphi_2) \vdash (\delta \Vdash \varphi_1) \vee (\delta \Vdash \varphi_2)} \qquad (\mathsf{C})\frac{\mathcal{C}(\delta) \qquad \delta \in \mathcal{L}^{\wedge}}{(\delta \Vdash \mathsf{False}) \vdash \mathsf{False}}$$

(e) Rules for the realizability implication $\Vdash$ (the predicate $\mathcal{C}$ is defined in Figure 5).

Figure 4 Deduction rules.

$$\frac{}{\mathcal{C}([\langle\rangle])} \qquad \frac{\mathcal{C}(\varphi)}{\mathcal{C}([\mathsf{fold}]\varphi)} \qquad \frac{\mathcal{C}(\varphi) \quad \mathcal{C}(\psi)}{\mathcal{C}([\pi_1]\varphi \wedge [\pi_2]\psi)} \qquad \frac{\mathcal{C}(\varphi)}{\mathcal{C}([\mathsf{in}_1]\varphi)} \qquad \frac{\mathcal{C}(\varphi)}{\mathcal{C}([\mathsf{in}_2]\varphi)}$$

$$\frac{}{\mathcal{C}(\mathsf{True})} \qquad \frac{\mathcal{C}(\psi) \quad \psi \vdash \varphi \quad \varphi \in \mathcal{L}^{\wedge}}{\mathcal{C}(\varphi)} \qquad \frac{\begin{array}{c} I \text{ finite and } \forall i \in I,\ \mathcal{C}(\psi_i) \text{ and } \mathcal{C}(\varphi_i); \\ \forall J \subseteq I,\ \bigwedge_{j\in J} \psi_j \vdash \mathsf{False} \ \text{ or } \ \mathcal{C}\left(\bigwedge_{j\in J} \varphi_j\right) \end{array}}{\mathcal{C}\left(\bigwedge_{i\in I}(\psi_i \parallel\!\!\rightarrow \varphi_i)\right)}$$

Figure 5 The consistency predicate $\mathcal{C}$.

▶ **Definition 3.5** (Deduction). *A* sequent *has the form* $\psi \vdash \varphi$ *where* $\varphi, \psi \in \mathcal{L}^s(\Sigma;\tau)$. *The* derivable *sequents are defined by the rules in Figure 4, using the* consistency predicate $\mathcal{C}$ *defined in Figure 5. Write* $\psi \dashv\vdash \varphi$ *when the sequents* $\psi \vdash \varphi$ *and* $\varphi \vdash \psi$ *are both derivable.*

We discuss some rules in Figure 4, heading to basic properties of the deduction system. We begin with a notion of normal form for $\mathcal{L}^{\omega}$. Let $\mathcal{L}^{\vee\wedge}(\tau)$ be the closure of $\mathcal{L}^{\wedge}(\tau)$ under disjunctions (but under no other rules in Figure 2, so e.g. $[\pi_1](\varphi \vee \psi)$ is *not* in $\mathcal{L}^{\vee\wedge}$).

As usual, the converse of the distributive rule (D) in Figure 4a is derivable, and so is the dual law $\psi \vee (\varphi \wedge \theta) \dashv\vdash (\psi \vee \varphi) \wedge (\psi \vee \theta)$. Moreover, given $\triangle \in \{\pi_i, \mathsf{in}_i, \mathsf{fold}\}$ we can derive $[\triangle](\varphi \vee \psi) \dashv\vdash [\triangle]\varphi \vee [\triangle]\psi$ (and similarly for $\wedge$). It follows that for each $\varphi \in \mathcal{L}^{\vee\wedge}$, there is some $\psi \in \mathcal{L}^{\vee\wedge}$ such that $[\triangle]\varphi \dashv\vdash \psi$. Using the rules $(\vee/\!\parallel\!\!\rightarrow)$ and $(\parallel\!\!\rightarrow/\vee)$ in Figure 4e, we can similarly obtain that for each $\varphi_1, \varphi_2 \in \mathcal{L}^{\vee\wedge}$ there is some $\psi \in \mathcal{L}^{\vee\wedge}$ such that $(\varphi_1 \parallel\!\!\rightarrow \varphi_2) \dashv\vdash \psi$ (but beware that the rule $(\parallel\!\!\rightarrow/\vee)$ requires δ to be in $\mathcal{L}^{\wedge}$). Hence:

▶ **Lemma 3.6.** *For each* $\varphi \in \mathcal{L}^{\omega}(\tau)$, *there is some* $\psi \in \mathcal{L}^{\vee\wedge}(\tau)$ *such that* $\varphi \dashv\vdash \psi$.

▶ Remark 3.7. The rule (C) in Figure 4e is a kind of a zero-ary case of $(\parallel\!\!\rightarrow/\vee)$. Since $\mathsf{True} \vdash (\mathsf{False} \parallel\!\!\rightarrow \mathsf{False})$, the sequent $(\psi \parallel\!\!\rightarrow \mathsf{False}) \vdash \mathsf{False}$ is derivable only when ψ is consistent. This is enforced by the consistency predicate $\mathcal{C}(\psi)$, see Theorem 5.13(1) in §5.3. Our predicate $\mathcal{C}$ is akin to [1, 32], but differs from the *coprimeness* predicates of [2, 7, 4]. Note that the clauses defining $\mathcal{C}$ are positive (compare with [7, Figure 3] and [4, Figure 10.3]).

At sum types, we have inconsistent formulae which do not mention False: The rule (I) in Figure 4b means that it is inconsistent to assert both $[\mathsf{in}_1]\mathsf{True}$ and $[\mathsf{in}_2]\mathsf{True}$. Hence, our sum types are *disjoint*, and we do not need the termination predicates of [1, 2].

Besides, note that the sequent $\mathsf{True} \vdash [\mathsf{in}_1]\mathsf{True} \vee [\mathsf{in}_2]\mathsf{True}$ is *not* derivable. ⌟

If t is a closed iteration term, then using the rules in Figure 4d we can derive

$$(\mu^t p)\varphi \quad \dashv\vdash \quad (p.\varphi)^n(\mathsf{False}) \qquad \text{and} \qquad (\nu^t p)\varphi \quad \dashv\vdash \quad (p.\varphi)^n(\mathsf{True}) \tag{5}$$

where t is the nth successor of 0 (cf Equation (2) in §3.1). This yields:

▶ **Lemma 3.8.** *For each* closed $\varphi \in \mathcal{L}^{\pm}(\tau)$, *there is some* $\psi \in \mathcal{L}^{\omega}(\tau)$ *such that* $\varphi \dashv\vdash \psi$.

The main fact of this §3.2 is that quantifiers can be assumed to be in prenex position. Quantifiers of the same type can be merged using the rules (∃M) and (∀M) in Figure 4c, themselves permitted by the polarity constraints in Figure 2d. Formally, the *positive*, resp. *negative*, *prenex forms* are the formulae $(\exists k)\psi$, resp. $(\forall k)\psi$, with $\psi \in \mathcal{L}^{\pm}$ neutral.

The rules (WF), (CC) in Figure 4e, which apply to positive formulae, are crucial to obtain prenex forms. Note also that the *Frobenius* rules (∃L) and (∀R) in Figure 4c make it possible to derive the following when $k \notin \mathrm{FV}(\psi)$:

$$\psi \wedge (\exists k)\varphi \quad \dashv\vdash \quad (\exists k)(\psi \wedge \varphi) \qquad \text{and} \qquad (\forall k)(\varphi \vee \psi) \quad \dashv\vdash \quad (\forall k)\varphi \vee \psi$$

▶ **Proposition 3.9.** *For each* $\varphi \in \mathcal{L}^s(\Sigma;\tau)$, *there is a prenex* $\psi \in \mathcal{L}^s(\Sigma;\tau)$ *such that* $\varphi \dashv\vdash \psi$.

$$\frac{}{T \preceq T} \qquad \frac{T \preceq U \qquad U \preceq V}{T \preceq V} \qquad \frac{}{T \preceq |T|} \qquad \frac{}{\tau \preceq \{\tau \mid \mathsf{True}\}} \qquad \frac{\psi \vdash \varphi}{\{\tau \mid \psi\} \preceq \{\tau \mid \varphi\}}$$

$$\frac{T \preceq T' \qquad U \preceq U'}{T \times U \preceq T' \times U'} \qquad \frac{}{\{\tau \mid \varphi\} \times \{\sigma \mid \psi\} \simeq \{\tau \times \sigma \mid [\pi_1]\varphi \wedge [\pi_2]\psi\}}$$

$$\frac{U' \preceq U \qquad T \preceq T'}{U \to T \preceq U' \to T'} \qquad \frac{}{\{\sigma \mid \psi\} \to \{\tau \mid \varphi\} \simeq \{\sigma \to \tau \mid \psi \Vdash \varphi\}}$$

Figure 6 Subtyping.

4 The Refinement Type System

We consider *neutral*, *positive* and *negative refinement types* (or *types*), notation T^s, U^s, etc. with $s = \pm, +, -$ respectively. They are given by the grammar

$$T^s, U^s \ ::= \ \tau \ \mid \ \{\tau \mid \varphi\} \ \mid \ T^s \times U^s \ \mid \ U^{-s} \to T^s$$

where τ is a pure type and where $\varphi \in \mathcal{L}^s(\tau)$ is *closed*. When possible, we write T for T^s.

▶ **Example 4.1.** For each $n \in \mathbb{N}$, the specification for $\mathsf{count} \circ \mathsf{bft}$ is given either by the positive refinement type $\{\mathsf{Tree\,Bool} \to \mathsf{Stream\,Nat} \mid \Phi^+\}$, where Φ^+ is the formula in Equation (4), Example 3.3, or by the positive refinement type S^+ defined as

$$\big\{ \mathsf{Tree\,Bool} \;\big|\; \forall\Box[\mathsf{label}](\langle\mathsf{tt}\rangle \vee \langle\mathsf{ff}\rangle) \ \wedge \ \exists\bigcirc\exists\Box[\mathsf{label}][\mathsf{tt}] \ \wedge \ \exists\bigcirc\exists\Box[\mathsf{label}][\mathsf{ff}] \big\} \ \longrightarrow \ \big\{ \mathsf{Stream\,Nat} \;\big|\; \Diamond[\mathsf{hd}][\geq \mathsf{n}][\mathsf{tot}] \big\}$$

⌟

We shall consider typing judgements of the form $\mathcal{E} \vdash M : T$, where $\mathcal{E}$ is allowed to mention refinement types. A judgement $M : \{\tau \mid \varphi\}$ is intended to mean that M is of pure type τ and satisfies φ. *Positive* typing judgements are of the form $\mathcal{E}^- \vdash M : T^+$, where only negative types are declared in $\mathcal{E}^-$.

We derive typing judgements $\mathcal{E} \vdash M : T$ using the rules in Figure 7 augmented with those in Figure 1 (§2). Deduction on formulae (§3.2) enters the type system in Figure 7c via the subtyping relation $U \preceq T$ defined in Figure 6, where $U \simeq T$ stands for the conjunction of $U \preceq T$ and $T \preceq U$. Note that for each τ we have $\tau \simeq \{\tau \mid \mathsf{True}\}$. Subtyping is extended to typing contexts: given $\mathcal{E} = x_1 : U_1, \ldots, x_n : U_n$ and $\mathcal{E}' = x_1 : U'_1, \ldots, x_n : U'_n$, we let $\mathcal{E} \preceq \mathcal{E}'$ when $U_i \preceq U'_i$ for all $i = 1, \ldots, n$. Note that if $\mathcal{E} \vdash M : T$ is derivable then so is $|\mathcal{E}| \vdash M : |T|$, where the *underlying pure type* $|T|$ of T is defined by induction from $|\tau| := \tau$ and $|\{\tau \mid \varphi\}| := \tau$, and where $|\mathcal{E}|$ is the extension of $|-|$ to $\mathcal{E}$.

The rules in Figures 6 and 7 are adaptations of those in [2, 7, 19, 32]. In particular, the crucial rule for $\mathsf{fix}\, x.M$ in Figure 7d comes from [2].

▶ **Example 4.2.** The bottom $\simeq$-rule in Figure 6 yields $S^+ \simeq \{\mathsf{Tree\,Bool} \to \mathsf{Stream\,Nat} \mid \Phi^+\}$ in Example 4.1, as well as the following derived typing rules.

$$\frac{\mathcal{E}, x : \{\sigma \mid \psi\} \vdash M : \{\tau \mid \varphi\}}{\mathcal{E} \vdash \lambda x.M : \{\sigma \to \tau \mid \psi \Vdash \varphi\}} \qquad \frac{\mathcal{E} \vdash M : \{\sigma \to \tau \mid \psi \Vdash \varphi\} \qquad \mathcal{E} \vdash N : \{\sigma \mid \psi\}}{\mathcal{E} \vdash MN : \{\tau \mid \varphi\}}$$

▶ **Lemma 4.3.** *For each type T^s, there is a $\varphi \in \mathcal{L}^s(|T|)$ such that $T^s \simeq \{|T^s| \mid \varphi\}$.*

$$\frac{(x\colon T)\in\mathcal{E}}{\mathcal{E}\vdash x\colon T}\qquad\frac{\mathcal{E},\,x\colon U\vdash M\colon T}{\mathcal{E}\vdash\lambda x.M\colon U\to T}\qquad\frac{\mathcal{E}\vdash M\colon U\to T\qquad\mathcal{E}\vdash N\colon U}{\mathcal{E}\vdash MN\colon T}$$

$$\frac{\mathcal{E}\vdash M\colon T\qquad\mathcal{E}\vdash N\colon U}{\mathcal{E}\vdash\langle M,N\rangle\colon T\times U}\qquad\frac{\mathcal{E}\vdash M\colon T\times U}{\mathcal{E}\vdash\pi_1(M)\colon T}\qquad\frac{\mathcal{E}\vdash M\colon T\times U}{\mathcal{E}\vdash\pi_2(M)\colon U}$$

(a) Basic rules.

$$\frac{\mathcal{E}\vdash M\colon\{\tau\mid\varphi_1\}\qquad\mathcal{E}\vdash M\colon\{\tau\mid\varphi_2\}}{\mathcal{E}\vdash M\colon\{\tau\mid\varphi_1\wedge\varphi_2\}}\qquad\frac{|\mathcal{E}|,\,x\colon\sigma,\,|\mathcal{E}'|\vdash M\colon|T|}{\mathcal{E},\,x\colon\{\sigma\mid\mathsf{False}\},\,\mathcal{E}'\vdash M\colon T}$$

$$\frac{\mathcal{E},\,x\colon\{\sigma\mid\psi_1\},\,\mathcal{E}'\vdash M\colon T\qquad\mathcal{E},\,x\colon\{\sigma\mid\psi_2\},\,\mathcal{E}'\vdash M\colon T}{\mathcal{E},\,x\colon\{\sigma\mid\psi_1\vee\psi_2\},\,\mathcal{E}'\vdash M\colon T}$$

(b) Propositional connectives.

$$\frac{\mathcal{E}'\preceq\mathcal{E}\qquad T\preceq T'\qquad\mathcal{E}\vdash M\colon T}{\mathcal{E}'\vdash M\colon T'}$$

(c) Subtyping (cf Figure 6).

$$\frac{\mathcal{E}\vdash\mathsf{fix}\,x.M\colon\{\tau\mid\psi\}\qquad\mathcal{E},x\colon\{\tau\mid\psi\}\vdash M\colon\{\tau\mid\varphi\}}{\mathcal{E}\vdash\mathsf{fix}\,x.M\colon\{\tau\mid\varphi\}}$$

(d) Fixpoints.

$$\frac{\mathcal{E}\vdash M\colon\{\tau_1\times\tau_2\mid[\pi_i]\varphi\}}{\mathcal{E}\vdash\pi_i(M)\colon\{\tau_i\mid\varphi\}}\qquad\frac{\mathcal{E}\vdash M_i\colon\{\tau_i\mid\varphi\}\qquad\mathcal{E}\vdash M_{3-i}\colon\tau_{3-i}}{\mathcal{E}\vdash\langle M_1,M_2\rangle\colon\{\tau_1\times\tau_2\mid[\pi_i]\varphi\}}$$

$$\frac{}{\mathcal{E}\vdash\langle\rangle\colon\{\mathsf{Unit}\mid[\langle\rangle]\}}\qquad\frac{\mathcal{E}\vdash M\colon\{\tau_i\mid\varphi\}}{\mathcal{E}\vdash\mathsf{in}_i(M)\colon\{\tau_1+\tau_2\mid[\mathsf{in}_i]\varphi\}}$$

$$\frac{\mathcal{E}\vdash M\colon\{\tau_1+\tau_2\mid[\mathsf{in}_i]\varphi\}\qquad\mathcal{E},x_i\colon\{\tau_i\mid\varphi\}\vdash N_i\colon T\qquad|\mathcal{E}|,x_{3-i}\colon\tau_{3-i}\vdash N_{3-i}\colon|T|}{\mathcal{E}\vdash\mathsf{case}\;M\;[x_1\mapsto N_1\mid x_2\mapsto N_2]\colon T}$$

$$\frac{\mathcal{E}\vdash M\colon\{\tau[\mathsf{rec}\,\alpha.\tau/\alpha]\mid\varphi\}}{\mathcal{E}\vdash\mathsf{fold}(M)\colon\{\mathsf{rec}\,\alpha.\tau\mid[\mathsf{fold}]\varphi\}}\qquad\frac{\mathcal{E}\vdash M\colon\{\mathsf{rec}\,\alpha.\tau\mid[\mathsf{fold}]\varphi\}}{\mathcal{E}\vdash\mathsf{unfold}(M)\colon\{\tau[\mathsf{rec}\,\alpha.\tau/\alpha]\mid\varphi\}}$$

(e) Modalities, where $i\in\{1,2\}$.

Figure 7 Typing rules.

▶ Remark 4.4. Refinement types are *not* closed under $+$ since in view of Remark 3.7, the equivalence $\{\tau\mid\varphi\}+\{\sigma\mid\psi\}\simeq\{\tau+\sigma\mid[\mathsf{in}_1]\varphi\vee[\mathsf{in}_2]\psi\}$ would be wrong, thus preventing from Lemma 4.3 to hold. ⌟

Our main result (Theorem 5.17) is that the *positive* fragment of this type system is (sound and) complete w.r.t. the usual Scott semantics: given $\vdash M\colon\tau$ and a *positive* $\varphi\in\mathcal{L}^+(\tau)$,

$$\vdash M\colon\{\tau\mid\varphi\}\qquad\text{if, and only if,}\qquad\varphi\text{ holds on }[\![M]\!]\text{ in the Scott semantics.}$$

In particular, the judgement $\vdash\mathsf{count}\circ\mathsf{bft}\colon S^+$ is derivable.

5 Semantics

We interpret pure types as Scott domains and λ-terms as Scott-continuous functions (§5.1). Using the well-known fact that Scott domains are spectral spaces (§5.2), this yields the semantics of our logic (§5.3) and of our refinement type system (§5.4). Our main result is the Positive Completeness Theorem 5.17 in §5.4.

We mostly use the terminology in [4, §1]. A *dcpo* is a poset with suprema of all directed subsets (a subset D of a poset is *directed* if all (possibly empty) finite $F \subseteq D$ have an upper bound in D; directed sets are non-empty). A *cpo* is a dcpo with a least element (often denoted $\bot$). A function between dcpos is *Scott-continuous* if it preserves the order (i.e. is monotone) as well as directed suprema.

▶ **Definition 5.1** (Scott Domain)**.** *A* Scott domain *is a bounded-complete algebraic cpo.* **Scott** *is the category of Scott domains and Scott-continuous functions.*

Recall that a cpo X is bounded-complete if any two $x, y \in X$ have a sup (or *least* upper bound) $x \vee y \in X$ whenever they have an upper bound.

An element x of a dcpo X is *finite* if for all directed $D \subseteq X$ such that $x \leq \bigvee D$, we have $x \leq d$ for some $d \in D$ (finite elements are called *compact* in [4]). Note that $\bot$ is always finite, and that if $d, d' \in X$ are finite, then $d \vee d'$ is finite whenever it exists. A dcpo X is *algebraic* if for each $x \in X$, the set $\{d \in X \mid d \text{ finite and } \leq x\}$ is directed and has sup x.

The category **Scott** is Cartesian-closed (see e.g. [3, Corollary 4.1.6]).

5.1 Semantics of the Pure System

Typed terms $\mathcal{E} \vdash M : \tau$ of the pure system (§2) are interpreted as morphisms $[\![M]\!] \colon [\![\mathcal{E}]\!] \to [\![\tau]\!]$ in **Scott**, where $[\![\mathcal{E}]\!] = \prod_{i=1}^{n} [\![\sigma_i]\!]$ when $\mathcal{E} = x_1 : \sigma_1, \dots, x_n : \sigma_n$. See Appendix 9 for details.

Product types $\tau \times \sigma$ are interpreted using the Cartesian product of **Scott**, i.e. the Cartesian product of sets equipped with component-wise order. Arrow types $\sigma \to \tau$ are interpreted using the closed structure of **Scott**, given by equipping each homset $\mathbf{Scott}(X, Y)$ with the pointwise order.

The interpretation of Unit is $\{\bot, \top\}$ with $[\![\langle\rangle]\!] \coloneqq \top$. We let $[\![\tau + \sigma]\!]$ be the *weak* coproduct $([\![\tau]\!] \amalg [\![\sigma]\!])_\bot$, i.e., the disjoint union of $[\![\tau]\!]$ and $[\![\sigma]\!]$ augmented with a new least element $\bot$ (**Scott** does not have finite coproducts [18]). We refer to [4, 3, 37] for the interpretation of recursive types $\mathsf{rec}\,\alpha.\tau$. Term-level fixpoints $\mathsf{fix}\,x.M$ are interpreted using the usual fixpoint combinators $\mathsf{Y} \colon (X \to X) \to X$ taking $f \colon X \to X$ to $\mathsf{Y}(f) \coloneqq \bigvee_{n \in \mathbb{N}} f^n(\bot)$.

▶ **Example 5.2.** The domain $[\![\mathsf{Stream}\,\sigma]\!]$ of streams is $[\![\sigma]\!]^{\mathbb{N}}$ equipped with the pointwise order, while the domain $[\![\mathsf{Tree}\,\sigma]\!]$ of trees is $[\![\sigma]\!]^{\{0,1\}^*}$. The finite elements are those $z \in [\![\sigma]\!]^I$ such that $z(p)$ is finite in $[\![\sigma]\!]$ for all $p \in I$, and $z(p) \neq \bot$ for at most finitely many $p \in I$.

Given $x \in [\![\mathsf{Stream}\,\sigma]\!]$ we have $[\![\mathsf{hd}]\!](x) = x(0)$, while $[\![\mathsf{tl}]\!](x)$ is the stream taking $n \in \mathbb{N}$ to $x(n+1) \in [\![\sigma]\!]$. Moreover, $x = [\![\mathsf{Cons}]\!]([\![\mathsf{hd}]\!](x), [\![\mathsf{tl}]\!](x))$. Similarly, if $y \in [\![\mathsf{Tree}\,\sigma]\!]$ then $[\![\mathsf{label}]\!](y) = y(\varepsilon)$ is the root label of y, while $[\![\mathsf{left}]\!](y)$ and $[\![\mathsf{right}]\!](y)$ are the left- and right-subtrees of y, respectively.

One can then check that $[\![\mathsf{bft}]\!] \colon [\![\mathsf{Tree}\,\sigma]\!] \to [\![\mathsf{Stream}\,\sigma]\!]$ indeed computes a breadth-first tree traversal (see Appendix 9.3.1). In particular, $[\![\mathsf{count}]\!] \circ [\![\mathsf{bft}]\!]$ satisfies its expected specification (cf Equation (4) in Example 3.3). ⌟

5.2 Scott Domains as Spectral Spaces

The semantics of formulae and refinement types involves some topology.

Let $(X, \leq)$ be a dcpo. A set $U \subseteq X$ is *Scott-open*, notation $U \in \mathcal{O}(X)$, if U is *saturated* (if $x \in U$ and $x \leq y$ in X, then $y \in U$), and if moreover U is inaccessible by directed sups, in the sense that if $\bigvee D \in U$ with $D \subseteq X$ directed, then $D \cap U \neq \emptyset$. This equips X with a topology, called the *Scott topology*. A function between dcpos is Scott-continuous precisely when it is continuous for the Scott topology. See e.g. [4, §1.2] or [3, §2.3]. The following is well-known (see e.g. [3, Theorem 7.2.29] or [12, Theorem 7.47]).

$$\begin{array}{lll}
[\![\varphi \wedge \psi]\!]\rho \coloneqq [\![\varphi]\!]\rho \cap [\![\psi]\!]\rho & [\![p]\!]\rho \coloneqq \rho(p) & [\![(\mu^t p)\varphi]\!]\rho \coloneqq [\![(p.\varphi)^{[\![t]\!]\rho}(\mathsf{False})]\!]\rho \\
[\![\varphi \vee \psi]\!]\rho \coloneqq [\![\varphi]\!]\rho \cup [\![\psi]\!]\rho & [\![[\langle\rangle]]\!]\rho \coloneqq \{\top\} & [\![(\nu^t p)\varphi]\!]\rho \coloneqq [\![(p.\varphi)^{[\![t]\!]\rho}(\mathsf{True})]\!]\rho \\
[\![\mathsf{True}]\!]\rho \coloneqq [\![\tau]\!] & [\![[\triangle]\varphi]\!]\rho \coloneqq [\![[\triangle]]\!]([\![\varphi]\!]\rho) & [\![(\exists k)\varphi]\!]\rho \coloneqq \bigcup_{n\in\mathbb{N}} [\![\varphi]\!]\rho[n/k] \\
[\![\mathsf{False}]\!]\rho \coloneqq \emptyset & [\![\psi \Vdash \varphi]\!]\rho \coloneqq [\![\psi]\!]\rho \Vdash [\![\varphi]\!]\rho & [\![(\forall k)\varphi]\!]\rho \coloneqq \bigcap_{n\in\mathbb{N}} [\![\varphi]\!]\rho[n/k]
\end{array}$$

Figure 8 Interpretation $[\![\varphi]\!]\rho \in \mathcal{P}([\![\tau]\!])$ of $\varphi \in \mathcal{L}(\Sigma;\tau)$, where $\triangle \in \{\pi_1, \pi_2, \mathsf{in}_1, \mathsf{in}_2, \mathsf{fold}\}$.

$$\begin{array}{rcll}
S \in \mathcal{P}([\![\tau_i]\!]) & \longmapsto & [\![[\pi_i]]\!](S) & \coloneqq \{x \in [\![\tau_1 \times \tau_2]\!] \mid [\![\pi_i]\!](x) \in S\} \\
S \in \mathcal{P}([\![\tau_i]\!]) & \longmapsto & [\![[\mathsf{in}_i]]\!](S) & \coloneqq \{[\![\mathsf{in}_i]\!](x) \in [\![\tau_1 + \tau_2]\!] \mid x \in S\} \\
S \in \mathcal{P}([\![\tau[\mathsf{rec}\,\alpha.\tau/\alpha]]\!]) & \longmapsto & [\![[\mathsf{fold}]]\!](S) & \coloneqq \{x \in [\![\mathsf{rec}\,\alpha.\tau]\!] \mid [\![\mathsf{unfold}]\!](x) \in S\} \\
S \in \mathcal{P}([\![\sigma]\!])\,,\, T \in \mathcal{P}([\![\tau]\!]) & \longmapsto & (S \Vdash T) & \coloneqq \{f \in [\![\sigma \to \tau]\!] \mid \forall x \in S,\ f(x) \in T\}
\end{array}$$

Figure 9 Semantic modalities, where $i = 1, 2$.

▶ **Theorem 5.3.** *The Scott topology of a Scott domain is spectral.*

There are different equivalent definitions of spectral spaces, see e.g. [11, 1.1.5], [12, §6.1] or [13, 9.5.1]. For now, a *spectral* space is T_0, well-filtered, and the compact-opens are stable under finite intersections and form a basis of the topology. See Appendix 10 for details.

Scott topologies are always T_0 since $x \leq y$ in a dcpo X if, and only if, $x \in U$ implies $y \in U$ for each Scott-open U. Given a finite $d \in X$, it is easily seen that $\uparrow d = \{x \in X \mid d \leq x\}$ is *compact-open* (i.e. compact and Scott-open). In fact, if X is algebraic then the Scott-opens are exactly the unions of $\uparrow d$'s, with d finite in X. Each cpo X is trivially compact since $\uparrow\bot = X$. More importantly, if X is a Scott domain then $\uparrow d \cap \uparrow d'$ is compact for all finite $d, d' \in X$ (by bounded-completeness, if $\uparrow d \cap \uparrow d'$ is non-empty, then $d \vee d'$ is defined, finite and such that $\uparrow(d \vee d') = \uparrow d \cap \uparrow d'$). It follows that the set $\mathcal{KO}(X)$ of compact-open subsets of X is stable under finite intersections and forms a basis of the Scott topology.

Proposition 5.4 below states that Scott domains are *well-filtered*. This will yield the topological classification of polarised formulae in Proposition 5.8 (§5.3). We let $\mathcal{KS}(X)$ be the set of *compact-saturated* (i.e. compact and saturated) subsets of X. Note that a set $S \subseteq X$ is saturated if, and only if, $S = \bigcap\{U \in \mathcal{O}(X) \mid S \subseteq U\}$. Proposition 5.4 implies that $Q \in \mathcal{KS}(X)$ if, and only if, $Q = \bigcap\{K \in \mathcal{KO}(X) \mid Q \subseteq K\}$, and thus that $\mathcal{KS}(X)$ is stable under all intersections. A subset $\mathcal{Q}$ of a poset P is *codirected* if $\mathcal{Q}$ is directed in P^{op}.

▶ **Proposition 5.4** (Well-Filteredness)**.** *Each Scott domain X is* well-filtered*: given a codirected set $\mathcal{Q} \subseteq \mathcal{KS}(X)$, if $\bigcap \mathcal{Q} \subseteq U$ with U Scott-open, then $Q \subseteq U$ for some $Q \in \mathcal{Q}$.*

▶ **Corollary 5.5.** *If X is a Scott-domain, then $\mathcal{KS}(X)$ is stable under all intersections.*

▶ Remark 5.6 (De Groot Duality). In a spectral space X, the compact-saturated sets are exactly the intersections of compact-opens, while the opens are exactly the unions of compact-opens. This symmetry between $\mathcal{KS}(X)$ and $\mathcal{O}(X)$ is formalised by *de Groot* duality. A nice panorama from the more general point of view of *stably compact spaces* is given in [24].[1] ⌟

5.3 Semantics of Formulae

A *valuation* of a fixpoint context Σ is a function ρ taking fixpoint variables p with $(p\colon \tau) \in \Sigma$ to sets $\rho(p) \in \mathcal{P}([\![\tau]\!])$, and iteration variables k to natural numbers $\rho(k) \in \mathbb{N}$. A valuation

[1] In a stably compact space X, $\mathcal{KO}(X)$ may not be a basis. A stably compact algebraic dcpo is spectral.

ρ thus interprets iteration terms t as natural numbers $[\![t]\!]\rho \in \mathbb{N}$. Figure 8 defines the interpretation $[\![\varphi]\!]\rho \in \mathcal{P}([\![\tau]\!])$ of a formula $\varphi \in \mathcal{L}(\Sigma;\tau)$ under a valuation ρ of Σ, where $(p.\varphi)^n(\mathsf{False})$ and $(p.\varphi)^n(\mathsf{True})$ are defined in Equation (2), §3.1, and where the semantic modalities $[\![[\triangle]]\!]$ and $(-) \Vdash\!\!\rightarrow (-)$ are defined in Figure 9. See Appendix 11 for details.

In view of Figure 9, it is clear that $[\![\mathsf{in}_1]\!](S_1) \cap [\![\mathsf{in}_2]\!](S_2)$ is always empty. This yields the soundness of the rule (I) in Figure 4b (cf Remark 3.7). The semantic realizability implication $(-) \Vdash\!\!\rightarrow (-)$ is clearly contravariant in its first argument and covariant in its second argument. Besides, the polarity constraints in Equation (1) (§3.1), as well as the rule (WF) in Figure 4e, come from the following crucial consequence of well-filteredness (Proposition 5.4).

▶ **Lemma 5.7.** *Given $V \in \mathcal{O}([\![\sigma]\!])$ and a codirected $\mathcal{Q} \subseteq \mathcal{KS}([\![\tau]\!])$, if $f \in \bigcap \mathcal{Q} \Vdash\!\!\rightarrow V$, then there is some $Q \in \mathcal{Q}$ such that $f \in Q \Vdash\!\!\rightarrow V$ already. Moreover, given $Q \in \mathcal{KS}([\![\tau]\!])$ we have $Q \Vdash\!\!\rightarrow V \in \mathcal{O}([\![\tau \to \sigma]\!])$ and $V \Vdash\!\!\rightarrow Q \in \mathcal{KS}([\![\sigma \to \tau]\!])$.*

▶ **Proposition 5.8.** *If $\varphi \in \mathcal{L}^+(\tau)$ then $[\![\varphi]\!]\rho \in \mathcal{O}([\![\tau]\!])$. If $\varphi \in \mathcal{L}^-(\tau)$ then $[\![\varphi]\!]\rho \in \mathcal{KS}([\![\tau]\!])$. In particular, if $\varphi \in \mathcal{L}^{\pm}(\tau)$ then $[\![\varphi]\!]\rho \in \mathcal{KO}([\![\tau]\!])$.*

Our polarised logical languages $\mathcal{L}^+$ and $\mathcal{L}^-$ are motivated by the topological classification in Proposition 5.8. In view of Figure 2d, the following Lemma 5.9 implies that the unions and intersections interpreting $(\exists k)\varphi$ and $(\forall k)\varphi$ are directed and codirected, respectively. It also yields the soundness of the rules ($\exists$M) and ($\forall$M) in Figure 4c.

▶ **Lemma 5.9.** *If k Pos φ (resp. k Neg φ) then the function $n \mapsto [\![\varphi]\!]\rho[n/k]$ is monotone (resp. antimonotone), that is, $n \leq m$ (resp. $m \leq n$) implies $[\![\varphi]\!]\rho[n/k] \subseteq [\![\varphi]\!]\rho[m/k]$.*

Formulae satisfy an important Scott-(co)continuity property w.r.t. their fixpoint variables. Given $\varphi \in \mathcal{L}(\Sigma;\tau)$ and a valuation ρ of Σ, for each $(p\colon \sigma) \in \Sigma$ we have a function

$$[\![p.\varphi]\!]\rho\colon \mathcal{P}([\![\sigma]\!]) \longrightarrow \mathcal{P}([\![\tau]\!]), \quad S \longmapsto [\![\varphi]\!]\rho[S/p]$$

Note that in the case of $\varphi \in \mathcal{L}(\Sigma, p\colon \tau;\tau)$, for all $\psi \in \mathcal{L}(\Sigma;\tau)$ and all $n \in \mathbb{N}$, we have $([\![p.\varphi]\!]\rho)^n([\![\psi]\!]\rho) = [\![(p.\varphi)^n(\psi)]\!]\rho$, where $(p.\varphi)^n(\psi)$ is the formula in Equation (2).

▶ **Proposition 5.10.** *Let ρ be a valuation of Σ, and let $(p\colon \sigma) \in \Sigma$.*

(1) *If $\varphi \in \mathcal{L}^s(\Sigma;\tau)$, then $[\![p.\varphi]\!]\rho$ is monotone.*

(2) *If $\varphi \in \mathcal{L}^+(\Sigma;\tau)$, then $[\![p.\varphi]\!]\rho$ is Scott-continuous (i.e. preserves directed unions).*

(3) *If $\varphi \in \mathcal{L}^-(\Sigma;\tau)$, then $[\![p.\varphi]\!]\rho$ is Scott-cocontinuous (i.e. preserves codirected intersections).*

▶ **Example 5.11.** Let $\varphi \in \mathcal{L}^+(\Sigma, p\colon \tau;\tau)$ and $\psi \in \mathcal{L}^-(\Sigma, p\colon \tau;\tau)$. It follows from Proposition 5.10 that the least fixpoint of $[\![p.\varphi]\!]\rho$ and the greatest fixpoint of $[\![p.\psi]\!]\rho$ are

$$[\![(\exists k)(\mu^k p)\varphi]\!]\rho = \bigcup_{n\in\mathbb{N}}([\![p.\varphi]\!]\rho)^n(\mathsf{False}) \quad \text{and} \quad [\![(\forall \ell)(\nu^\ell p)\psi]\!]\rho = \bigcap_{n\in\mathbb{N}}([\![p.\psi]\!]\rho)^n(\mathsf{True})$$

On streams, we get that $\Diamond\varphi$ defines an *open* set, while $\Box\psi$ defines a *compact-saturated* set (with φ positive and ψ negative). For instance, $[\![\Box[\mathsf{hd}][\mathsf{tt}]]\!] = \{[\![\mathsf{tt}]\!]^\omega\}$ is compact-saturated in $[\![\mathsf{Stream\ Bool}]\!] = [\![\mathsf{Bool}]\!]^\omega$. This contrasts with the case of ω-words, for which $\{\mathsf{tt}^\omega\}$ is a *closed* subset of $\{\mathsf{tt}, \mathsf{ff}\}^\omega$ for the product topology.[2] Besides, ω-words over a finite alphabet form a Stone (Boolean) space, i.e. a spectral space in which the compact-saturated sets coincide

[2] See [33] for a comparison of the Scott topology on streams with the product topology on ω-words.

with the closed ones. But in the case of the Scott domain of streams, proper compact-saturated sets (such as $\{[\![\mathsf{tt}]\!]^\omega\}$) are never closed. We think this makes the case of de Groot duality (Remark 5.6) for temporal properties on streams.

In the context of Example 3.3, the significance of well-filteredness (Proposition 5.4) in Lemma 5.7 is that a Scott-continuous function $f\colon [\![\mathsf{Tree}\,\mathsf{Bool} \to \mathsf{Stream}\,\mathsf{Nat}]\!]$, say $[\![\mathsf{count} \circ \mathsf{bft}]\!]$, satisfies the formula in Equation (4) *if, and only if,* there is some $m \in \mathbb{N}$ such that for each totally defined $t \in [\![\mathsf{Tree}\,\mathsf{Bool}]\!]$, the length-$m$ prefix of $f(t)$ contains a totally defined number $\geq n$ provided that in the depth-m truncation of t, the children of the root have a maximal path consisting of tt's and one consisting of ff's. Hence, only a depth-m exploration of $f(t)$ is needed to establish the property, where $m \in \mathbb{N}$ depends on f but *not* on t. ⌟

Proposition 5.10 yields the rules $(\iota/\exists)$ and $(\forall/\iota)$ in Figure 4c. The following Proposition 5.12 is crucial for the rules $(\Vdash\!\!\to/\vee)$ and (CC) in Figure 4e, as well as for completeness.

▶ **Proposition 5.12.** *Given $\delta \in \mathcal{L}^\wedge(\tau)$, if $[\![\delta]\!] \neq \emptyset$ then $[\![\delta]\!] = {\uparrow}d$ for some finite $d \in [\![\tau]\!]$. Conversely, if $d \in [\![\tau]\!]$ is finite, then ${\uparrow}d = [\![\delta]\!]$ for some $\delta \in \mathcal{L}^\wedge(\tau)$.*

▶ **Theorem 5.13** (Soundness of Deduction)**.**
(1) *For all $\delta \in \mathcal{L}^\wedge(\tau)$, if $\mathcal{C}(\delta)$ is derivable, then $[\![\delta]\!] \neq \emptyset$.*
(2) *For all $\varphi, \psi \in \mathcal{L}^s(\Sigma;\tau)$, if $\psi \vdash \varphi$ is derivable, then $[\![\psi]\!]\rho \subseteq [\![\varphi]\!]\rho$.*

The following is the key logical fact toward completeness. It also entails the completeness of the neutral fragment (which we actually do not need for our main Theorem 5.17).

▶ **Theorem 5.14** (Completeness of $\mathcal{L}^\wedge$)**.** *Let $\varphi, \psi \in \mathcal{L}^\wedge(\tau)$.*
(1) *If $[\![\varphi]\!] \neq \emptyset$, then $\mathcal{C}(\varphi)$ is derivable.*
(2) *If $[\![\psi]\!] \subseteq [\![\varphi]\!]$, then $\psi \vdash \varphi$ is derivable.*

5.4 Semantics of the Refinement Type System

The interpretation $[\![T]\!] \subseteq [\![|T|]\!]$ of a refinement type T is defined as $[\![\{\tau \mid \varphi\}]\!] \coloneqq [\![\varphi]\!]$ and $[\![U \to T]\!] \coloneqq [\![U]\!] \Vdash\!\!\to [\![T]\!]$ and $[\![T \times U]\!] \coloneqq [\![[\pi_1]]\!]([\![T]\!]) \cap [\![[\pi_2]]\!]([\![U]\!])$.

▶ **Definition 5.15** (Sound Judgement)**.** *A judgement $\mathcal{E} \vdash M\colon T$ with $\mathcal{E} = x_1\colon U_1, \dots, x_n\colon U_n$ is* sound *if $|\mathcal{E}| \vdash M\colon |T|$ is derivable and if moreover $[\![M]\!](u_1, \dots, u_n) \in [\![T]\!]$ whenever $u_i \in [\![U_i]\!]$ for all $i = 1, \dots, n$.*

▶ **Theorem 5.16** (Soundness of Typing)**.** *If $\mathcal{E} \vdash M\colon T$ is derivable then it is sound.*

It follows from Examples 5.2 and 5.11 that $\vdash \mathsf{count} \circ \mathsf{bft}\colon T^+$ is sound, where T^+ is either type in Example 4.1. This judgement is also derivable, thanks to our main result:

▶ **Theorem 5.17** (Positive Completeness)**.** *If $\mathcal{E}^- \vdash M\colon T^+$ is sound, then it is derivable.*

Write $T^\wedge$ for a refinement type which only contains formulae in $\mathcal{L}^\wedge$,
and similarly for $\mathcal{E}^\wedge$. We obtain Theorem 5.17 by reduction to judgements of the form $\mathcal{E}^\wedge \vdash M\colon T^\wedge$, for which completeness essentially amounts to [2] (through formally our system differs).

▶ **Theorem 5.18.** *If $\mathcal{E}^\wedge \vdash M\colon T^\wedge$ is sound, then it is derivable.*

The reduction of Theorem 5.17 to Theorem 5.18 is an adaptation of [7, 32]. Consider first the case of a sound $\vdash M\colon T$, where T is positive. By Lemma 4.3 and Proposition 3.9, let $\varphi \in \mathcal{L}^{\pm}(|T|)$ such that $T \simeq \{|T| \mid (\exists k)\varphi\}$. Since the judgement $\vdash M\colon T$ is sound, there

is some $m \in \mathbb{N}$ such that $\vdash M\colon \{|T| \mid \varphi[m/k]\}$ is also sound. By Lemmas 3.8 and 3.6, the closed formula $\varphi[m/k] \in \mathcal{L}^{\pm}(|T|)$ is equivalent to a disjunction of formulae $\gamma \in \mathcal{L}^{\wedge}(|T|)$. But there must be at least one of these γ's such that the judgement $\vdash M\colon \{|T| \mid \gamma\}$ is sound, and we can apply Theorem 5.18. Since $\{|T| \mid \gamma\} \preceq \{|T| \mid \varphi[m/k]\} \preceq \{|T| \mid (\exists k)\varphi\}$, we obtain a derivation of $\vdash M\colon \{|T| \mid (\exists k)\varphi\}$ from a derivation of $\vdash M\colon \{|T| \mid \gamma\}$.

Consider now the case of $x\colon U \vdash M\colon T$, where U is negative. Let $\psi \in \mathcal{L}^{\pm}(|U|)$ such that $U \simeq \{|U| \mid (\forall \ell)\psi\}$. Since $[\![T]\!]$ is Scott-open and $[\![M]\!]$ is Scott-continuous, well-filteredness (Proposition 5.4) yields some $m \in \mathbb{N}$ such that $[\![\psi[m/\ell]]\!] \subseteq [\![M]\!]^{-1}([\![T]\!])$, i.e. such that $x\colon \{|U| \mid \psi[m/\ell]\} \vdash M\colon T$ is sound. We can similarly write $\psi[m/\ell]$ as a disjunction of $\delta \in \mathcal{L}^{\wedge}(|U|)$, and each disjunct δ leads to a sound judgement $x\colon \{|U| \mid \delta\} \vdash M\colon T$. If $\delta \not\vdash \mathsf{False}$, then Proposition 5.12 gives a finite $d \in [\![|U|]\!]$ such that $[\![\delta]\!] = {\uparrow}d$. Since $[\![T]\!]$ is saturated, we have ${\uparrow}d \subseteq [\![M]\!]^{-1}([\![T]\!])$ if, and only if, $[\![M]\!](d) \in [\![T]\!]$. Unfolding T similarly as above, the condition $[\![M]\!](d) \in [\![T]\!]$ yields some $\gamma \in \mathcal{L}^{\wedge}(|T|)$ with $\{|T| \mid \gamma\} \preceq T$ and such that $x\colon \{|U| \mid \delta\} \vdash M\colon \{|T| \mid \gamma\}$ is sound. See Appendix 12 for details.

▶ Remark 5.19 (Undecidability). Positive judgements are semi-decidable but not decidable: by Turing-completeness, there is a $K\colon \mathsf{Nat} \to \mathsf{Nat}$ such that $K\mathsf{n}$ evaluates the nth partial recursive function on input n. Hence the total function which given $n \in \mathbb{N}$ decides the derivability (or soundness) of $\vdash K\mathsf{n}\colon \{\mathsf{Nat} \mid [\mathsf{tot}]\}$, is not recursive ([30, Theorem II.2.3]). ⌟

6 Conclusion

We presented a finitary refinement type system which is complete for a set of Scott-open specifications strictly extending [1, 2]. We moreover noted that these specifications are not decidable for the typed λ-calculus under consideration (Remark 5.19 above).

An important direction of future work is to characterise the expressiveness of our logic. Example 5.11 means that our positive (resp. negative) fragment corresponds to a modal μ-calculus on (finitary) polynomial types, restricted to negation-free formulae with least (resp. greatest) fixpoints. However, we cannot yet be as precise for higher-order types (with formulae involving the realizability implication $\Vdash\!\!\rightarrow$).

An other direction for future work is to extend this system to liveness properties (as in Remark 3.4), while still targeting a complete finitary type system (in contrast with [19, 32]).

Our system has sum types (so as to have a recursive type of natural numbers), but the category **Scott** does not have finite coproducts. Working with Call-By-Push-Value (CBPV) [25, 26], for the usual adjunction between dcpos and cpos with strict functions, may lead to a cleaner situation. In the long run, it would be nice if this basis could extend to enriched models of CBPV, so as to handle further computational effects. Print and global store are particularly relevant, as an important trend in proving temporal properties considers programs generating streams of events. Major works in this line include [35, 15, 16, 28, 23, 38, 28, 34]. In contrast with ours, these approaches are based on trace semantics of syntactic expressions rather than denotational domains.

In a different direction, we think the approach of this paper could extend to linear types [17, 29, 41], in particular for the λ-calculus HOPLA, and possibly relying on the categorical study of [9].

7 Proofs of §3 (The Logic)

Figure 10 gathers some derivable sequents.

▶ **Lemma 7.1.** *The sequents in Figure 10 are all derivable.*

$$\psi \wedge (\varphi \vee \theta) \dashv\vdash (\psi \wedge \varphi) \vee (\psi \wedge \theta) \qquad \psi \vee (\varphi \wedge \theta) \dashv\vdash (\psi \vee \varphi) \wedge (\psi \vee \theta)$$

(a) Distributive laws.

$$[\triangle](\varphi \vee \psi) \dashv\vdash [\triangle]\varphi \vee [\triangle]\psi \qquad [\triangle](\varphi \wedge \psi) \dashv\vdash [\triangle]\varphi \wedge [\triangle]\psi$$

(b) Modalities $\triangle \in \{\pi_1, \pi_2, \mathsf{in}_1, \mathsf{in}_2, \mathsf{fold}\}$.

$$(\psi_1 \vee \psi_2) \Vdash \varphi \dashv\vdash (\psi_1 \Vdash \varphi) \wedge (\psi_2 \Vdash \varphi)$$

$$\psi \Vdash (\varphi_1 \wedge \varphi_2) \dashv\vdash (\psi \Vdash \varphi_1) \wedge (\psi \Vdash \varphi_2)$$

$$\delta \Vdash (\varphi_1 \vee \varphi_2) \dashv\vdash (\delta \Vdash \varphi_1) \vee (\delta \Vdash \varphi_2)$$

(c) The realizability implication $\Vdash$, where $\delta \in \mathcal{L}^{\wedge}$.

$$(\mu^0 p)\varphi \dashv\vdash \mathsf{False} \qquad (\mu^{t+1} p)\varphi \dashv\vdash \varphi[(\mu^t p)\varphi/p]$$

$$(\nu^0 p)\varphi \dashv\vdash \mathsf{True} \qquad (\nu^{t+1} p)\varphi \dashv\vdash \varphi[(\nu^t p)\varphi/p]$$

(d) Bounded iteration.

Figure 10 Some derivable sequents.

Proof. We discuss each subfigure separately.

Case of Figure 10a. The sequent $\psi \wedge (\varphi \vee \theta) \vdash (\psi \wedge \varphi) \vee (\psi \wedge \theta)$ is given by the rule (D) in Figure 4a. We derive its converse with

$$\dfrac{\dfrac{\dfrac{}{\psi \vdash \psi}}{\psi \wedge \varphi \vdash \psi} \quad \dfrac{\dfrac{}{\psi \vdash \psi}}{\psi \wedge \theta \vdash \psi}}{(\psi \wedge \varphi) \vee (\psi \wedge \theta) \vdash \psi} \quad \dfrac{\dfrac{\dfrac{}{\varphi \vdash \varphi}}{\psi \wedge \varphi \vdash \varphi} \quad \dfrac{\dfrac{}{\theta \vdash \theta}}{\psi \wedge \theta \vdash \theta}}{(\psi \wedge \varphi) \vee (\psi \wedge \theta) \vdash \varphi \vee \theta}$$

$$(\psi \wedge \varphi) \vee (\psi \wedge \theta) \vdash \psi \wedge (\varphi \vee \theta)$$

We similarly obtain $\psi \vee (\varphi \wedge \theta) \vdash (\psi \vee \varphi) \wedge (\psi \vee \theta)$. For the converse, note that (D) yields

$$(\psi \vee \varphi) \wedge (\psi \vee \theta) \vdash \big((\psi \vee \varphi) \wedge \psi\big) \vee \big((\psi \vee \varphi) \wedge \theta\big)$$

and thus

$$(\psi \vee \varphi) \wedge (\psi \vee \theta) \vdash \psi \vee \big((\psi \vee \varphi) \wedge \theta\big)$$

Now, using (D) again yields

$$(\psi \vee \varphi) \wedge (\psi \vee \theta) \vdash \psi \vee (\psi \wedge \theta) \vee (\varphi \wedge \theta)$$

Since $\psi \vee (\psi \wedge \theta) \vdash \psi$, we obtain

$$(\psi \vee \varphi) \wedge (\psi \vee \theta) \vdash \psi \vee (\varphi \wedge \theta)$$

Case of Figure 10b. The sequent $[\triangle](\varphi \vee \psi) \vdash [\triangle]\varphi \vee [\triangle]\psi$ is given by Figure 4b. We derive its converse with

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\varphi_1 \vdash \varphi_1}}{\varphi_1 \vdash \varphi_1 \vee \varphi_2}}{[\triangle]\varphi_1 \vdash [\triangle](\varphi_1 \vee \varphi_2)} \quad \dfrac{\dfrac{\dfrac{}{\varphi_2 \vdash \varphi_2}}{\varphi_2 \vdash \varphi_1 \vee \varphi_2}}{[\triangle]\varphi_2 \vdash [\triangle](\varphi_1 \vee \varphi_2)}}{[\triangle]\varphi_1 \vee [\triangle]\varphi_2 \vdash [\triangle](\varphi_1 \vee \varphi_2)}$$

The law for $\wedge$ is derived similarly.

Case of Figure 10c. The sequent $(\psi_1 \Vdash \varphi) \wedge (\psi_1 \Vdash \varphi) \vdash (\psi_1 \vee \psi_2) \Vdash \varphi$ is given by the rule $(\vee/\Vdash)$ in Figure 4e. We derive its converse with

$$\dfrac{\dfrac{\dfrac{}{\psi_1 \vdash \psi_1}}{\psi_1 \vdash \psi_1 \vee \psi_2} \quad \dfrac{}{\varphi \vdash \varphi}}{(\psi_1 \vee \psi_2) \Vdash \varphi \vdash \psi_1 \Vdash \varphi} \quad \dfrac{\dfrac{\dfrac{}{\psi_2 \vdash \psi_2}}{\psi_2 \vdash \psi_1 \vee \psi_2} \quad \dfrac{}{\varphi \vdash \varphi}}{(\psi_1 \vee \psi_2) \Vdash \varphi \vdash \psi_2 \Vdash \varphi}}{(\psi_1 \vee \psi_2) \Vdash \varphi \vdash (\psi_1 \Vdash \varphi) \wedge (\psi_2 \Vdash \varphi)}$$

The law $\psi \Vdash (\varphi_1 \wedge \varphi_2) \dashv\vdash (\psi \Vdash \varphi_1) \wedge (\delta \wedge \varphi_2)$ is derived similarly.
The sequent $\delta \Vdash (\varphi_1 \vee \varphi_2) \vdash (\delta \Vdash \varphi_1) \vee (\delta \Vdash \varphi_2)$ with $\delta \in \mathcal{L}^{\wedge}$ is given by the rule $(\Vdash/\vee)$ in Figure 4e. We derive its converse with

$$\dfrac{\dfrac{\dfrac{}{\delta \vdash \delta} \quad \dfrac{\dfrac{}{\varphi_1 \vdash \varphi_1}}{\varphi_1 \vdash \varphi_1 \vee \varphi_2}}{\delta \Vdash \varphi_1 \vdash \delta \Vdash (\varphi_1 \vee \varphi_2)} \quad \dfrac{\dfrac{}{\delta \vdash \delta} \quad \dfrac{\dfrac{}{\varphi_2 \vdash \varphi_2}}{\varphi_2 \vdash \varphi_1 \vee \varphi_2}}{\delta \Vdash \varphi_2 \vdash \delta \Vdash (\varphi_1 \vee \varphi_2)}}{(\delta \Vdash \varphi_1) \vee (\delta \Vdash \varphi_2) \vdash \delta \Vdash (\varphi_1 \vee \varphi_2)}$$

Case of Figure 10d. We derive $(\mu^0 p)\varphi \dashv\vdash \mathsf{False}$ using the rules

$$\dfrac{}{(\mu^0 p)\varphi \vdash \mathsf{False}} \qquad \dfrac{}{\mathsf{False} \vdash \psi}$$

We moreover derive $(\mu^{t+1} p)\varphi \dashv\vdash \varphi[(\mu^t p)\varphi/p]$ from the rules

$$\dfrac{(\mu^t p)\varphi \vdash \psi}{(\mu^{t+1} p)\varphi \vdash \varphi[\psi/p]} \qquad \dfrac{}{\varphi[(\mu^t p)\varphi/p] \vdash (\mu^{t+1} p)\varphi}$$

The laws for $(\nu^t p)$ are obtained similarly. ◀

We now turn to Lemma 3.6. It uses the following fact.

▶ **Lemma 7.2.** **(1)** *For each $\varphi \in \mathcal{L}^{\vee\wedge}$, there is some $\psi \in \mathcal{L}^{\vee\wedge}$ such that $[\triangle]\varphi \dashv\vdash \psi$.*
(2) *For each $\delta \in \mathcal{L}^{\wedge}$ and each $\varphi \in \mathcal{L}^{\vee\wedge}$, there is some $\psi \in \mathcal{L}^{\vee\wedge}$ such that $(\delta \Vdash \varphi) \dashv\vdash \psi$*
(3) *For each $\varphi_1, \varphi_2 \in \mathcal{L}^{\vee\wedge}$, there is some $\psi \in \mathcal{L}^{\vee\wedge}$ such that $(\varphi_1 \Vdash \varphi_2) \dashv\vdash \psi$*

Proof. **(1)** By induction on $\varphi \in \mathcal{L}^{\vee\wedge}$, using Figure 10b.
(2) By induction on $\varphi \in \mathcal{L}^{\vee\wedge}$, using the last law in Figure 10c.
(3) By induction on $\varphi_1 \in \mathcal{L}^{\vee\wedge}$, using item (2) in the base case of $\varphi_1 \in \mathcal{L}^{\wedge}$, and using the first law in Figure 10c together with distributive laws (Figure 10a) in the induction step (φ_1 of the form $\varphi \vee \varphi'$). ◀

We can now prove Lemma 3.6.

▶ **Lemma 3.6.** *For each $\varphi \in \mathcal{L}^{\omega}(\tau)$, there is some $\psi \in \mathcal{L}^{\vee\wedge}(\tau)$ such that $\varphi \dashv\vdash \psi$.*

Proof. We reason by induction on $\varphi \in \mathcal{L}^{\omega}(\tau)$.
- The cases of $[\langle\rangle]$, True and False are trivial.
- The case of $\varphi \vee \psi$ follows directly from the induction hypothesis. As for $\varphi \wedge \psi$, we use the induction hypothesis and conclude with distributive laws (Figure 10a).
- The cases of $[\triangle]\varphi$ follow from the induction hypothesis and Lemma 7.2(1).
- The case of $\psi \Vdash \varphi$ follows from the induction hypothesis and Lemma 7.2(3). ◀

As for Lemma 3.8, we use the following.

▶ **Lemma 7.3.** *Let the closed iteration term t be the nth successor of 0. Then:*

(1) $(\mu^t p)\varphi \dashv\Vdash (p.\varphi)^n(\mathsf{False})$
(2) $(\nu^t p)\varphi \dashv\Vdash (p.\varphi)^n(\mathsf{True})$

Proof.
(1) By induction on $n \in \mathbb{N}$, using Figure 10d and Equation (2) in §3.1.
(2) Similar to item (1). ◀

We now prove Lemma 3.8.

▶ **Lemma 3.8.** *For each* closed $\varphi \in \mathcal{L}^{\pm}(\tau)$*, there is some* $\psi \in \mathcal{L}^{\omega}(\tau)$ *such that* $\varphi \dashv\Vdash \psi$.

Proof. We prove a more general statement. Let $\varphi \in \mathcal{L}^{\pm}(\Sigma; \tau)$, without iteration variables, but with $\mathcal{E} = p_1 \colon \tau_1, \ldots, p_n \colon \tau_n$. Then for all $\theta_1, \ldots, \theta_n$ with $\theta_i \in \mathcal{L}^{\omega}(\tau_i)$, there is some $\psi \in \mathcal{L}^{\omega}(\tau)$ such that

$$\varphi[\theta_1/p_1, \ldots, \theta_n/p_n] \quad \dashv\Vdash \quad \psi$$

We reason by induction on $\varphi \in \mathcal{L}^{\pm}(\Sigma; \tau)$.
- In the case of $p_j \in \mathcal{L}^{\pm}(\Sigma; \tau)$, we take ψ to be θ_j.
- In the cases of $[\langle\rangle]$, True and False, we can take ψ to be φ.
- The cases of $\varphi_1 \vee \varphi_2$, $\varphi_1 \wedge \varphi_2$, $[\triangle]\varphi$ and $\varphi_1 \Vdash \varphi_2$ all follow from the induction hypothesis.
- In the case of $(\mu^t p)\varphi$, let $\theta_1, \ldots, \theta_n$ be as in the statement. Note that $\varphi \in \mathcal{L}^{\pm}(\Sigma, p \colon \tau; \tau)$. We show that for each $m \in \mathbb{N}$, there is some $\psi_m \in \mathcal{L}^{\omega}(\tau)$ such that

$$(p.\varphi[\theta_1/p_1, \ldots, \theta_n/p_n])^m(\mathsf{False}) \quad \dashv\Vdash \quad \psi_m$$

We reason by induction on $m \in \mathbb{N}$. In the base case $m = 0$, we can take ψ_0 to be False. As for the induction step, we can take

$$\psi_{m+1} \quad \dashv\Vdash \quad \varphi[\theta_1/p_1, \ldots, \theta_n/p_n, \psi_m/p]$$

obtained by applying the induction hypothesis on φ with $\theta_1, \ldots, \theta_n$ and ψ_m. We can then conclude using Lemma 7.3(1), taking ψ to be ψ_m with $m \in \mathbb{N}$ such that the *closed* iteration term t is the mth successor of 0.
- The case of $(\nu^t p)\varphi$ is similar, using item (2) of Lemma 7.3 instead of item (1). ◀

We finally turn to the crucial Proposition 3.9. Figure 11 gathers some derivable quantifier laws.

▶ **Lemma 7.4.** *The sequents in Figure 11 are all derivable.*

Proof. We discuss each subfigure separately.
Case of Figure 11a. We derive $(\exists k)\varphi \vee \psi \dashv\Vdash (\exists k)(\varphi \vee \psi)$ as

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\varphi \vdash \varphi}}{\varphi \vdash \varphi \vee \psi}}{\varphi \vdash (\exists k)(\varphi \vee \psi)}}{(\exists k)\varphi \vdash (\exists k)(\varphi \vee \psi)} \qquad \dfrac{\dfrac{\dfrac{}{\psi \vdash \psi}}{\psi \vdash \varphi \vee \psi}}{\psi \vdash (\exists k)(\varphi \vee \psi)}}{(\exists k)\varphi \vee \psi \vdash (\exists k)(\varphi \vee \psi)}$$

and

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\varphi \vdash \varphi}}{\varphi \vdash (\exists k)\varphi}}{\varphi \vdash (\exists k)\varphi \vee \psi} \qquad \dfrac{\dfrac{}{\psi \vdash \psi}}{\psi \vdash (\exists k)\varphi \vee \psi}}{\dfrac{\varphi \vee \psi \vdash (\exists k)\varphi \vee \psi}{(\exists k)(\varphi \vee \psi) \vdash (\exists k)\varphi \vee \psi}}$$

$$(\exists k)\varphi \vee \psi \dashv\vdash (\exists k)(\varphi \vee \psi) \qquad (\exists k)\varphi \wedge \psi \dashv\vdash (\exists k)(\varphi \wedge \psi)$$

$$(\forall k)\varphi \wedge \psi \dashv\vdash (\forall k)(\varphi \wedge \psi) \qquad (\forall k)\varphi \vee \psi \dashv\vdash (\forall k)(\varphi \vee \psi)$$

(a) Commutation over propositional connectives, where $k \notin \mathrm{FV}(\psi)$.

$$[\triangle](\exists k)\varphi \dashv\vdash (\exists k)[\triangle]\varphi \qquad [\triangle](\forall k)\varphi \dashv\vdash (\forall k)[\triangle]\varphi$$

(b) Commutation over modalities.

$$(\forall \ell)\psi \Vdash\!\!\rightarrow \varphi \dashv\vdash (\exists \ell)(\psi \Vdash\!\!\rightarrow \varphi) \qquad \theta \Vdash\!\!\rightarrow (\exists k)\varphi \dashv\vdash (\exists k)(\theta \Vdash\!\!\rightarrow \varphi)$$

$$(\exists \ell)\psi \Vdash\!\!\rightarrow \varphi \dashv\vdash (\forall \ell)(\psi \Vdash\!\!\rightarrow \varphi) \qquad \psi \Vdash\!\!\rightarrow (\forall k)\varphi \dashv\vdash (\forall k)(\psi \Vdash\!\!\rightarrow \varphi)$$

(c) Commutation over the realizability implication, where $\ell \notin \mathrm{FV}(\varphi)$, $k \notin \mathrm{FV}(\psi, \theta)$ and $\theta \in \mathcal{L}^{\pm}$.

$$(\iota^t p)(\exists k)\varphi \dashv\vdash (\exists k)(\iota^t p)\varphi \qquad (\iota^t p)(\forall k)\varphi \dashv\vdash (\forall k)(\iota^t p)\varphi$$

(d) Commutation over bounded iteration, where $\iota \in \{\mu, \nu\}$ and $k \notin \mathrm{FV}(t)$.

$$(\exists k)\varphi[k/\ell] \dashv\vdash (\exists k)(\exists \ell)\varphi \qquad (\forall k)(\forall \ell)\varphi \dashv\vdash (\forall k)\varphi[k/\ell]$$

(e) Merge laws.

Figure 11 Quantifier laws.

The rule ($\exists$L) in Figure 4c yields $(\exists k)\varphi \wedge \psi \vdash (\exists k)(\varphi \wedge \psi)$ and we derive $(\exists k)(\varphi \wedge \psi) \vdash (\exists k)\varphi \wedge \psi$ as

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\varphi \vdash \varphi}}{\varphi \wedge \psi \vdash \varphi}}{\varphi \wedge \psi \vdash (\exists k)\varphi}}{(\exists k)(\varphi \wedge \psi) \vdash (\exists k)\varphi} \qquad \dfrac{\dfrac{\dfrac{}{\psi \vdash \psi}}{\varphi \wedge \psi \vdash \psi}}{(\exists k)(\varphi \wedge \psi) \vdash \psi}}{(\exists k)(\varphi \wedge \psi) \vdash (\exists k)\varphi \wedge \psi}$$

The laws for $(\forall k)$ are obtained similarly.

Case of Figure 11b. We have $[\triangle](\exists k)\varphi \vdash (\exists k)[\triangle]\varphi$ from Figure 4b, and we derive the converse as

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\varphi \vdash \varphi}}{\varphi \vdash (\exists k)\varphi}}{[\triangle]\varphi \vdash [\triangle](\exists k)\varphi}}{(\exists k)[\triangle]\varphi \vdash [\triangle](\exists k)\varphi}$$

The laws for $(\forall k)$ are obtained similarly.

Case of Figure 11c. We begin with $(\forall \ell)\psi \Vdash\!\!\rightarrow \varphi \dashv\vdash (\exists \ell)(\psi \Vdash\!\!\rightarrow \varphi)$. We have $(\forall \ell)\psi \Vdash\!\!\rightarrow \varphi \vdash (\exists \ell)(\psi \Vdash\!\!\rightarrow \varphi)$ by rule (WF) in Figure 4e, and we derive its converse as

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\psi \vdash \psi}}{(\forall \ell)\psi \vdash \psi} \qquad \dfrac{}{\varphi \vdash \varphi}}{\psi \Vdash\!\!\rightarrow \varphi \vdash (\forall \ell)\psi \Vdash\!\!\rightarrow \varphi}}{(\exists \ell)(\psi \Vdash\!\!\rightarrow \varphi) \vdash (\forall \ell)\psi \Vdash\!\!\rightarrow \varphi}$$

The other laws are obtained similarly.

Case of Figure 11d. We have $(\iota^t p)(\exists k)\varphi \vdash (\exists k)(\iota^t p)\varphi$ from Figure 4c and we derive its

converse as

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\varphi \vdash \varphi}}{\varphi \vdash (\exists k)\varphi}}{(\iota^t p)\varphi \vdash (\iota^t p)(\exists k)\varphi}}{(\exists k)(\iota^t p)\varphi \vdash (\iota^t p)(\exists k)\varphi}$$

The laws for $(\forall k)$ are handled similarly.

Case of Figure 11e. We derive

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\varphi[k/\ell] \vdash \varphi[k/\ell]}}{\varphi[k/\ell] \vdash (\exists \ell)\varphi}}{\varphi[k/\ell] \vdash (\exists k)(\exists \ell)\varphi}}{(\exists k)\varphi[k/\ell] \vdash (\exists k)(\exists \ell)\varphi} \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{\varphi[k/\ell] \vdash \varphi[k/\ell]}}{(\forall \ell)\varphi \vdash \varphi[k/\ell]}}{(\forall k)(\forall \ell)\varphi \vdash \varphi[k/\ell]}}{(\forall k)(\forall \ell)\varphi \vdash (\forall k)\varphi[k/\ell]}$$

and we obtain the converses from the rules $(\exists \mathsf{M})$ and $(\forall \mathsf{M})$ in Figure 4c. ◀

We can finally prove Proposition 3.9.

▶ **Proposition 3.9.** *For each $\varphi \in \mathcal{L}^s(\Sigma;\tau)$, there is a prenex $\psi \in \mathcal{L}^s(\Sigma;\tau)$ such that $\varphi \dashv\vdash \psi$.*

Proof. We reason by induction on $\varphi \in \mathcal{L}^s(\Sigma;\tau)$.

- In the case of p with $(p\colon \sigma) \in \Sigma$, we take ψ to be p.
- In the cases of $[\langle\rangle]$, True and False, we can take ψ to be φ.
- The cases of $(\exists k)\varphi$ and $(\forall k)\varphi$ follow from the induction hypothesis and Figure 11e.
- The cases of $\varphi_1 \vee \varphi_2$ and $\varphi_1 \wedge \varphi_2$ are dealt-with using the induction hypothesis and Figures 11a and 11e.
- The cases of $[\triangle]\varphi$ are dealt-with using the induction hypothesis and Figure 11b.
- In the case of a *negative* realizability implication $\varphi_1 \Vdash\!\!\mapsto \varphi_2$, we conclude by induction hypothesis and Figure 11c.
  Consider now the case of a *positive* realizability implication $\varphi_1 \Vdash\!\!\mapsto \varphi_2$. Using the induction hypothesis and the top left law in Figure 11c, we can reduce to the case of $(\exists k)(\theta \Vdash\!\!\mapsto \varphi_2)$ with $\theta \in \mathcal{L}^{\pm}$. We can the conclude using the top right law in Figure 11c.
- The cases of $(\mu^t p)\varphi$ and $(\nu^t p)\varphi$ are dealt-with using the induction hypothesis and Figure 11d. ◀

8 Proofs of §4 (The Refinement Type System)

We prove Lemma 4.3.

▶ **Lemma 4.3.** *For each type T^s, there is a $\varphi \in \mathcal{L}^s(|T|)$ such that $T^s \simeq \{|T^s| \mid \varphi\}$.*

Proof. The proof is by induction on T^s. The base case of $\{\tau \mid \varphi\}$ with $\varphi \in \mathcal{L}^s(\tau)$ is trivial. In the base case of τ, one can take $\varphi \coloneqq \mathsf{True} \in \mathcal{L}^s(\tau)$. In the cases of $T^s \times U^s$ and $V^{-s} \to T^s$, by induction hypotheses we get $\varphi \in \mathcal{L}^s(|T|)$, $\psi \in \mathcal{L}^s(|U|)$, and $\theta \in \mathcal{L}^{-s}(|V|)$ such that $T^s \simeq \{|T| \mid \varphi\}$, $U^s \simeq \{|U| \mid \varphi\}$ and $V^{-s} \simeq \{|V| \mid \varphi\}$. We then conclude with

$$\begin{array}{rcl} T^s \times U^s & \simeq & \{|T| \times |U| \mid [\pi_1]\varphi \wedge [\pi_2]\psi\} \\ V^{-s} \to T^s & \simeq & \{|V| \to |T| \mid \theta \Vdash\!\!\mapsto \varphi\} \end{array}$$

◀

9 Proofs of §5.1 (Semantics of the Pure System)

9.1 Solutions of Recursive Domain Equations

We review the usual solution of recursive domain equations. We refer to [4, 3, 37].

9.1.1 Categories of Domains

In the following, **DCPO** is the category of those posets with all directed suprema, and with Scott-continuous functions as morphisms. **CPO** is the full subcategory of **DCPO** on posets with a least element. Note that **Scott** is a full subcategory of **CPO**.

Recall that **DCPO**, **CPO** and **Scott** are Cartesian-closed categories. Finite products are given by Cartesian products of sets equipped with the component-wise order. The closed structures are given by equipping each homset with the pointwise order. See [3, Theorem 3.3.3, Theorem 3.3.5 and Corollary 4.1.6]. Hence for each $n \in \mathbb{N}$, the categories $\mathbf{Scott}^n$, $\mathbf{CPO}^n$ and $\mathbf{DCPO}^n$ are (not full) subcategories of **Scott**, **CPO** and **DCPO** respectively.

▶ **Lemma 9.1.** *If $n \in \mathbb{N}$ then $\mathbf{DCPO}^n$, $\mathbf{CPO}^n$, $\mathbf{Scott}^n$ are enriched in* **DCPO**.

Proof. The result for $n = 1$ follows from the Cartesian-closure of **DCPO**, **CPO** and **Scott**. In the cases of $n \neq 1$, the result follows from the fact that in $\mathbf{DCPO}, \mathbf{CPO}, \mathbf{Scott}$, finite products are Cartesian products of sets equipped with the component-wise order. ◀

Let $\mathcal{C}$ be a category enriched over **DCPO**. Given objects $X, Y \in \mathcal{C}$, an *embedding-projection* pair $X \to Y$ is a pair of morphisms $f^\star : X \rightleftarrows Y : f_\star$ where $f_\star \circ f^\star = \mathrm{id}_X$ and $f^\star \circ f_\star \leq \mathrm{id}_Y$. The morphism $f^\star$ is an *embedding* (it reflects (as well as preserves) the order), while $f_\star$ is called a *projection*. Note that if X (resp. Y) has a least element, then so does Y (resp. X) and $f^\star$ (resp $f_\star$) is strict.[3] It is well-known that $f^\star$ completely determines $f_\star$ and reciprocally, see [4, §7.1] (cf. also [3, §3.1.4] and [37, §9]). Given an embedding e (resp. a projection p), we write $e_\star$ (resp. $p^\star$) for the corresponding projection (resp. embedding).

We write $\mathcal{C}^{\mathbf{e}}$ for the category with the same objects as $\mathcal{C}$, and with embedding-projection pairs as morphisms. Note that we have faithful functors $(-)^\star : \mathcal{C}^{\mathbf{e}} \to \mathcal{C}$ and $(-)_\star : \mathcal{C}^{\mathbf{e}} \to \mathcal{C}^{\mathrm{op}}$, taking $(f^\star, f_\star)$ to $f^\star$ and to $f_\star$, respectively. Given a functor H of codomain $\mathcal{C}^{\mathbf{e}}$, we write $H_\star$ for $(-)_\star \circ H$, and similarly for $H^\star$.

9.1.2 The Limit-Colimit Coincidence

The following (crucial and) well-known fact is [4, Theorem 7.1.10] (see also [3, Theorem 3.3.7]).

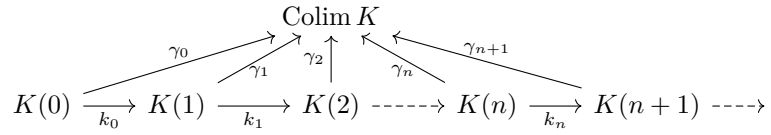
▶ **Theorem 9.2.** *Let $K : \omega \to \mathcal{C}^{\mathbf{e}}$ be a functor, where $\mathcal{C}$ is enriched over* **DCPO**. *Each limiting cone $\varpi : \operatorname{Lim} K_\star \to K_\star$ for $K_\star : \omega^{\mathrm{op}} \to \mathcal{C}$ consists of projections, and the $((\varpi_n)^\star, \varpi_n)_n$ form a colimiting cocone $K \to \operatorname{Colim} K$ in $\mathcal{C}^{\mathbf{e}}$.*

Proof. Let $K : \omega \to \mathcal{C}^{\mathbf{e}}$ and consider a limiting cone in $\mathcal{C}$:

$$\begin{array}{ccccccccccc} & & & & \operatorname{Lim} K_\star & & & & & & \\ & & \overset{\varpi_0}{\swarrow} & & \overset{\varpi_1}{\swarrow} & \downarrow{\scriptstyle \varpi_2} & \overset{\varpi_n}{\searrow} & & \overset{\varpi_{n+1}}{\searrow} & & \\ K_\star(0) & \underset{(k_0)_\star}{\leftarrow} & K_\star(1) & \underset{(k_1)_\star}{\leftarrow} & K_\star(2) & \leftarrow\!\!\text{---} & K_\star(n) & \underset{(k_n)_\star}{\leftarrow} & K_\star(n+1) & \leftarrow\!\!\text{---} & \end{array} \tag{6}$$

[3] A Scott-continuous function is *strict* if it preserves least elements.

The components of the colimiting cocone in $\mathcal{C}^{\mathbf{e}}$



(7)

are given by $(\gamma_n)_\star = \varpi_n$ for projections.

Concerning embeddings, for each $n \in \mathbb{N}$ we build a cone with vertex $K(n) = K_\star(n)$. Given $m \in \mathbb{N}$, we have a morphism $h_{n,m} \colon K(n) \to K(m)$ obtained by composing $(k_i)_\star$'s or $(k_i)^\star$'s according to whether $m \leq n$ or $n \leq m$. The $h_{n,m}$'s can be made so that $h_{n,m} = (k_m)_\star \circ h_{n,m+1}$. The universal property of limits in $\mathcal{C}$ then yields a unique morphism c_n from $K(n) = K_\star(n)$ to $\operatorname{Lim} K_\star(n)$ such that $\varpi_m \circ c_n = h_{n,m}$ for all $m \in \mathbb{N}$.

We are going to show that $c_n = (\varpi_n)^\star$. Note that $\varpi_n \circ c_n$ is the identity by definition of c_n. It remains to show that $c_n \circ \varpi_n \leq \mathrm{id}_{\operatorname{Lim} K_\star}$. We first show that $(c_n \circ \varpi_n)_n$ forms an increasing sequence in $\mathcal{C}(\operatorname{Lim} K_\star, \operatorname{Lim} K_\star)$. To this end, note that $\varpi_n = (k_n)_\star \circ \varpi_{n+1}$ (since ϖ is a cone). We moreover have $c_n = c_{n+1} \circ (k_n)^\star$ since $\varpi_m \circ c_{n+1} \circ (k_n)^\star = h_{n+1,m} \circ (k_n)^\star = h_{n,m}$ for all $m \in \mathbb{N}$. We compute

$$c_n \circ \varpi_n \quad = \quad c_{n+1} \circ (k_n)^\star \circ (k_n)_\star \circ \varpi_{n+1} \quad \leq \quad c_{n+1} \circ \varpi_{n+1}$$

Let $\ell = \bigvee_n (c_n \circ \varpi_n)$. We now claim that ℓ is the identity. This will yield that $c_n \circ \varpi_n \leq \mathrm{id}_{\operatorname{Lim} K_\star}$. In order to show that $\ell = \mathrm{id}_{\operatorname{Lim} K_\star}$, we show that $\varpi_m \circ \ell = \varpi_m$ for all $m \in \mathbb{N}$, and use the universal property of limits in $\mathcal{C}$. We have

$$\begin{aligned} \varpi_m \circ \ell &= \varpi_m \circ \textstyle\bigvee_n (c_n \circ \varpi_n) \\ &= \varpi_m \circ \textstyle\bigvee_{n \geq m} (c_n \circ \varpi_n) \\ &= \textstyle\bigvee_{n \geq m} (\varpi_m \circ c_n \circ \varpi_n) \\ &= \textstyle\bigvee_{n \geq m} (h_{n,m} \circ \varpi_n) \end{aligned}$$

But by definition of $h_{n,m}$, we have $h_{n,m} \circ \varpi_n = \varpi_m$ when $m \leq n$.

We can thus set $\gamma_n = (c_n, \varpi_n)$. Moreover, $\gamma = (\gamma_n)_n$ is indeed a cocone since $c_n = c_{n+1} \circ (k_n)^\star$ (see above).

We now claim that $\gamma \colon K \to \operatorname{Lim} K_\star$ is colimiting. To this end, consider a cocone $\tau \colon K \to C$. We thus get a cone $\tau_\star \colon C_\star \to K_\star$ in $\mathcal{C}$, and the universal property of limits yields a unique $p \colon C_\star \to \operatorname{Lim} K_\star$ such that $\varpi_n \circ p = (\tau_n)_\star$ for all $n \in \mathbb{N}$. We show that p is a projection. We define a morphism $e \colon \operatorname{Lim} K_\star \to C$ as $e = \bigvee_n ((\tau_n)^\star \circ \varpi_n)$. We have

$$\begin{aligned} e \circ p &= \left(\textstyle\bigvee_n (\tau_n)^\star \circ \varpi_n\right) \circ p \\ &= \textstyle\bigvee_n (\tau_n)^\star \circ (\tau_n)_\star \\ &\leq \mathrm{id}_C \end{aligned}$$

On the other hand, given $m \in \mathbb{N}$ we have

$$\begin{aligned} \varpi_m \circ p \circ e &= \textstyle\bigvee_n (\tau_m)_\star \circ (\tau_n)^\star \circ \varpi_n \\ &= \textstyle\bigvee_{n \geq m} (\tau_m)_\star \circ (\tau_n)^\star \circ \varpi_n \\ &= \textstyle\bigvee_{n \geq m} h_{n,m} \circ (\tau_n)_\star \circ (\tau_n)^\star \circ \varpi_n \\ &= \textstyle\bigvee_{n \geq m} h_{n,m} \circ \varpi_n \\ &= \textstyle\bigvee_{n \geq m} \varpi_m \\ &= \varpi_m \end{aligned}$$

so that $p \circ e = \mathrm{id}_{\operatorname{Lim} K_\star}$ by the universal property of limits in $\mathcal{C}$.

Moreover, for all $n \in \mathbb{N}$ we have

$$\begin{array}{rcl} e \circ c_n & = & \bigvee_m (\tau_m)^\star \circ \varpi_m \circ c_n \\ & = & \bigvee_m (\tau_m)^\star \circ h_{n,m} \\ & = & \bigvee_{m \geq n} (\tau_m)^\star \circ h_{n,m} \\ & = & \bigvee_{m \geq n} (\tau_n)^\star \\ & = & (\tau_n)^\star \end{array}$$

Consider now a morphism $\ell \colon \operatorname{Lim} K_\star \to C$ in $\mathcal{C}^{\mathbf{e}}$ such that $\varpi_n \circ \ell_\star = (\tau_n)_\star$ and $\ell^\star \circ c_n = (\tau_n)^\star$ for all $n \in \mathbb{N}$. The universal property of limits in $\mathcal{C}$ yields $\ell_\star = p$, so that $\ell^\star = e$ since e is uniquely determined from p. ◀

9.1.3 Solutions of Domain Equations

We shall use Theorem 9.2 in the following situation. Consider a functor

$$G \colon \quad \mathcal{D}^{\mathbf{e}} \times \mathcal{C}^{\mathbf{e}} \quad \longrightarrow \quad \mathcal{C}^{\mathbf{e}}$$

where $\mathcal{C}$ and $\mathcal{D}$ are enriched over **DCPO**. We moreover assume that $\mathcal{C}$ has a terminal object $\mathbf{1}$ which is initial in $\mathcal{C}^{\mathbf{e}}$. We are going to define a functor

$$K \colon \quad \mathcal{D}^{\mathbf{e}} \times \omega \quad \longrightarrow \quad \mathcal{C}^{\mathbf{e}}$$

Given an object B of $\mathcal{D}^{\mathbf{e}}$, $K(B,-)$ is the ω-chain in $\mathcal{C}^{\mathbf{e}}$ obtained by iterating $G_B = G(B,-)$ from the initial object $\mathbf{1}$ of $\mathcal{C}^{\mathbf{e}}$:

$$\mathbf{1} \xrightarrow{\mathbf{1}} G_B(\mathbf{1}) \xrightarrow{G_B(\mathbf{1})} G_B^2(\mathbf{1}) \dashrightarrow G_B^n(\mathbf{1}) \xrightarrow{G_B^n(\mathbf{1})} G_B^{n+1}(\mathbf{1}) \dashrightarrow \tag{8}$$

Given a morphism $f \colon B \to B'$ in $\mathcal{D}^{\mathbf{e}}$, $K(f,-)$ is obtained by commutativity of the following.

$$\begin{array}{ccccccccc} \mathbf{1} & \xrightarrow{\mathbf{1}} & G_B(\mathbf{1}) & \xrightarrow{G_B(\mathbf{1})} & G_B^2(\mathbf{1}) & \dashrightarrow & G_B^n(\mathbf{1}) & \xrightarrow{G_B^n(\mathbf{1})} & G_B^{n+1}(\mathbf{1}) \dashrightarrow \\ \downarrow{\scriptstyle \mathbf{1}} & & \downarrow{\scriptstyle G_f(\mathbf{1})} & & \downarrow{\scriptstyle G_f^2(\mathbf{1})} & & \downarrow{\scriptstyle G_f^n(\mathbf{1})} & & \downarrow{\scriptstyle G_f^{n+1}(\mathbf{1})} \\ \mathbf{1} & \xrightarrow[\mathbf{1}]{} & G_{B'}(\mathbf{1}) & \xrightarrow[G_{B'}(\mathbf{1})]{} & G_{B'}^2(\mathbf{1}) & \dashrightarrow & G_{B'}^n(\mathbf{1}) & \xrightarrow[G_{B'}^n(\mathbf{1})]{} & G_{B'}^{n+1}(\mathbf{1}) \dashrightarrow \end{array} \tag{9}$$

Assume now that $\mathcal{C}$ has limits of ω^{op}-chains of projections. Then Theorem 9.2 yields that each $K(B,-)$ has a colimit in $\mathcal{C}^{\mathbf{e}}$. Since K is a functor $\mathcal{D}^{\mathbf{e}} \times \omega \to \mathcal{C}^{\mathbf{e}}$, it follows from [27, Theorem V.3.1] that these colimits assemble into a functor

$$\begin{array}{rccc} \operatorname{Fix} G \colon & \mathcal{D}^{\mathbf{e}} & \longrightarrow & \mathcal{C}^{\mathbf{e}} \\ & B & \longmapsto & \operatorname{Colim}_{n \in \omega} K(B,n) \end{array}$$

If $G(B,-)$ preserves colimits of ω-chains, then the universal property of colimits gives an isomorphism $\mathsf{fold}^{\mathbf{e}} : G(B, \operatorname{Fix} G(B)) \rightleftarrows \operatorname{Fix} G(B) : \mathsf{unfold}^{\mathbf{e}}$ in $\mathcal{C}^{\mathbf{e}}$.

We are going to prove the following.

▶ **Proposition 9.3.** *If $G \colon \mathcal{D}^{\mathbf{e}} \times \mathcal{C}^{\mathbf{e}} \to \mathcal{C}^{\mathbf{e}}$ preserves colimits of ω-chains, then so does $\operatorname{Fix} G \colon \mathcal{D}^{\mathbf{e}} \to \mathcal{C}^{\mathbf{e}}$.*

The proof of Proposition 9.3 is split into the following lemmas. Fix a functor $G \colon \mathcal{D}^{\mathbf{e}} \times \mathcal{C}^{\mathbf{e}} \to \mathcal{C}^{\mathbf{e}}$ which preserves colimits of ω-chains.

▶ **Lemma 9.4.** *The diagonal functor $\triangle\colon \mathcal{D}^{\mathbf{e}} \to \mathcal{D}^{\mathbf{e}} \times \mathcal{D}^{\mathbf{e}}$ preserves colimits of ω-chains.*

Proof. Since colimits are pointwise in functor categories ([27, Corollary V.3]).[4] ◀

Lemma 9.4 entails in particular that each functor $G^n_{(-)}(\mathbf{1})\colon \mathcal{D}^{\mathbf{e}} \to \mathcal{C}^{\mathbf{e}}$ preserves colimits of ω-chains ($G^{n+1}_{(-)}(\mathbf{1})$ is $G(-, G^n_{(-)}(\mathbf{1})) \circ \triangle$).

Proposition 9.3 relies on the fact that the functor $K\colon \mathcal{D}^{\mathbf{e}} \to [\omega, \mathcal{C}^{\mathbf{e}}]$ preserves colimits of ω-chains. This involves some notation.

Let $W\colon \omega \to \mathcal{D}^{\mathbf{e}}$ be an ω-chain, with colimiting cocone $\gamma\colon W \to \operatorname{Colim} W$. In the following, we write $w_m\colon W(m) \to W(m+1)$ for the connecting morphisms of W. The cocone $K\gamma\colon K(W) \to K(\operatorname{Colim} W)$ has component at $m \in \mathbb{N}$ the commutative diagram in (9) where one takes $\gamma_m\colon W(m) \to \operatorname{Colim} W$ for $f\colon B \to B'$.

▶ **Lemma 9.5.** *The cocone $K\gamma\colon K(W) \to K(\operatorname{Colim} W)$ is colimiting.*

Proof. First, it follows from the above that each $G^n_\gamma(\mathbf{1})\colon G^n_W(\mathbf{1}) \to G^n_{\operatorname{Colim} W}(\mathbf{1})$ is colimiting.

Consider now a cocone $\tau\colon K(W) \to H$ in $[\omega, \mathcal{C}^{\mathbf{e}}]$. For each $m \in \mathbb{N}$, we have $\tau_m = \tau_{m+1} \circ K(w_m)$, that is

$$\begin{array}{ccccccccc}
\mathbf{1} & \xrightarrow{\mathbf{1}} & G_B(\mathbf{1}) & \xrightarrow{G_B(\mathbf{1})} & G^2_B(\mathbf{1}) & \dashrightarrow & G^n_B(\mathbf{1}) & \xrightarrow{G^n_B(\mathbf{1})} & G^{n+1}_B(\mathbf{1}) \dashrightarrow \\
\downarrow{\scriptstyle \mathbf{1}} & & \downarrow{\scriptstyle G_{w_m}(\mathbf{1})} & & \downarrow{\scriptstyle G^2_{w_m}(\mathbf{1})} & & \downarrow{\scriptstyle G^n_{w_m}(\mathbf{1})} & & \downarrow{\scriptstyle G^{n+1}_{w_m}(\mathbf{1})} \\
\mathbf{1} & \xrightarrow{\mathbf{1}} & G_{B'}(\mathbf{1}) & \xrightarrow{G_{B'}(\mathbf{1})} & G^2_{B'}(\mathbf{1}) & \dashrightarrow & G^n_{B'}(\mathbf{1}) & \xrightarrow{G^n_{B'}(\mathbf{1})} & G^{n+1}_{B'}(\mathbf{1}) \dashrightarrow \\
\downarrow{\scriptstyle (\tau_{m+1})_0} & & \downarrow{\scriptstyle (\tau_{m+1})_1} & & \downarrow{\scriptstyle (\tau_{m+1})_2} & & \downarrow{\scriptstyle (\tau_{m+1})_n} & & \downarrow{\scriptstyle (\tau_{m+1})_{n+1}} \\
H(0) & \xrightarrow[h(0)]{} & H(1) & \xrightarrow[h(1)]{} & H(2) & \dashrightarrow & H(n) & \xrightarrow[h(n)]{} & H(n+1) \dashrightarrow
\end{array} \tag{10}$$

where B is $W(m)$, B' is $W(m+1)$ and the $h(n)\colon H(n) \to H(n+1)$ are the connecting morphisms of H. In particular, for each $m \in \mathbb{N}$ and each $n \in \mathbb{N}$, we have $(\tau_m)_n = (\tau_{m+1})_n \circ G^n_{w_m}(\mathbf{1})$. Hence, for each $n \in \mathbb{N}$ we have a cocone $((\tau_m)_n)_m\colon G^n_{W(-)}(\mathbf{1}) \to H(n)$, and the universal property of $G^n_\gamma(\mathbf{1})$ gives a unique morphism $\ell_n\colon G^n_{\operatorname{Colim} W}(\mathbf{1}) \to H(n)$ such that $(\tau_m)_n = \ell_n \circ G^n_{\gamma_m}(\mathbf{1})$ for all $m \in \mathbb{N}$.

We show that the ℓ_n's assemble into a morphism $\ell\colon K(\operatorname{Colim} W) \to H$ in $[\omega, \mathcal{C}^{\mathbf{e}}]$. We thus have to show that the following commutes, where B' is $\operatorname{Colim} W$:

$$\begin{array}{ccccccccc}
\mathbf{1} & \xrightarrow{\mathbf{1}} & G_{B'}(\mathbf{1}) & \xrightarrow{G_{B'}(\mathbf{1})} & G^2_{B'}(\mathbf{1}) & \dashrightarrow & G^n_{B'}(\mathbf{1}) & \xrightarrow{G^n_{B'}(\mathbf{1})} & G^{n+1}_{B'}(\mathbf{1}) \dashrightarrow \\
\downarrow{\scriptstyle \ell_0} & & \downarrow{\scriptstyle \ell_1} & & \downarrow{\scriptstyle \ell_2} & & \downarrow{\scriptstyle \ell_n} & & \downarrow{\scriptstyle \ell_{n+1}} \\
H(0) & \xrightarrow[h(0)]{} & H(1) & \xrightarrow[h(1)]{} & H(2) & \dashrightarrow & H(n) & \xrightarrow[h(n)]{} & H(n+1) \dashrightarrow
\end{array}$$

We show that $\ell_{n+1} \circ G^n_{B'}(\mathbf{1}) = h(n) \circ \ell_n$ for all $n \in \mathbb{N}$. For each $m \in \mathbb{N}$, by commutativity of (9) and (10) we have the following, where B is $W(m)$:

$$\begin{array}{rcl}
\ell_{n+1} \circ G^n_{B'}(\mathbf{1}) \circ G^n_{\gamma_m}(\mathbf{1}) & = & \ell_{n+1} \circ G^{n+1}_{\gamma_m}(\mathbf{1}) \circ G^n_B(\mathbf{1}) \\
& = & (\tau_m)_{n+1} \circ G^n_B(\mathbf{1}) \\
& = & h(n) \circ (\tau_m)_n \\
& = & h(n) \circ \ell_n \circ G^n_{\gamma_m}(\mathbf{1})
\end{array}$$

[4] Note that [27, Corollary V.3] only gives the result for limits. But recall that the opposite of a functor category $[\mathcal{C}, \mathcal{D}]$ is the functor category $[\mathcal{C}^{\mathrm{op}}, \mathcal{D}^{\mathrm{op}}]$.

Then we are done by the universal property of $G^n_{\gamma_m}(\mathbf{1})$.

Consider finally a morphism $f\colon K(\operatorname{Colim} W) \to H$ in $[\omega, \mathcal{C}^{\mathbf{e}}]$ such that $f \circ K(\gamma) = \tau$. Then for all $m \in \mathbb{N}$ we have $f \circ K(\gamma_m) = \tau_m$, and thus $f_n \circ G^n_{\gamma_m}(\mathbf{1}) = (\tau_m)_n$ for all $n \in \mathbb{N}$. It follows that $f_n = \ell_n$, so that $f = \ell$. ◀

We can now prove Proposition 9.3.

Proof of Proposition 9.3. Let $W\colon \omega \to \mathcal{D}^{\mathbf{e}}$ be an ω-chain. By Lemma 9.5, and since colimits always commute over colimits, we have

$$\begin{array}{rcl} \operatorname{Fix} G(\operatorname{Colim} W) & = & \operatorname{Colim}_{n\in\omega} K(\operatorname{Colim} W, n) \\ & \cong & \operatorname{Colim}_{n\in\omega} \operatorname{Colim}_{m\in\omega} K(W(m), n) \\ & \cong & \operatorname{Colim}_{m\in\omega} \operatorname{Colim}_{n\in\omega} K(W(m), n) \\ & \cong & \operatorname{Colim}_{m\in\omega} \operatorname{Fix} G(W(m)) \end{array}$$

◀

9.1.4 Local Continuity

Functors $G\colon \mathcal{D}^{\mathbf{e}} \times \mathcal{C}^{\mathbf{e}} \to \mathcal{E}^{\mathbf{e}}$ will be obtained from "mixed-variance" functors

$$F\colon \quad \mathcal{D}^{\mathrm{op}} \times \mathcal{C} \quad \longrightarrow \quad \mathcal{E}$$

where $\mathcal{D}, \mathcal{C}, \mathcal{E}$ are enriched over **DCPO**.

▶ **Definition 9.6.** *We say that F is* locally monotone *(resp.* continuous*) if each hom-function*

$$\begin{array}{rcl} \mathcal{D}(B', B) \times \mathcal{C}(A, A') & \longrightarrow & \mathcal{C}(F(B, A), F(B', A')) \\ (g, f) & \longmapsto & F(g, f) \end{array}$$

is monotone (resp. Scott-continuous).

We refer to [3, Definition 5.2.5], [4, Definition 7.1.15] and [37, Definition 9.1]. The following is a straightforward adaptation of [4, Proposition 7.1.19] (see also [3, Proposition 5.2.6]).

▶ **Lemma 9.7.** *Let $F\colon \mathcal{D}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{E}$ be locally monotone. Then F lifts to a covariant functor*

$$F^{\mathbf{e}}\colon \quad \mathcal{D}^{\mathbf{e}} \times \mathcal{C}^{\mathbf{e}} \quad \longrightarrow \quad \mathcal{E}^{\mathbf{e}}$$

with $F^{\mathbf{e}}(B, A) = F(B, A)$ on objects and $F^{\mathbf{e}}(g, f) = (F(g_\star, f^\star)\,,\, F(g^\star, f_\star))$ on morphisms.

If moreover F is locally continuous, then $F^{\mathbf{e}}$ preserves colimits of ω-chains.

9.2 Interpretation of Pure Types

A *pure type expression* is a possibly open production of the grammar of pure types (§2), namely

$$\tau \ ::= \ \mathsf{Unit} \ \mid \ \tau \times \tau \ \mid \ \tau \to \tau \ \mid \ \tau + \tau \ \mid \ \alpha \ \mid \ \mathsf{rec}\,\alpha.\tau$$

where α is a type variable, and where $\mathsf{rec}\,\alpha.\tau$ binds α in τ.

Consider a pure type expression τ with free type variables $\vec{\alpha} = \alpha_1, \ldots, \alpha_n$. We are going to interpret τ as a functor

$$[\![\tau]\!]\colon \quad (\mathbf{Scott}^{\mathbf{e}})^n \quad \longrightarrow \quad \mathbf{Scott}^{\mathbf{e}}$$

which preserves colimits of ω-chains.

Since $\mathbf{Scott}^{\mathbf{e}}$ has the same objects as **Scott**, this thus provides the interpretation $[\![\tau]\!] \in$ **Scott** of each (pure) *type* τ (i.e. each type expression τ without free type variables α).

9.2.1 Preliminaries

Recall that the category **Scott** is Cartesian-closed (products and homsets are equipped with pointwise orders), see [3, Corollary 4.1.6] or [4, §1.4]. This yields functors

$$\begin{array}{rrcl} \mathbf{Scott}(-,-)\colon & \mathbf{Scott}^{\mathrm{op}} \times \mathbf{Scott} & \longrightarrow & \mathbf{Scott} \\ (-)\times(-)\colon & \mathbf{Scott} \times \mathbf{Scott} & \longrightarrow & \mathbf{Scott} \end{array}$$

These functors are locally continuous ([4, Example 7.1.16]). By Lemma 9.7 we obtain functors

$$(\mathbf{Scott}(-,-))^{\mathbf{e}}\,, ((-)\times(-))^{\mathbf{e}}\colon \quad \mathbf{Scott}^{\mathbf{e}} \times \mathbf{Scott}^{\mathbf{e}} \quad\longrightarrow\quad \mathbf{Scott}^{\mathbf{e}}$$

which preserve colimits of ω-chains.

Moreover, **Scott** has limits of ω^{op}-chains of projections (in the embedding-projection sense), see [3, Theorem 3.3.7, Theorem 3.3.11 and Proposition 4.1.3]. More precisely, the full inclusion $\mathbf{Scott} \hookrightarrow \mathbf{DCPO}$ creates limits for ω^{op}-chains of projections.[5] In particular, **Scott** is closed in **DCPO** under limits of ω^{op}-chains of projections. Note that the category **DCPO** has all limits, and that they are created by the forgetful functor to the category of posets (and monotone functions), see [3, Theorem 3.3.1]. It follows that given $K\colon \omega \to \mathbf{Scott}^{\mathbf{e}}$, the limit of $K_\star\colon \omega^{\mathrm{op}} \to \mathbf{Scott}$ is

$$\left\{ (x_i)_i \in \prod_{i\in\mathbb{N}} K(i) \;\middle|\; K_\star(i \le j)(x_j) = x_i \right\}$$

equipped with the pointwise order. Moreover, the limiting cone $\operatorname{Lim} K_\star \to K_\star$ consists in set-theoretic projections.[6] In view of Theorem 9.2, we also get that **Scott** is closed in **DCPO** under colimits of ω-chains of embeddings.

The terminal object $\mathbf{1} = \{\bot\}$ of **Scott** is initial in $\mathbf{Scott}^{\mathbf{e}}$ ([4, Proposition 7.1.9]).

9.2.2 Interpretation of (weak) sum types

We now discuss the interpretation of $\tau + \sigma$. It is well-known that the categories **Scott** and **CPO**, as opposed to **DCPO**, do not have coproducts [18]. We shall thus interpret $\tau + \sigma$ as a *weak* coproduct, satisfying a weakening of the universal property of coproducts: we do not require the "universal" morphisms to be unique.

Consider first the composite functor

$$\mathbf{CPO} \times \mathbf{CPO} \xrightarrow{\boldsymbol{\iota}\times\boldsymbol{\iota}} \mathbf{DCPO} \times \mathbf{DCPO} \xrightarrow{(-)\amalg(-)} \mathbf{DCPO} \xrightarrow{(-)_\bot} \mathbf{CPO} \tag{11}$$

Here, $\boldsymbol{\iota}\colon \mathbf{CPO} \hookrightarrow \mathbf{DCPO}$ is the full inclusion, while $(-)\amalg(-)$ is the coproduct in **DCPO** (disjoint union with the pointwise order), and $(-)_\bot\colon \mathbf{DCPO} \to \mathbf{CPO}$ adds a new least element.

Note that the inclusion $\iota_\bot\colon X \hookrightarrow X_\bot$ is Scott-continuous (since directed sets are non-empty) and an order embedding. Also, given $f\colon X \to Y$ and $g\colon X' \to Y'$ we have

$$\begin{array}{rrcl} (f \amalg g)_\bot\colon & (X \amalg Y)_\bot & \longrightarrow & (X' \amalg Y')_\bot \\ & \bot & \longmapsto & \bot \\ & (\iota_\bot \circ \kappa_1)(x) & \longmapsto & (\iota_\bot \circ \kappa_1)(f(x)) \\ & (\iota_\bot \circ \kappa_2)(y) & \longmapsto & (\iota_\bot \circ \kappa_2)(g(y)) \end{array}$$

[5] The notion of creation of limits has to be understood in the usual sense of [27, Definition V.1].
[6] These are also projections in the embedding-projection sense by Theorem 9.2.

where κ_1 and κ_2 are the coprojection for the coproduct $(-) \amalg (-)$ in **DCPO**.

▶ **Lemma 9.8.** *Given $X, Y \in$ **Scott**, we have $(X \amalg Y)_\perp \in$ **Scott**.*

Proof. We have seen that $(X \amalg Y)_\perp$ is a cpo. It is algebraic since the only finite approximation of $\perp \in (X \amalg Y)_\perp$ is $\perp$, while given $x \in X$, the finite approximations of $(\iota_\perp \circ \kappa_1)(x)$ are $\perp$ and the $(\iota_\perp \circ \kappa_1)(d)$'s where d is a finite approximation of x in X (and similarly with $(\iota_\perp \circ \kappa_2)(y)$ for $y \in Y$).

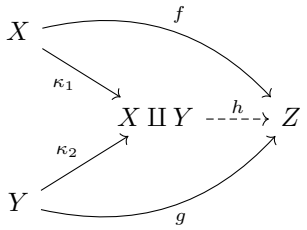
Moreover, $(X \amalg Y)_\perp$ is bounded-complete since if a, b have an upper bound, then either both a and b are in the same components, or at least one of them is $\perp$. ◀

▶ **Definition 9.9.** *Given $X, Y \in$ **Scott**, we let $X + Y \coloneqq (X \amalg Y)_\perp \in$ **Scott**.*

Let $X, Y \in$ **Scott**. The "coprojections" $\mathsf{in}_1 \colon X \to X + Y$ and $\mathsf{in}_2 \colon Y \to X + Y$ are the order embeddings defined as the composites

$$X \xrightarrow{\kappa_1} X \amalg Y \xhookrightarrow{\iota_\perp} X + Y \qquad \text{and} \qquad Y \xrightarrow{\kappa_2} X \amalg Y \xhookrightarrow{\iota_\perp} X + Y$$

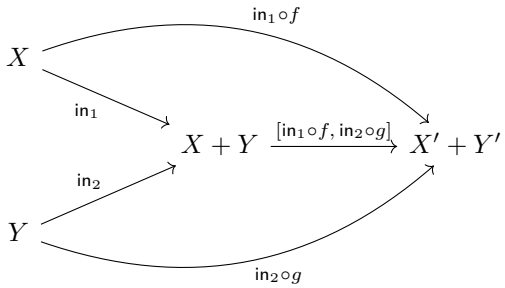
Further, consider some $Z \in$ **Scott**, together with $f \colon X \to Z$ and $g \colon Y \to Z$. Let the dashed arrow h in the diagram below be given by the universal property of coproducts in **DCPO**:



We define $[f, g] \colon X + Y \to Z$ to be the extension of h with $[f, g](\perp) = \perp_Z$.

The function $[f, g]$ is Scott-continuous. Indeed, if $a \leq b$ in $X + Y$, then either $a = \perp$ and $[f, g](a) = \perp_Z \leq [f, g](b)$, or both a and b are in the same component. Moreover, given a directed $D \subseteq X + Y$, if $D \setminus \{\perp\}$ is non-empty, then it is a directed set either in the component X or in the component Y, and $h(\bigvee D) = h(\bigvee (D \setminus \{\perp\})) = \bigvee h(D \setminus \{\perp\}) = \bigvee [f, g](D)$ since $[f, g](\perp) = \perp_Z$.

Let now $f \colon X \to Y$ and $g \colon X' \to Y'$ in **Scott**, and consider $[\mathsf{in}_1 \circ f, \mathsf{in}_2 \circ g]$ in



Then $[\mathsf{in}_1 \circ f, \mathsf{in}_2 \circ g] = f + g$, where $f + g = (f \amalg g)_\perp$ (cf Equation (11)). Indeed, we have $[\mathsf{in}_1 \circ f, \mathsf{in}_2 \circ g](\perp) = \perp = (f + g)(\perp)$, while $[\mathsf{in}_1 \circ f, \mathsf{in}_2 \circ g](\mathsf{in}_1(x)) = \mathsf{in}_1(f(x)) = (\iota_\perp \circ \kappa_1)(f(x)) = (f \amalg g)_\perp((\iota_\perp \circ \kappa_1)(x)) = (f + g)(\mathsf{in}_1(x))$, and similarly for $\mathsf{in}_2(y)$.

▶ **Lemma 9.10.** *The functor $(-) + (-) \colon$ **Scott** $\times$ **Scott** $\to$ **Scott** is locally continuous.*

Proof. Let $X, Y, X', Y' \in \mathbf{Scott}$, and consider the hom-function

$$\begin{array}{rcl} \mathbf{Scott}(X,Y) \times \mathbf{Scott}(X',Y') & \longrightarrow & \mathbf{Scott}(X+Y,\, X'+Y') \\ (f,g) & \longmapsto & f+g \end{array}$$

This function is monotone: given $f \leq f'$ and $g \leq g'$ we have $f+g \leq f'+g'$ since $(f+g)(\bot) = \bot = (f'+g')(\bot)$, while $(f+g)(\mathsf{in}_1(x)) \leq (f'+g')(\mathsf{in}_1(x))$ since $(f+g)(\mathsf{in}_1(x)) = [\mathsf{in}_1 \circ f, \mathsf{in}_2 \circ g](\mathsf{in}_1(x)) = \mathsf{in}_1(f(x))$ for $x \in X$, and similarly for $y \in Y$.

This function is moreover Scott-continuous. Recall (from e.g. [4, Proposition 1.4.3]) that Scott-continuity in several arguments is equivalent to Scott-continuity in each argument separately. Let $F \subseteq \mathbf{CPO}(X,Y)$ be directed, and let $g \in \mathbf{CPO}(X',Y')$. Since directed sups of Scott-continuous functions are pointwise, we have $((\bigvee F)+g)(\bot) = \bot = (\bigvee_{f\in F}(f+g))(\bot)$ and $((\bigvee F)+g)(\mathsf{in}_2(y)) = \mathsf{in}_2(g(y)) = (\bigvee_{f\in F}(f+g))(\mathsf{in}_2(y))$, while

$$\begin{array}{rcl} ((\bigvee F)+g)(\mathsf{in}_1(x)) & = & \mathsf{in}_1((\bigvee F)(x)) \\ & = & \mathsf{in}_1(\bigvee_{f\in F} f(x)) \\ & = & \bigvee_{f\in F}(\mathsf{in}_1(f(x))) \\ & = & \bigvee_{f\in F}((f+g)(\mathsf{in}_1(x))) \\ & = & (\bigvee_{f\in F}(f+g))(\mathsf{in}_1(x)) \end{array}$$

The case of a directed sup in the second argument of $(-)+(-)$ is similar. ◀

By combining Lemma 9.10 with Lemma 9.7, we obtain a functor

$$((-)+(-))^{\mathbf{e}} : \quad \mathbf{Scott}^{\mathbf{e}} \times \mathbf{Scott}^{\mathbf{e}} \quad \longrightarrow \quad \mathbf{Scott}^{\mathbf{e}}$$

which preserves colimits of ω-chains.

9.2.3 Definition of the Interpretation

Let τ be a (pure) type expression with free type variables $\vec{\alpha} = \alpha_1, \ldots, \alpha_n$. The interpretation $[\![\tau]\!] \colon (\mathbf{Scott}^{\mathbf{e}})^n \to \mathbf{Scott}^{\mathbf{e}}$ is defined by induction on τ.

- In the case of $\tau = \alpha_i$, we let $[\![\tau]\!]$ take $\vec{X} = X_1, \ldots, X_n$ to X_i.
- In the case of $\tau = \mathsf{Unit}$, we let $[\![\tau]\!](\vec{X})$ be $\{\bot, \top\}$ with $\bot \leq \top$.
  Moreover, we let $[\![\langle\rangle]\!] \coloneqq \top \in [\![\mathsf{Unit}]\!](\vec{X})$.
- In the cases of $\tau \times \sigma$ and $\sigma \to \tau$, the induction hypotheses give us functors

  $$[\![\tau]\!], [\![\sigma]\!] \colon \quad (\mathbf{Scott}^{\mathbf{e}})^n \quad \longrightarrow \quad \mathbf{Scott}^{\mathbf{e}}$$

  which preserve colimits of ω-chains. We can thus set

  $$\begin{array}{rcl} [\![\sigma \to \tau]\!](\vec{X}) & = & \Big(\mathbf{Scott}\left([\![\sigma]\!](\vec{X}),\, [\![\tau]\!](\vec{X})\right)\Big)^{\mathbf{e}} \\ [\![\tau \times \sigma]\!](\vec{X}) & = & \Big([\![\tau]\!](\vec{X}) \times [\![\sigma]\!](\vec{X})\Big)^{\mathbf{e}} \end{array}$$

- In the case of $\tau + \sigma$, we similarly let

  $$[\![\tau+\sigma]\!](\vec{X}) \quad = \quad \Big([\![\tau]\!](\vec{X}) + [\![\sigma]\!](\vec{X})\Big)^{\mathbf{e}}$$

  where $(-)+(-)$ is the function in Definition 9.9 (§9.2.2).
  Moreover, we let $[\![\mathsf{in}_1]\!] \colon [\![\tau]\!](\vec{X}) \to [\![\tau]\!](\vec{X}) + [\![\sigma]\!](\vec{X})$ and $[\![\mathsf{in}_2]\!] \colon [\![\sigma]\!](\vec{X}) \to [\![\tau]\!](\vec{X}) + [\![\sigma]\!](\vec{X})$ be the morphisms in_1 and in_2 defined in §9.2.2.
  Further, the terms $\mathsf{case}\ M\ [x_1 \mapsto N_1 \mid x_2 \mapsto N_2]$ are interpreted using the operation $[-,-]$, also defined in §9.2.2.

$$\frac{}{\bot \in \mathrm{Fin}([\![\tau]\!](\vec{X}))} \qquad \frac{d \text{ finite in } X_i}{d \in \mathrm{Fin}([\![\alpha_i]\!](\vec{X}))}$$

$$\frac{}{\top \in \mathrm{Fin}([\![\mathsf{Unit}]\!](\vec{X}))} \qquad \frac{d \in \mathrm{Fin}([\![\tau]\!](\vec{X}))}{[\![\mathsf{in}_1]\!](d) \in \mathrm{Fin}([\![\tau+\sigma]\!](\vec{X}))} \qquad \frac{d \in \mathrm{Fin}([\![\sigma]\!](\vec{X}))}{[\![\mathsf{in}_2]\!](d) \in \mathrm{Fin}([\![\tau+\sigma]\!](\vec{X}))}$$

$$\frac{d \in \mathrm{Fin}([\![\tau]\!](\vec{X})) \qquad e \in \mathrm{Fin}([\![\sigma]\!](\vec{X}))}{(d,e) \in \mathrm{Fin}([\![\tau \times \sigma]\!](\vec{X}))} \qquad \frac{d \in \mathrm{Fin}([\![\tau[\mathsf{rec}\,\alpha.\tau/\alpha]]\!](\vec{X}))}{[\![\mathsf{fold}]\!](d) \in \mathrm{Fin}([\![\mathsf{rec}\,\alpha.\tau]\!](\vec{X}))}$$

$$\frac{\begin{array}{l}\text{for all } i \in I,\ d_i \in \mathrm{Fin}([\![\tau]\!])(\vec{X}) \text{ and } e_i \in \mathrm{Fin}([\![\sigma]\!])(\vec{X})\ ; \\ \text{for all } J \subseteq I,\ \bigvee_{j\in J} d_j \text{ defined in } [\![\tau]\!](\vec{X}) \Rightarrow \bigvee_{j\in J} e_j \text{ defined in } [\![\sigma]\!](\vec{X})\end{array}}{\bigvee_{i\in I}(d_i \searrow e_i) \in \mathrm{Fin}([\![\tau \to \sigma]\!](\vec{X}))}\ (I \text{ finite})$$

Figure 12 Inductive description of the finite elements of $[\![\tau]\!](\vec{X})$.

In the case of $\mathsf{rec}\,\alpha.\tau$, the induction hypothesis gives a functor

$$[\![\tau]\!]\colon \quad (\mathbf{Scott}^{\mathbf{e}})^n \times \mathbf{Scott}^{\mathbf{e}} \quad \longrightarrow \quad \mathbf{Scott}^{\mathbf{e}}$$

which preserves colimits of ω-chains. Theorem 9.2 gives a functor

$$\begin{array}{rccl}[\![\mathsf{rec}\,\alpha.\tau]\!]\colon & (\mathbf{Scott}^{\mathbf{e}})^n & \longrightarrow & \mathbf{Scott}^{\mathbf{e}} \\ & \vec{X} & \longmapsto & \mathrm{Fix}([\![\tau]\!](\vec{X}))\end{array}$$

This functor preserves colimits of ω-chains by Proposition 9.3. Moreover, since $[\![\tau]\!]$ preserves colimits of ω-chains, we obtain canonical isomorphisms

$$[\![\mathsf{fold}]\!]\colon\ [\![\tau[\mathsf{rec}\,\alpha.\tau/\alpha]]\!](\vec{X}) \quad \rightleftarrows \quad [\![\mathsf{rec}\,\alpha.\tau]\!](\vec{X})\ :[\![\mathsf{unfold}]\!]$$

by taking $[\![\mathsf{fold}]\!] = (\mathsf{fold}^{\mathbf{e}})^\star$ and $[\![\mathsf{unfold}]\!] = (\mathsf{unfold}^{\mathbf{e}})^\star$.

9.2.4 Description of the Finite Elements

For each (pure) type expression τ with free variables $\vec{\alpha} = \alpha_1, \ldots, \alpha_n$, we define a set $\mathrm{Fin}([\![\tau]\!](\vec{X}))$. The definition is by induction on derivations with the rules in Figure 12. The set $\mathrm{Fin}([\![\tau]\!](\vec{X}))$ describes the finite elements of $[\![\tau]\!](\vec{X})$. This relies on the following.

First, the least element $\bot$ is finite in X for each $X \in \mathbf{Scott}$. Moreover, $\top$ is finite in $[\![\mathsf{Unit}]\!] = \{\bot, \top\}$.

Let $X, Y \in \mathbf{Scott}$. The finite elements in the weak coproduct $X + Y$ are $\bot$, as well as the $[\![\mathsf{in}_1]\!](d)$'s and the $[\![\mathsf{in}_2]\!](e)$'s where d is finite in X and e is finite in Y. The finite elements in the product $X \times Y$ are exactly the pairs of finite elements. The finite elements of $\mathbf{Scott}(X, Y)$ are exactly the finite sups of step functions. Given finite $d \in X$ and $e \in Y$, the *step function* $(d \searrow e)\colon X \to Y$ is defined as $(d \searrow e)(x) = e$ if $x \geq d$ and $(d \searrow e)(x) = \bot$ otherwise. Recall that the sup $\bigvee_{i\in I}(d_i \searrow e_i)$ of a finite family of step functions exists if, and only if, for every $J \subseteq I$, the set $\{e_j \mid j \in J\}$ has an upper bound whenever so does $\{d_j \mid j \in J\}$. See [4, Theorem 1.4.12].

Concerning recursive types, let

$$G\colon \quad \mathbf{Scott}^{\mathbf{e}} \quad \longrightarrow \quad \mathbf{Scott}^{\mathbf{e}}$$

be a functor which preserves colimits of ω-chains. Recall that $\operatorname{Fix} G$ is the colimit in (7) where $K\colon \omega \to \mathbf{Scott}^{\mathbf{e}}$ takes n to $G^n(\mathbf{1})$ (similarly as in (8)). We have seen that **Scott** is closed in **DCPO** under colimits of ω-chains of embeddings. Hence it follows from [3, Theorem 3.3.11] that the finite elements of $\operatorname{Fix} G$ are the images of the finite elements of the $G^n(\mathbf{1})$'s under the components of the colimiting cocone $\gamma\colon K \to \operatorname{Fix} G$.

We thus have the following.

▶ **Proposition 9.11.** $\mathrm{Fin}([\![\tau]\!](\vec{X}))$ *is the set of finite elements of* $[\![\tau]\!](\vec{X})$.

9.3 Proofs of Example 5.2

We now provide some details on $[\![\mathsf{Stream}\,\sigma]\!]$ and $[\![\mathsf{Tree}\,\sigma]\!]$ (Example 5.2), where σ is a pure type. We handle streams and binary trees uniformly by considering the covariant functor

$$\begin{array}{rccc} F\colon & \mathbf{Scott} & \longrightarrow & \mathbf{Scott} \\ & X & \longmapsto & [\![\sigma]\!] \times X^{\mathbf{k}} \end{array}$$

where $\mathbf{k}$ is a finite set. In view of Theorem 9.2, $\operatorname{Fix} F$ is the limit of the ω^{op}-chain

$$\mathbf{1} \xleftarrow{\mathbf{1}} F(\mathbf{1}) \xleftarrow{F(\mathbf{1})} F^2(\mathbf{1}) \longleftarrow\!\!\!\!\!\!\cdots\; F^n(\mathbf{1}) \xleftarrow{F^n(\mathbf{1})} F^{n+1}(\mathbf{1}) \longleftarrow\!\!\!\!\!\!\cdots$$

where $\mathbf{1}$ is the terminal Scott domain $\{\bot\}$. Hence, $\operatorname{Fix} F$ is

$$\left\{ x \in \textstyle\prod_{n\in\mathbb{N}} F^n(\mathbf{1}) \;\middle|\; x(n) = F^n(\mathbf{1})(x(n+1)) \right\}$$

equipped with the pointwise order. In order to see that $\operatorname{Fix} F$ is isomorphic to $[\![\sigma]\!]^{\mathbf{k}^*}$, define for each $n \in \mathbb{N}$ an isomorphism $\iota_n\colon [\![\sigma]\!]^{\mathbf{k}^n} \to F^n(\mathbf{1})$ as $\iota_0 = \mathbf{1}\colon \mathbf{1} \to \mathbf{1}$ and

$$\begin{array}{rccc} \iota_{n+1}\colon & [\![\sigma]\!]^{\mathbf{k}^{n+1}} & \longrightarrow & F^{n+1}(\mathbf{1}) = [\![\sigma]\!] \times (F^n(\mathbf{1}))^{\mathbf{k}} \\ & T & \longmapsto & (T(\varepsilon),\ (\iota_n(u \mapsto T(d \cdot u)))_{d\in\mathbf{k}}) \end{array}$$

Then observe that the following commutes

$$\begin{array}{ccc} [\![\sigma]\!]^{\mathbf{k}^{n+1}} & \xrightarrow{\iota_{n+1}} & F^{n+1}(\mathbf{1}) \\ {\scriptstyle T\mapsto T\restriction \mathbf{k}^n}\downarrow & & \downarrow{\scriptstyle F^n(\mathbf{1})} \\ [\![\sigma]\!]^{\mathbf{k}^n} & \xrightarrow[\iota_n]{} & F^n(\mathbf{1}) \end{array}$$

The characterisation of the finite elements then follows from Proposition 9.11.

9.3.1 Martin Hofmann's Breadth-First Tree Traversal

We now show that the function $\mathsf{bft}\colon \mathsf{Tree}\,\sigma \to \mathsf{Stream}\,\sigma$ in Table 1 (§2) indeed computes a tree breadth-first traversal. We work at the denotational level.

It will be convenient to see $[\![\mathsf{Tree}\,\sigma]\!]$ as $(1 \cdot \{0,1\}^*) \to [\![\sigma]\!]$ rather than $\{0,1\}^* \to [\![\sigma]\!]$ (via the bijection $\{0,1\}^* \to 1 \cdot \{0,1\}^*$ which prepends a 1). Likewise, $[\![\mathsf{Stream}\,\sigma]\!]$ will be seen as $\mathbb{N}_{>0} \to [\![\sigma]\!]$ rather than $\mathbb{N} \to [\![\sigma]\!]$. With these conventions, the *breadth-first traversal* of a tree $t \in [\![\mathsf{Tree}\,\sigma]\!]$ is the stream $x \in [\![\mathsf{Stream}\,\sigma]\!]$ such that $x(i) = t(b(i))$, where $b(i)$ is the binary representation of $i \in \mathbb{N}_{>0}$ with most significant digit first (hence $b(1) = 1$ and $b(2) = 10$). Note that in a tree t, the children of the node $b(i)$ (of label $t(b(i))$) are the nodes $b(2i)$ and $b(2i+1)$ (of labels $t(b(2i))$ and $t(b(2i+1))$).

Fix a tree $t \in [\![\mathsf{Tree}\,\sigma]\!]$, and let $x \in [\![\mathsf{Stream}\,\sigma]\!]$ be its breadth-first traversal.

Given a binary string $b \in 1 \cdot \{0,1\}^*$, we write $t\restriction b$ for the tree taking $1 \cdot b' \in 1 \cdot \{0,1\}^*$ to $t(b \cdot b')$ (thus dropping the first 1 of $1 \cdot b'$). Note that $t\restriction 1 = t$.

For notational simplicity, we drop the brackets $[\![-]\!]$ around interpretations of terms (so that e.g. bftrec stands for $[\![\mathsf{bftrec}]\!]$). Given $i \in \mathbb{N}_{>0}$ and $n \in \mathbb{N}$, let

$$\begin{array}{rcl} a(i) & = & \mathsf{bftrec}\ (t\restriction b(i)) \\ c_n & = & a(2^n) \circ \cdots \circ a(2^{n+1}-1) \end{array}$$

Note that $a(i), c_n \colon [\![\mathsf{Cont}(\mathsf{Stream}\,\sigma)]\!] \to [\![\mathsf{Cont}(\mathsf{Stream}\,\sigma)]\!]$ and that $c_0 = a(1) = (\mathsf{bftrec}\ \ t)$.

▶ **Lemma 9.12.** *For all $n \in \mathbb{N}$, we have*

$$\mathsf{extract}\ (c_n\,\mathsf{Over}) \quad = \quad x(2^n) :: \ldots :: x(2^{n+1}-1) :: (\mathsf{extract}\ (c_{n+1}\,\mathsf{Over}))$$

Proof. Note that given $c \in [\![\mathsf{Cont}(\mathsf{Stream}\,\sigma)]\!]$ and $i \in \mathbb{N}_{>0}$, since $\mathsf{left}(t\restriction b(i)) = t\restriction b(2i)$ and $\mathsf{right}(t\restriction b(i)) = t\restriction b(2i+1)$, we have

$$\begin{array}{rcl} \mathsf{unfold}\ (a(i)\ c) & = & \lambda k.\ t(b(i)) :: \mathsf{unfold}\, c\ (k \circ a(2i) \circ a(2i+1)) \\ & = & \lambda k.\ x(i) :: \mathsf{unfold}\, c\ (k \circ a(2i) \circ a(2i+1)) \end{array}$$

and thus

$$\mathsf{extract}\ (a(2^n)\ c) \quad = \quad x(2^n) :: \mathsf{unfold}\, c\ (\mathsf{extract} \circ a(2^{n+1}) \circ a(2^{n+1}+1))$$

It follows that

$$\begin{array}{rcl} \mathsf{extract}\ (c_n\,\mathsf{Over}) & = & \mathsf{extract}\ \big((a(2^n) \circ \cdots \circ a(2^{n+1}-1))\,\mathsf{Over}\big) \\ & = & \mathsf{extract}\ \big(a(2^n)\ \underbrace{((a(2^n+1) \circ \cdots \circ a(2^{n+1}-1))\,\mathsf{Over})}_{c}\big) \\ & = & x(2^n) :: \mathsf{unfold}\ \ c\ (\mathsf{extract} \circ a(2^{n+1}) \circ a(2^{n+1}+1)) \end{array}$$

But

$$\begin{array}{rcl} c & = & (a(2^n+1) \circ \cdots \circ a(2^{n+1}-1))\,\mathsf{Over} \\ & = & a(2^n+1)\ \underbrace{\big((a(2^n+2) \circ \cdots \circ a(2^{n+1}-1))\,\mathsf{Over}\big)}_{c'} \end{array}$$

so that

$$\begin{array}{rcl} & & \mathsf{unfold}\ \ c\ (\mathsf{extract} \circ a(2^{n+1}) \circ a(2^{n+1}+1)) \\ & = & x(2^n+1) :: \mathsf{unfold}\ \ c'\ \big(\mathsf{extract} \circ a(2^{n+1}) \circ a(2^{n+1}+1) \circ a(2(2^n+1)) \circ a(2(2^n+1)+1)\big) \\ & = & x(2^n+1) :: \mathsf{unfold}\ \ c'\ \big(\mathsf{extract} \circ a(2^{n+1}) \circ a(2^{n+1}+1) \circ a(2^{n+1}+2) \circ a(2^{n+1}+3)\big) \end{array}$$

Hence, $\mathsf{extract}\ (c_n\,\mathsf{Over})$ is

$$x(2^n) :: x(2^n+1) :: \mathsf{unfold}\ \ c'\ \big(\mathsf{extract} \circ a(2^{n+1}) \circ a(2^{n+1}+1) \circ a(2^{n+1}+2) \circ a(2^{n+1}+3)\big)$$

By iteratively unfolding c' as much as possible, we obtain

$$\begin{array}{rcl} \mathsf{extract}\ (c_n\,\mathsf{Over}) & = & x(2^n) :: \ldots :: x(2^{n+1}-1) :: \mathsf{unfold}\ \ \mathsf{Over}\ (\mathsf{extract} \circ c_{n+1}) \\ & = & x(2^n) :: \ldots :: x(2^{n+1}-1) :: (\mathsf{extract}\ (c_{n+1}\,\mathsf{Over})) \end{array}$$

◀

Since $(\mathsf{bft}\ \ t)$ is $(\mathsf{extract}\ (c_0\ \ \mathsf{Over}))$, we thus obtain from Lemma 9.12 that $(\mathsf{bft}\,t)(i) = x(i)$ for all $i \in \mathbb{N}_{>0}$.

10 Proofs of §5.2 (Scott Domains as Spectral Spaces)

We provide some additional well-known material on spectral spaces (§5.2). Besides [11, 12, 13], see also [24] on the more general *stably compact* spaces.

Topological Spaces. A *topological space* is a pair $(X, \mathcal{O}(X))$ of a set X and a collection $\mathcal{O}(X)$ of subsets of X, called *open sets*. The collection $\mathcal{O}(X)$ is called a *topology* on X, and is asked to be stable under arbitrary unions and under finite intersections. In particular, $\emptyset$ and X are open in X (respectively as the empty union and the empty intersection).

A set $C \subseteq X$ is *closed* if its complement $X \setminus C$ is open. Closed sets are stable under finite unions and arbitrary intersections. Hence, any $S \subseteq X$ is contained in a least closed set $\overline{S} \subseteq X$. Each space $(X, \mathcal{O}(X))$ is equipped with a *specialisation* (pre)order $\leq_{\mathcal{O}(X)}$ on X, defined as

$$x \leq_{\mathcal{O}(X)} y \quad \text{iff} \quad (\forall U \in \mathcal{O}(X))\,(x \in U \implies y \in U)$$

Given $x \in X$, we have $\overline{\{x\}} = {\downarrow}x := \{y \in X \mid y \leq_{\mathcal{O}(X)} x\}$ ([13, Lemma 4.2.7]). A topology $\mathcal{O}(X)$ is T_0 when $\leq_{\mathcal{O}(X)}$ is a partial order ([13, Proposition 4.2.3]).

A subset S of a T_0 space $(X, \mathcal{O}(X))$ is *saturated* if S is upward-closed w.r.t. the specialisation order $\leq_{\mathcal{O}(X)}$, or equivalently if $S = \bigcap\{U \in \mathcal{O}(X) \mid S \subseteq U\}$ ([13, Proposition 4.2.9]).

In the case of the Scott topology $\mathcal{O}(X)$ on a dcpo $(X, \leq_X)$, the specialisation (pre)order $\leq_{\mathcal{O}(X)}$ coincides with $\leq_X$ ([4, Lemma 1.2.3]). Hence $\leq_{\mathcal{O}(X)}$ is a partial order and $(X, \mathcal{O}(X))$ is T_0. Moreover, a set $S \subseteq X$ is saturated precisely when S is upward-closed w.r.t. $\leq_X$.

Spectral Spaces. We take [11, Definition 1.1.5] as our official definition of spectral space.

▶ **Definition 10.1.** *A space $(X, \mathcal{O}(X))$ is* spectral *if all the following conditions are satisfied.*

(i) $(X, \mathcal{O}(X))$ *is T_0.*

(ii) $\mathcal{KO}(X)$ *is stable under finite intersections and form a basis of $\mathcal{O}(X)$.*

(iii) $(X, \mathcal{O}(X))$ *is sober.*

Items (i) and (ii) in Definition 10.1 were explained above and in §5.2. Note that item (ii) implies that $X \in \mathcal{KO}(X)$ (as the empty intersection). Spectral spaces are therefore compact.

Regarding item (iii), let us say that a closed set F is *irreducible* if given a (possibly empty) finite set of closed sets $\mathcal{C}$, if $F \subseteq \bigcup \mathcal{C}$, then $F \subseteq C$ for some $C \in \mathcal{C}$. Hence irreducible closed sets are non-empty. It is clear that each $\overline{\{x\}}$ is irreducible. A space $(X, \mathcal{O}(X))$ is *sober* if for each irreducible closed set F, there is exactly one $x \in X$ such that $F = \overline{\{x\}}$. Note that if X is T_0, then $\overline{\{x\}} = \overline{\{y\}}$ implies $x = y$.

It is well-known that if X is an algebraic dcpo, then the Scott topology on X is sober.

▶ **Lemma 10.2.** *If X is an algebraic dcpo, then $(X, \mathcal{O}(X))$ is a sober space.*

Proof. Let C be an irreducible closed set. We have to provide some $x \in X$ such that $C = \overline{\{x\}}$. Let D be the set of all finite $d \in X$ such that $d \in C$.

We show that D is directed. Since C is non-empty, let $x \in C$. Since X is algebraic, there is some finite $d \leq x$, and $d \in C$ since C is Scott-closed. Hence D is non-empty. Let now $d, d' \in D$. Assume toward a contradiction that $\uparrow d \cap \uparrow d' \cap C = \emptyset$, and consider the closed sets $F := X \setminus \uparrow d$ and $F' := X \setminus \uparrow d'$. Then $C \subseteq F \cup F'$, but $C \not\subseteq F, F'$ since $d, d' \in C$. Hence there is some $x \in \uparrow d \cap \uparrow d' \cap C$, and d, d' are two finite approximations of x. But since X

is algebraic, the set of finite approximations of x is directed. Hence there is some finite d'' such that $d, d' \leq d'' \leq x$, and we get $d'' \in C$ since $x \in C$ with C Scott-closed.

Let now $x = \bigvee D$. Since $D \subseteq C$ with C Scott-closed, we have $x \in C$.

We now show that $C = \overline{\{x\}}$, i.e. that for all $y \in X$, we have $y \in C$ if, and only if, $y \leq x$. If $y \leq x$, then $y \in C$ since $x \in C$ with C Scott-closed. Assume now $y \in C$. Given a finite $d \leq y$, we have $d \in C$ since C is Scott-closed, and thus $d \in D$ and $d \leq x$. By algebraicity, it follows that $y \leq x$. ◀

Equivalent Definitions of Spectral Spaces. It follows from Lemma 10.2 that if X is a Scott domain, then $(X, \mathcal{O}(X))$ is spectral in the sense of Definition 10.1. But the notion of spectral space we gave in §5.2 differs from Definition 10.1, since item (iii) is replaced with the condition that X is well-filtered (cf Proposition 5.4). It is well-known that the two conditions are equivalent for T_0 spaces which satisfy a weakening of condition (ii) in Definition 10.1. We present here a reformulation of proofs in [13, §8.3]. See also [12, §6.1 and Exercise 6.3.6].

A space $(X, \mathcal{O}(X))$ is *locally* compact if for each $x \in X$ and each open $U \in \mathcal{O}(X)$ such that $x \in U$, there is an open $V \in \mathcal{O}(X)$ and a compact K such that

$$x \subseteq V \subseteq K \subseteq U$$

It is clear that condition (ii) in Definition 10.1 implies local compactness. The following is e.g. [13, Proposition 8.3.8].

▶ **Proposition 10.3.** *If $(X, \mathcal{O}(X))$ is T_0, locally compact and well-filtered, then $(X, \mathcal{O}(X))$ is sober.*

Proof. Let C be an irreducible closed set. We have to show that $C = \overline{\{x\}}$ for some $x \in X$. Let $\mathcal{K}_C$ be the set of all compacts K such that $U \subseteq K$ for some open U with $U \cap C \neq \emptyset$.

We first show that $\mathcal{K}_C$ is codirected. First, since C is not empty, there is some $x \in C$. By local compactness (applied to $x \in X$ with X open), there is an open U and a compact K such that $x \in U \subseteq K$. But $x \in U \cap C$ implies $U \cap C \neq \emptyset$. Hence $K \in \mathcal{K}_C$. Let now $K, L \in \mathcal{K}_C$. Hence, there are opens U, V such that $U \subseteq K$, $V \subseteq L$ and $U \cap C \neq \emptyset$ and $V \cap C \neq \emptyset$. We have $(U \cap V) \cap C \neq \emptyset$, since otherwise we would have $C \subseteq (X \setminus U) \cup (X \setminus V)$, which is impossible since C is irreducible with $C \not\subseteq X \setminus U, X \setminus V$. Hence, there is some $x \in (U \cap V) \cap C$. By local compactness, there is some open W and some compact M such that

$$x \in W \subseteq M \subseteq U \cap V$$

Since $x \in C$, we have $W \cap C \neq \emptyset$, and we are done with $M \in \mathcal{K}_C$ since $M \subseteq U \cap V \subseteq K, L$.

We are going to apply well-filteredness to a codirected family of compact-saturated induced by $\mathcal{K}_C$. Note that for each compact K, the set $\uparrow K$ is also compact (since opens are saturated, given $U \in \mathcal{O}(X)$ we have $K \subseteq U$ if, and only if $\uparrow K \subseteq U$).

Let $\mathcal{Q} \subseteq \mathcal{KS}(X)$ be the set of all $\uparrow K$ for $K \in \mathcal{K}_C$. Note that $\mathcal{Q}$ is codirected since so is $\mathcal{K}_C$. Assume toward a contradiction that $\bigcap \mathcal{Q} \cap C = \emptyset$. Hence $\bigcap \mathcal{Q} \subseteq X \setminus C$, and by well-filteredness there is some $\uparrow K \in \mathcal{Q}$ such that $\uparrow K \subseteq X \setminus C$. But then $K \cap C = \emptyset$, which contradicts $K \in \mathcal{K}_C$.

Hence $\mathcal{Q} \cap C \neq \emptyset$. Let $x \in \mathcal{Q} \cap C$. We show that $C = \overline{\{x\}}$. We have to show that for all $y \in X$, we have $y \in C$ if, and only if, $y \leq_{\mathcal{O}(X)} x$. If $y \leq_{\mathcal{O}(X)} x$ then we must have $y \in C$ since C otherwise y belongs to the open $X \setminus C$ with $x \in C$. Assume conversely that $y \in C$,

and let U be an open such that $y \in U$. By local compactness there is an open V and a compact K such that

$$y \in V \subseteq K \subseteq U$$

Since $y \in V \cap C$, we have $K \in \mathcal{K}_C$. Hence $x \in \uparrow K$ since $x \in \bigcap \mathcal{Q}$, and $x \in U$ since $K \subseteq U$ implies $\uparrow K \subseteq U$. ◀

The Hofmann-Mislove Theorem. Conversely, sober spaces are well-filtered. This is a consequence of the famous Hofmann-Mislove (or Scott-open filter) theorem. The following is e.g. [13, Theorem 8.3.2] or [3, Theorem 7.2.9]. A *filter* in a poset is a codirected set which is also upward-closed.

▶ **Theorem 10.4** (Hofmann-Mislove). *Let $(X, \mathcal{O}(X))$ be a sober space, and let $\mathcal{U} \subseteq \mathcal{O}(X)$ be a Scott-open filter. If $\bigcap \mathcal{U} \subseteq V$ for some $V \in \mathcal{O}(X)$, then $V \in \mathcal{U}$.*

Proof. Let $\mathcal{U} \subseteq \mathcal{O}(X)$ be Scott-open (w.r.t. inclusion), and let $V_0 \in \mathcal{O}(X)$ such that $\bigcap \mathcal{U} \subseteq V_0$. Assume toward a contradiction that $V_0 \notin \mathcal{U}$.

We shall apply Zorn's lemma to the poset

$$\left(\left\{V \in \mathcal{O}(X) \mid \bigcap \mathcal{U} \subseteq V \text{ and } V \notin \mathcal{U}\right\}, \subseteq\right)$$

This poset is non-empty since V_0 belongs to it. Moreover, it is stable by unions of non-empty chains: if $\mathcal{V}$ is a non-empty chain of opens $V \supseteq \bigcap \mathcal{U}$, then $\bigcup \mathcal{V} \in \mathcal{U}$ would imply $V \in \mathcal{U}$ for some $V \in \mathcal{V}$ since $\mathcal{U}$ is Scott-open w.r.t. inclusion.

Hence by Zorn's lemma there is a maximal open V such that $\bigcap \mathcal{U} \subseteq V$ and $V \notin \mathcal{U}$. Let $C \coloneqq X \setminus V$.

We claim that C is irreducible. First, $C \neq \emptyset$ since otherwise we would have $V = X$, contradicting $X \in \mathcal{U}$ ($\mathcal{U}$ is a filter of opens). Assume now $C \subseteq F \cup G$ with F, G closed, and consider the opens $A \coloneqq X \setminus F$ and $B \coloneqq X \setminus G$. We have $A \cap B \subseteq V$. Assume that the opens $V \cup A$ and $V \cup B$ both belong to $\mathcal{U}$. Since $\mathcal{U}$ is a filter of opens, we thus get $(V \cup A) \cap (V \cup B) \in \mathcal{U}$. But $(V \cup A) \cap (V \cup B) = V \cup (A \cap B)$, while $V \cup (A \cap B) = V$ since $A \cap B \subseteq V$. Hence $V \in \mathcal{U}$, a contradiction. We thus have (say) $V \cup A \notin \mathcal{U}$, which by maximality of V implies $A \subseteq V$, that is, $C \subseteq F$.

Since X is sober, there is some $x \in X$ such that $C = \overline{\{x\}}$. We of course have $x \notin V$. Moreover, if $x \notin U$ with U open, then $U \cap C = \emptyset$ (since $y \in U \cap C$ would imply $y \leq_{\mathcal{O}(X)} x$ and $x \in U$), so that $U \subseteq V$ and $U \notin \mathcal{U}$ since $\mathcal{U}$ is a filter with $V \notin \mathcal{U}$. Hence $x \in U$ for all $U \in \mathcal{U}$. We thus have some $x \in \bigcap \mathcal{U}$ with $x \notin V$, contradicting $\bigcap \mathcal{U} \subseteq V$. ◀

The Hofmann-Mislove Theorem 10.4 yields that sober spaces are well-filtered. The following is e.g. [13, Proposition 8.3.5] or [3, Corollary 7.2.11].

▶ **Proposition 10.5.** *If X is sober, then X is well-filtered.*

Proof. Let $\mathcal{Q}$ be a codirected set of compact-saturated sets, and let $V \in \mathcal{O}(X)$ such that $\bigcap \mathcal{Q} \subseteq V$. Let $\mathcal{U}$ be the set of all opens U such that $Q \subseteq U$ for some $Q \in \mathcal{Q}$.

We claim that $\mathcal{U}$ is a Scott-open filter. First, if $U \in \mathcal{U}$ and $U \subseteq U'$ with U' open, then we of course have $U' \in \mathcal{U}$. Moreover, $\mathcal{U}$ is not empty since $X \in \mathcal{U}$ as $\mathcal{Q}$ is not empty. Let now $U, U' \in \mathcal{U}$. Hence there are $Q, Q' \in \mathcal{Q}$ such that $Q \subseteq U$ and $Q' \subseteq U'$. Since $\mathcal{Q}$ is codirected, there is $Q'' \in \mathcal{Q}$ such that $Q'' \subseteq Q$ and $Q'' \subseteq Q'$. Hence $Q'' \subseteq U \cap U'$ and $U \cap U' \in \mathcal{U}$. It follows that $\mathcal{U}$ is a filter.

It remains to show that $\mathcal{U}$ is Scott-open. Let $\bigcup \mathcal{D} \in \mathcal{U}$ with $\mathcal{D} \subseteq \mathcal{O}(X)$ directed. Hence there is some $Q \in \mathcal{Q}$ such that $Q \subseteq \bigcup \mathcal{D}$. But since Q is compact and $\mathcal{D}$ is a directed set of opens, there is some $U \in \mathcal{D}$ such that $Q \subseteq U$. Hence $U \in \mathcal{U}$.

For each $Q \in \mathcal{Q}$, since Q is saturated we have $Q = \bigcap\{U \in \mathcal{O}(X) \mid Q \subseteq U\}$. Hence $Q = \bigcap\{U \in \mathcal{U} \mid Q \subseteq U\}$ and $\bigcap \mathcal{Q} = \bigcap \mathcal{U}$.

We now apply the Hofmann-Mislove Theorem 10.4 to $\mathcal{U}$ and V, to the effect that $V \in \mathcal{U}$. Hence there is some $Q \in \mathcal{Q}$ such that $Q \subseteq V$, as required. ◀

Combining Proposition 10.5 with Lemma 10.2 yields Proposition 5.4:

▶ **Proposition 5.4** (Well-Filteredness). *Each Scott domain X is* well-filtered*: given a codirected set $\mathcal{Q} \subseteq \mathcal{KS}(X)$, if $\bigcap \mathcal{Q} \subseteq U$ with U Scott-open, then $Q \subseteq U$ for some $Q \in \mathcal{Q}$.*

Moreover, Propositions 10.5 and 10.3 imply that Definition 10.1 is equivalent to the definition of spectral space in §5.2. Hence, with Lemma 10.2 we have proved Theorem 5.3.

▶ **Theorem 5.3.** *The Scott topology of a Scott domain is spectral.*

Intersections of Compact-Saturated Sets. Proposition 10.5 also yields the following, from which one easily obtains Corollary 5.5.

▶ **Lemma 10.6.** *Let X be a spectral space. Given $S \subseteq X$, we have $S \in \mathcal{KS}(X)$ if, and only if, $S = \bigcap\{K \in \mathcal{KO}(X) \mid S \subseteq K\}$.*

Proof. Fix $S \subseteq X$ with X spectral.

First, note that the set $\{K \in \mathcal{KO}(X) \mid S \subseteq K\}$ is a codirected family of compact-saturated sets. Indeed, since X is compact we have $X \in \mathcal{KO}(X)$, so that $\{K \in \mathcal{KO}(X) \mid S \subseteq K\}$ is not empty. Moreover, given $K, L \in \mathcal{KO}(X)$ such that $S \subseteq K, L$, we have $S \subseteq K \cap L$ with $K \cap L \in \mathcal{KO}(X)$.

We first show that if $S \in \mathcal{KS}(X)$, then $S = \bigcap\{K \in \mathcal{KO}(X) \mid S \subseteq K\}$. We have $S \subseteq \bigcap\{K \in \mathcal{KO}(X) \mid S \subseteq K\}$ since $S \subseteq X$ with $X \in \mathcal{KO}(X)$. Moreover, we have seen above that if S is saturated, then $S = \bigcap\{U \in \mathcal{O}(X) \mid S \subseteq U\}$. But $\mathcal{KO}(X)$ is a basis of the topology since X is spectral. Hence, if $S \subseteq X$ is compact and included in an open U, then S is included in a compact-open $K \subseteq U$. It follows that

$$\bigcap\{K \in \mathcal{KO}(X) \mid S \subseteq K\} \quad \subseteq \quad \bigcap\{U \in \mathcal{O}(X) \mid S \subseteq U\}$$

and we are done.

Well-filteredness (Proposition 10.5) yields the converse, namely that $S \in \mathcal{KS}(X)$ whenever $S = \bigcap\{K \in \mathcal{KO}(X) \mid S \subseteq K\}$. Indeed, if $S = \bigcap\{K \in \mathcal{KO}(X) \mid S \subseteq K\}$, then S is saturated (as an intersection of saturated sets). The set S is moreover compact. Let $\mathcal{D} \subseteq \mathcal{O}(X)$ be a directed family of opens. Since $\{K \in \mathcal{KO}(X) \mid S \subseteq K\}$ is a codirected family of compact-saturated sets, we get from Proposition 10.5 that if $S = \bigcap\{K \in \mathcal{KO}(X) \mid S \subseteq K\} \subseteq \bigcup \mathcal{D}$, then $K \subseteq \bigcup \mathcal{D}$ for some $K \in \mathcal{KO}(X)$ such that $S \subseteq K$. But since K is compact, we have $K \subseteq U$ for some $U \in \mathcal{D}$. Hence $S \subseteq U$. ◀

▶ **Lemma 10.7.** *If X is a spectral space, then $\mathcal{KS}(X)$ is stable under finite intersections.*

Proof. We of course have $X \in \mathcal{KS}(X)$.

Let $Q_1, Q_2 \in \mathcal{KS}(X)$. We have $Q_i = \bigcap\{K_i \in \mathcal{KO}(X) \mid Q_i \subseteq K_i\}$ by Lemma 10.6. Hence

$$Q_1 \cap Q_2 \quad = \quad \bigcap\{K_1 \cap K_2 \mid Q_1 \subseteq K_1 \text{ and } Q_2 \subseteq K_2 \text{ with } K_1, K_2 \in \mathcal{KO}(X)\}$$

Since $\mathcal{KO}(X)$ is stable under binary intersections (Definition 10.1(ii)), it follows that $Q_1 \cap Q_2 = \bigcap\{K \in \mathcal{KO}(X) \mid Q_1 \cap Q_2 \subseteq K\}$, and we get $Q_1 \cap Q_2 \in \mathcal{KS}(X)$ by Lemma 10.6. ◀

Corollary 5.5 is given by Theorem 5.3 and the following.

▶ **Corollary 10.8.** *If X is a spectral space, then $\mathcal{KS}(X)$ is stable under all intersections.*

An other important consequence of well-filteredness is Lemma 5.7 (§5.3).

11 Proofs of §5.3 (Semantics of Formulae)

11.1 Properties of Semantic Modalities

We begin with Lemma 5.7.

▶ **Lemma 5.7.** *Given $V \in \mathcal{O}([\![\sigma]\!])$ and a codirected $\mathcal{Q} \subseteq \mathcal{KS}([\![\tau]\!])$, if $f \in \bigcap \mathcal{Q} \Vdash V$, then there is some $Q \in \mathcal{Q}$ such that $f \in Q \Vdash V$ already. Moreover, given $Q \in \mathcal{KS}([\![\tau]\!])$ we have $Q \Vdash V \in \mathcal{O}([\![\tau \to \sigma]\!])$ and $V \Vdash Q \in \mathcal{KS}([\![\sigma \to \tau]\!])$.*

Proof. For the first part of the statement, since $f : [\![\tau]\!] \to [\![\sigma]\!]$ is Scott-continuous, we have that $f^{-1}(V)$ is open in $[\![\tau]\!]$. Since $\bigcap \mathcal{Q} \subseteq f^{-1}(V)$ with $\mathcal{Q}$ a codirected family of compact-saturated sets, it follows from Proposition 5.4 (well-filteredness) that $Q \subseteq f^{-1}(V)$ for some $Q \in \mathcal{Q}$. Hence $f \in Q \Vdash V$.

We now show that $Q \Vdash V$ is open in $[\![\tau]\!] \to [\![\sigma]\!]$ when V is open in $[\![\sigma]\!]$ and Q is compact-saturated in $[\![\tau]\!]$. We have seen Lemma 10.6 (Appendix 10) that Q is the codirected intersection of the set of all compact-opens K such that $Q \subseteq K$. It thus follows from the first part of the statement that

$$Q \Vdash V \quad \subseteq \quad \bigcup \{K \Vdash V \mid Q \subseteq K \text{ and } K \text{ compact open}\}$$

The converse inclusion follows from the fact that if $Q \subseteq K$, then $K \Vdash V \subseteq Q \Vdash V$.

Consider now some $K \in \mathcal{KO}([\![\tau]\!])$. There are finite $d_1, \ldots, d_n \in [\![\tau]\!]$ such that $K = \uparrow d_1 \cup \cdots \cup \uparrow d_n$. Hence $K \Vdash V = \bigcap_{1 \leq i \leq n} (\uparrow d_i \Vdash V)$. Moreover, given $f \in \uparrow d_i \Vdash V$, we have $f(d_i) \in V$. Hence there is some finite $e \in V$ such that $e \leq f(d_i)$. We thus have $(d_i \searrow e) \leq f$, where the step function $(d_i \searrow e)$ is finite in $[\![\tau]\!] \to [\![\sigma]\!]$ (Proposition 9.11, §9.2.4). It follows that

$$K \Vdash V \quad = \quad \textstyle\bigcap_{1 \leq i \leq n} \bigcup \{\uparrow (d_i \searrow e) \mid e \in V \text{ finite}\}$$

Hence $K \Vdash V$ is open as a finite intersection of unions of basic opens $\uparrow(d_i \searrow e)$, and $Q \Vdash V$ is open as a union of opens.

We now show that $V \Vdash Q$ is compact-saturated in $[\![\sigma]\!] \to [\![\tau]\!]$. Since

$$\begin{array}{rcl} V & = & \bigcup \{\uparrow e \mid e \in V \text{ finite}\} \\ Q & = & \bigcap \{K \in \mathcal{KO}([\![\tau]\!]) \mid Q \subseteq K\} \end{array}$$

we have

$$V \Vdash Q \quad = \quad \bigcap \{\uparrow e \Vdash K \mid e \in V \text{ finite and } K \in \mathcal{KO}([\![\tau]\!]) \text{ with } Q \subseteq K\}$$

Now, writing again $K \in \mathcal{KO}([\![\tau]\!])$ as $K = \uparrow d_1 \cup \cdots \cup \uparrow d_n$ with $d_1, \ldots, d_n \in [\![\tau]\!]$ finite, we have that $\uparrow e \Vdash K = \bigcup_{1 \leq i \leq n} \uparrow (e \searrow d_i)$ is compact-open. Hence $V \Vdash Q$ is compact-saturated as an intersection of compact-opens (Corollary 5.5). ◀

We now turn to simple but fundamental facts on the semantic modalities $[\![[\triangle]]\!]$.

▶ **Lemma 11.1.** *If $\triangle$ is either π_1, π_2 or* fold*, then $[\![[\triangle]]\!]$ (is monotone and) preserves all unions and all intersections.*

If $i \in \{1,2\}$, then $[\![[\mathsf{in}_i]]\!]$ (is monotone and) preserves all unions all non-empty *intersections.*

Proof. In the case of $\triangle \in \{\pi_1, \pi_2, \mathsf{fold}\}$, the semantic modality $[\![[\triangle]]\!]$ acts by inverse image, and thus preserves all unions and all intersections.

The semantic modalities $[\![[\mathsf{in}_i]]\!]$ act by *direct* image, and thus preserves all unions (direct images are lower adjoints to inverse images). It also preserves *non-empty* intersections since the morphism $[\![\mathsf{in}_i]\!]$ is injective (but $[\![[\mathsf{in}_i]]\!]$ does not preserve empty intersections since $[\![\mathsf{in}_i]\!]$ is not surjective). ◀

▶ **Lemma 11.2.**

(1) *Given $x \in [\![\tau[\mathsf{rec}\,\alpha.\tau/\alpha]]\!]$, we have $[\![[\mathsf{fold}]]\!](\uparrow x) = \uparrow[\![\mathsf{fold}]\!](x)$.*

(2) *Given $x \in [\![\tau]\!]$ and $y \in [\![\sigma]\!]$, we have $[\![[\pi_1]]\!](\uparrow x) = \uparrow(x, \bot)$ and $[\![[\pi_2]]\!](\uparrow y) = \uparrow(\bot, y)$.*

(3) *Given $x \in [\![\tau_i]\!]$, we have $[\![[\mathsf{in}_i]]\!](\uparrow x) = \uparrow[\![\mathsf{in}_i]\!](x)$.*

Proof.

(1) Since $[\![\mathsf{fold}]\!]$ is an iso with inverse $[\![\mathsf{unfold}]\!]$, we have

$$\begin{array}{rcl} [\![[\mathsf{fold}]]\!](\uparrow x) & = & \{y \in [\![\mathsf{rec}\,\alpha.\tau]\!] \mid [\![\mathsf{unfold}]\!](y) \geq x\} \\ & = & \{y \in [\![\mathsf{rec}\,\alpha.\tau]\!] \mid y \geq [\![\mathsf{fold}]\!](x)\} \end{array}$$

(2) We only discuss the case of $x \in [\![\tau]\!]$. Since the order in $[\![\tau]\!] \times [\![\sigma]\!]$ is pointwise, we have

$$\begin{array}{rcl} [\![[\pi_1]]\!](\uparrow x) & = & \{p \in [\![\tau]\!] \times [\![\sigma]\!] \mid [\![\pi_1]\!](p) \geq x\} \\ & = & \{(a,b) \in [\![\tau]\!] \times [\![\sigma]\!] \mid a \geq x\} \\ & = & \uparrow(x, \bot) \end{array}$$

(3) Since $[\![\mathsf{in}_i]\!]$ is an order embedding (§9.2.3 and §9.2.2), we have

$$\begin{array}{rcl} [\![[\mathsf{in}_i]]\!](\uparrow x) & = & \{[\![\mathsf{in}_i]\!](y) \mid y \geq x\} \\ & = & \{z \mid z \geq [\![\mathsf{in}_i]\!](x)\} \\ & = & \uparrow[\![\mathsf{in}_i]\!](x) \end{array}$$

◀

▶ **Lemma 11.3.** *Given $\triangle \in \{\pi_1, \pi_2, \mathsf{in}_1, \mathsf{in}_2, \mathsf{fold}\}$, the semantic modality $[\![[\triangle]]\!]$ preserves compact-opens, opens and compact-saturated sets.*

Proof. Let $\triangle \in \{\pi_1, \pi_2, \mathsf{in}_1, \mathsf{in}_2, \mathsf{fold}\}$.

Recall that (compact) opens are (finite) unions of sets of the form $\uparrow d$ with d finite. By combining Lemma 11.2 with Proposition 9.11, we get that $[\![[\triangle]]\!](\uparrow d)$ is of the form $\uparrow e$ for some finite e. Hence, Lemma 11.1 gives the result for (compact) opens.

Regarding compact saturated sets, recall from §5.2 that given a spectral space X, each compact saturated set $Q \in \mathcal{KS}(X)$ is the intersection of all compact opens $K \in \mathcal{KO}(X)$ such that $Q \subseteq K$. But since spectral spaces are compact, we have $X \in \mathcal{KO}(X)$, and Q is always a *non-empty* intersection of compact opens. Hence the result by Lemma 11.1 (again) and Corollary 5.5. ◀

11.2 Semantic Properties of Formulae

We begin with Proposition 5.8.

▶ **Proposition 5.8.** *If $\varphi \in \mathcal{L}^+(\tau)$ then $[\![\varphi]\!]\rho \in \mathcal{O}([\![\tau]\!])$. If $\varphi \in \mathcal{L}^-(\tau)$ then $[\![\varphi]\!]\rho \in \mathcal{KS}([\![\tau]\!])$. In particular, if $\varphi \in \mathcal{L}^\pm(\tau)$ then $[\![\varphi]\!]\rho \in \mathcal{KO}([\![\tau]\!])$.*

Proof. We prove a more general statement. Let ρ be a valuation of $\Sigma = p_1 : \sigma_1, \dots, p_n : \sigma_n$.
(1) If $\varphi \in \mathcal{L}^+(\Sigma, \tau)$, then the function $[\![p_1.\dots.p_n.\varphi]\!]\rho$ restricts to

$$\begin{array}{rccc} [\![p_1.\cdots.p_n.\varphi]\!]\rho\colon & \mathcal{O}([\![\sigma_1]\!]) \times \cdots \times \mathcal{O}([\![\sigma_n]\!]) & \longrightarrow & \mathcal{O}([\![\tau]\!]) \\ & (S_1, \dots, S_n) & \longmapsto & [\![\varphi]\!]\rho[S_1/p_1, \dots, S_n/p_n] \end{array}$$

(2) If $\varphi \in \mathcal{L}^-(\Sigma, \tau)$, then the function $[\![p_1.\dots.p_n.\varphi]\!]\rho$ restricts to

$$\begin{array}{rccc} [\![p_1.\cdots.p_n.\varphi]\!]\rho\colon & \mathcal{KS}([\![\sigma_1]\!]) \times \cdots \times \mathcal{KS}([\![\sigma_n]\!]) & \longrightarrow & \mathcal{KS}([\![\tau]\!]) \\ & (S_1, \dots, S_n) & \longmapsto & [\![\varphi]\!]\rho[S_1/p_1, \dots, S_n/p_n] \end{array}$$

We reason by induction on φ.

- The case of $p_i \in \mathcal{L}^\pm(\Sigma; \tau)$ is trivial.
- The case of $[\langle\rangle] \in \mathcal{L}^\pm(\Sigma; \mathbf{1})$ follows from the description of finite elements in Figure 12 (Proposition 9.11, §9.2.4).
- The cases of True, $\varphi \wedge \psi$, False and $\varphi \vee \psi$ all follow from the fact that $\mathcal{O}(\tau)$ and $\mathcal{KS}(\tau)$ are closed under finite unions and finite intersections (using Corollary 5.5 for finite intersections of compact-saturated sets).
- The cases of $[\triangle]\varphi$ follow from Lemma 11.3, and that of the realizability implication $\|\!\mapsto$ follows from Lemma 5.7 (note that $\psi \|\!\mapsto \varphi$ has no free fixpoint variables).
- Finally, the case of $(\exists k)\varphi \in \mathcal{L}^+(\Sigma; \tau)$ is trivial (since $\mathcal{O}([\![\tau]\!])$ is stable under unions), while that of $(\forall k)\varphi \in \mathcal{L}^-(\Sigma; \tau)$ follows from Corollary 5.5 (closure of $\mathcal{KS}([\![\tau]\!])$ under all intersections). ◀

11.2.1 Proofs of Lemma 5.9 and Proposition 5.10

We now discuss Lemma 5.9 and Proposition 5.10. Lemma 5.9 uses the following, which is actually item (1) of Proposition 5.10.

▶ **Lemma 11.4.** *Let ρ be a valuation of $\Sigma = p_1 : \sigma_1, \dots, p_n : \sigma_n$. If $\varphi \in \mathcal{L}(\Sigma; \tau)$, then the function*

$$\begin{array}{rccc} [\![p_1.\cdots.p_n.\varphi]\!]\rho\colon & \mathcal{P}([\![\sigma_1]\!]) \times \cdots \times \mathcal{P}([\![\sigma_n]\!]) & \longrightarrow & \mathcal{P}([\![\tau]\!]) \\ & (S_1, \dots, S_n) & \longmapsto & [\![\varphi]\!]\rho[S_1/p_1, \dots, S_n/p_n] \end{array}$$

is monotone.

Proof. We reason by induction on $\varphi \in \mathcal{L}(\Sigma; \tau)$.

- The case of $p_i \in \mathcal{L}^\pm(\Sigma; \tau)$ is trivial.
- The case of $[\langle\rangle] \in \mathcal{L}(\Sigma; \mathbf{1})$ is trivial.
- The case of $\psi \|\!\mapsto \varphi$ is trivial since ψ and φ have no free fixpoint variable.
- The cases of $[\triangle]$ follow from Lemma 11.1.
- The cases of True, $\varphi \wedge \psi$, False, $\varphi \vee \psi$, $(\exists k)\varphi$ and $(\forall k)\varphi$ all follow from the fact that unions and intersections are monotone operations.

- In the cases of $(\mu^t q)\varphi$ and $(\nu^t q)\psi$, let $m \coloneqq [\![t]\!]\rho$ and note that

$$[\![(\mu^t q)\varphi]\!]\rho = ([\![q.\varphi]\!]\rho)^m(\emptyset) \qquad \text{and} \qquad [\![(\nu^t q)\psi]\!]\rho = ([\![q.\psi]\!]\rho)^m([\![\tau]\!])$$

The induction hypothesis yields that the functions $[\![p_1.\cdots.p_n.q.\varphi]\!]\rho$ and $[\![p_1.\cdots.p_n.q.\psi]\!]\rho$ are monotone. An induction on $i \leq m$ then shows that the functions

$$\begin{array}{rcl} \mathcal{P}([\![\sigma_1]\!]) \times \cdots \times \mathcal{P}([\![\sigma_n]\!]) & \longrightarrow & \mathcal{P}([\![\tau]\!]) \\ (S_1, \ldots, S_n) & \longmapsto & ([\![q.\varphi]\!]\rho[S_1/p_1, \ldots, S_n/p_n])^i(\emptyset) \\ (S_1, \ldots, S_n) & \longmapsto & ([\![q.\psi]\!]\rho[S_1/p_1, \ldots, S_n/p_n])^i([\![\tau]\!]) \end{array}$$

are both monotone. ◀

We can now prove Lemma 5.9.

▶ **Lemma 5.9.** *If k Pos φ (resp. k Neg φ) then the function $n \mapsto [\![\varphi]\!]\rho[n/k]$ is monotone (resp. antimonotone), that is, $n \leq m$ (resp. $m \leq n$) implies $[\![\varphi]\!]\rho[n/k] \subseteq [\![\varphi]\!]\rho[m/k]$.*

Proof. The proof is by mutual induction on the predicates k Pos φ and k Neg φ.
Cases of

$$\frac{k \notin \mathrm{FV}(\varphi)}{k \text{ Pos } \varphi} \qquad \frac{k \notin \mathrm{FV}(\varphi)}{k \text{ Neg } \varphi}$$

Trivial.
Cases of

$$\frac{k \text{ Pos } \varphi}{k \text{ Pos } [\triangle]\varphi} \qquad \frac{k \text{ Neg } \varphi}{k \text{ Neg } [\triangle]\varphi}$$

By induction hypothesis and Lemma 11.1.
Cases of

$$\frac{k \text{ Neg } \psi \qquad k \text{ Pos } \varphi}{k \text{ Pos } \psi \Vdash \varphi} \qquad \frac{k \text{ Pos } \psi \qquad k \text{ Neg } \varphi}{k \text{ Neg } \psi \Vdash \varphi}$$

By induction hypothesis, since $\Vdash$ is contravariant in its first argument and covariant in its second argument.
Cases of

$$\frac{k \text{ Pos } \varphi, \psi}{k \text{ Pos } \varphi \vee \psi} \qquad \frac{k \text{ Pos } \varphi, \psi}{k \text{ Pos } \varphi \wedge \psi} \qquad \frac{k \text{ Neg } \varphi, \psi}{k \text{ Neg } \varphi \vee \psi} \qquad \frac{k \text{ Neg } \varphi, \psi}{k \text{ Neg } \varphi \wedge \psi}$$

$$\frac{k \text{ Pos } \varphi}{k \text{ Pos } (\exists \ell)\varphi} \qquad \frac{k \text{ Neg } \varphi}{k \text{ Neg } (\forall \ell)\varphi}$$

By induction hypothesis, since unions and intersections are monotone operations.
Cases of

$$\frac{k \text{ Pos } \varphi \qquad k \notin \mathrm{FV}(t)}{k \text{ Pos } (\nu^t p)\varphi} \qquad \frac{k \text{ Neg } \psi \qquad k \notin \mathrm{FV}(t)}{k \text{ Neg } (\mu^t p)\psi}$$

Let $n \leq m$ in $\mathbb{N}$. Note that $[\![t]\!]\rho[n/k] = [\![t]\!]\rho[m/k] = [\![t]\!]\rho$ since $k \notin \mathrm{FV}(t)$. Moreover, $[\![\varphi]\!]\rho[n/k] \subseteq [\![\varphi]\!]\rho[m/k]$ and $[\![\psi]\!]\rho[m/k] \subseteq [\![\psi]\!]\rho[n/k]$ by induction hypothesis.
Note that we have

$$[\![(\mu^t p)\varphi]\!]\rho = ([\![p.\varphi]\!]\rho)^{[\![t]\!]\rho}(\emptyset) \qquad \text{and} \qquad [\![(\nu^t p)\psi]\!]\rho = ([\![p.\psi]\!]\rho)^{[\![t]\!]\rho}([\![\tau]\!])$$

By monotonicity of $[\![p.\varphi]\!]\rho$ and $[\![p.\psi]\!]\rho$ (Lemma 11.4), an induction on $i \in \mathbb{N}$ yields

$$\begin{array}{rrcl} & ([\![p.\varphi]\!]\rho[n/k])^i(\emptyset) & \subseteq & ([\![p.\varphi]\!]\rho[m/k])^i(\emptyset) \\ \text{and} & ([\![p.\psi]\!]\rho[m/k])^i([\![\tau]\!]) & \subseteq & ([\![p.\psi]\!]\rho[n/k])^i([\![\tau]\!]) \end{array}$$

Hence we are done with

$$\begin{array}{rrcl} & ([\![p.\varphi]\!]\rho[n/k])^{[\![t]\!]}(\emptyset) & \subseteq & ([\![p.\varphi]\!]\rho[m/k])^{[\![t]\!]}(\emptyset) \\ \text{and} & ([\![p.\psi]\!]\rho[m/k])^{[\![t]\!]}([\![\tau]\!]) & \subseteq & ([\![p.\psi]\!]\rho[n/k])^{[\![t]\!]}([\![\tau]\!]) \end{array}$$

Cases of

$$\frac{k\ \mathsf{Pos}\ \varphi}{k\ \mathsf{Pos}\ (\mu^t p)\varphi} \qquad \frac{k\ \mathsf{Neg}\ \psi}{k\ \mathsf{Neg}\ (\nu^t p)\psi}$$

Let $n \leq m$ in $\mathbb{N}$. We have $[\![t]\!]\rho[n/k] \leq [\![t]\!]\rho[m/k]$. Moreover, $[\![\varphi]\!]\rho[n/k] \subseteq [\![\varphi]\!]\rho[m/k]$ and $[\![\psi]\!]\rho[m/k] \subseteq [\![\psi]\!]\rho[n/k]$ by induction hypothesis.
Note that we have

$$[\![(\mu^t p)\varphi]\!]\rho \quad = \quad ([\![p.\varphi]\!]\rho)^{[\![t]\!]\rho}(\emptyset) \qquad \text{and} \qquad [\![(\nu^t p)\psi]\!]\rho \quad = \quad ([\![p.\psi]\!]\rho)^{[\![t]\!]\rho}([\![\tau]\!])$$

By monotonicity of $[\![p.\varphi]\!]\rho$ and $[\![p.\psi]\!]\rho$ (Lemma 11.4), an induction on $i \in \mathbb{N}$ yields

$$\begin{array}{rrcl} & ([\![p.\varphi]\!]\rho[n/k])^i(\emptyset) & \subseteq & ([\![p.\varphi]\!]\rho[m/k])^i(\emptyset) \\ \text{and} & ([\![p.\psi]\!]\rho[m/k])^i([\![\tau]\!]) & \subseteq & ([\![p.\psi]\!]\rho[n/k])^i([\![\tau]\!]) \end{array}$$

so that

$$\begin{array}{rrcl} & ([\![p.\varphi]\!]\rho[n/k])^{[\![t]\!]\rho[n/k]}(\emptyset) & \subseteq & ([\![p.\varphi]\!]\rho[m/k])^{[\![t]\!]\rho[n/k]}(\emptyset) \\ \text{and} & ([\![p.\psi]\!]\rho[m/k])^{[\![t]\!]\rho[n/k]}([\![\tau]\!]) & \subseteq & ([\![p.\psi]\!]\rho[n/k])^{[\![t]\!]\rho[n/k]}([\![\tau]\!]) \end{array}$$

Again using the monotonicity of $[\![p.\varphi]\!]\rho$ and $[\![p.\psi]\!]\rho$ (Lemma 11.4), we get

$$\begin{array}{rrcl} & ([\![p.\varphi]\!]\rho[m/k])^{[\![t]\!]\rho[n/k]}(\emptyset) & \subseteq & ([\![p.\varphi]\!]\rho[m/k])^{[\![t]\!]\rho[m/k]}(\emptyset) \\ \text{and} & ([\![p.\psi]\!]\rho[m/k])^{[\![t]\!]\rho[m/k]}([\![\tau]\!]) & \subseteq & ([\![p.\psi]\!]\rho[m/k])^{[\![t]\!]\rho[n/k]}([\![\tau]\!]) \end{array}$$

and we are done. ◀

We finally turn to Proposition 5.10. This requires some preparatory material on Scott (co)continuity. Given complete lattices A and B, a function $f\colon A \to B$ is *Scott-cocontinuous* if it preserves infs of codirected sets. We use the well-known fact that Scott-(co)continuity in several arguments is equivalent to Scott-(co)continuity in each argument separately.

▶ **Lemma 11.5.** *Let A, B, C be complete lattices, and let $f\colon A \times B \to C$ be a function.*
Then f is Scott-(co)continuous if, and only if, for all $a \in A$ and all $b \in B$, the functions $y \mapsto f(a,y)$ and $x \mapsto f(x,b)$ are Scott-(co)continuous.

Proof. For Scott-continuity, this is e.g. [4, Proposition 1.4.3]. Regarding Scott-cocontinuity, since A, B and C are assumed to be *complete* lattices, one sees that f is Scott-cocontinuous precisely when f is Scott-continuous as a function $A^{\mathrm{op}} \times B^{\mathrm{op}} \to C^{\mathrm{op}}$. Hence the result. ◀

Given a function $f\colon A \times B \to C$ the function $(x,y) \mapsto f(x,-)^n(y)$ is defined as

$$f(x,-)^0(y) \quad \coloneqq \quad y \qquad \text{and} \qquad f(x,-)^{n+1}(y) \quad \coloneqq \quad f(x,\, f(x,-)^n(y))$$

▶ **Lemma 11.6.** *Let A and B be complete lattices, and let $f\colon A \times B \to B$ be a function.*
If f is Scott-(co)continuous, then so is the function $(x,y) \mapsto f(x,-)^n(y)$ for each $n \in \mathbb{N}$.

Proof. The proof is by induction on $n \in \mathbb{N}$. In the base case $n = 0$, the function $(x, y) \mapsto f(x, -)^n(y)$ is the second projection. This function is Scott-continuous and cocontinuous by Lemma 11.5.

Assume now that $(x, y) \mapsto f(x, -)^n(y)$ is Scott-(co)continuous. The function $(x, y) \mapsto f(x, -)^{n+1}(y)$ is the composite

$$A \times B \xrightarrow{\langle \pi_1, \mathrm{id} \rangle} A \times (A \times B) \xrightarrow{\mathrm{id} \times (x,y) \mapsto f(x,-)^n(y)} A \times B \xrightarrow{f} B$$

which is Scott-continuous thanks to Lemma 11.5. ◀

▶ **Lemma 11.7.** *Let A and B be complete lattices, let $(f_i)_{i \in I}$ be a family of functions $f_i \colon A \to B$.*

(1) *If each f_i is Scott-continuous, then $x \mapsto \bigvee_i f_i(x)$ is Scott-continuous.*

(2) *Assume that sups distribute over finite infs in B. If each f_i is Scott-continuous and I is finite, then $x \mapsto \bigwedge_i f_i(x)$ is Scott-continuous.*

(3) *If each f_i is Scott-cocontinuous, then $x \mapsto \bigwedge_i f_i(x)$ is Scott-cocontinuous.*

(4) *Assume that infs distribute over finite sups in B. If each f_i is Scott-cocontinuous and I is finite, then $x \mapsto \bigvee_i f_i(x)$ is Scott-cocontinuous.*

Proof. By duality (see the proof of Lemma 11.5), it is sufficient to consider the case of Scott-continuity.

(1) Given a directed $D \subseteq A$, since sups commute over sups, we have

$$\begin{array}{rcl} \bigvee_{i \in I} f_i(\bigvee D) & = & \bigvee_{i \in I} \bigvee_{d \in D} f_i(d) \\ & = & \bigvee_{d \in D} \bigvee_{i \in I} f_i(d) \end{array}$$

and we are done.

(2) Let $D \subseteq A$ be directed. Since the f_i's are Scott continuous, we have

$$\bigwedge_{i \in I} f_i(\bigvee D) \quad = \quad \bigwedge_{i \in I} \bigvee_{d \in D} f_i(d)$$

Now, we have $\bigwedge_{i \in I} f_i(d) \leq \bigwedge_{i \in I} \bigvee_{d \in D} f_i(d)$ for each $d \in D$, and thus

$$\bigvee_{d \in D} \bigwedge_{i \in I} f_i(d) \quad \leq \quad \bigwedge_{i \in I} \bigvee_{d \in D} f_i(d)$$

For the converse inequality, since I is finite and sups distribute over finite infs in B, we have

$$\bigwedge_{i \in I} \bigvee_{d \in D} f_i(d) \quad \leq \quad \bigvee_{g \in D^I} \bigwedge_{i \in I} f_i(g(i))$$

Let $g \in D^I$. Since I is finite and D is directed, there is some $d \in D$ such that $g(i) \leq d$ for all $i \in I$. By monotonicity of the f_i's we have $f_i(g(i)) \leq f_i(d)$ for all $i \in I$, and thus $\bigwedge_{i \in I} f_i(g(i)) \leq \bigwedge_{i \in I} f_i(d)$. It follows that

$$\bigwedge_{i \in I} f_i(g(i)) \quad \leq \quad \bigvee_{d \in D} \bigwedge_{i \in I} f_i(d)$$

Since this holds for each $g \in D^I$, we obtain

$$\bigvee_{g \in D^I} \bigwedge_{i \in I} f_i(g(i)) \quad \leq \quad \bigvee_{d \in D} \bigwedge_{i \in I} f_i(d)$$

◀

We can finally prove Proposition 5.10.

▶ **Proposition 5.10.** *Let ρ be a valuation of Σ, and let $(p\colon \sigma) \in \Sigma$.*
(1) *If $\varphi \in \mathcal{L}^s(\Sigma;\tau)$, then $[\![p.\varphi]\!]\rho$ is monotone.*
(2) *If $\varphi \in \mathcal{L}^+(\Sigma;\tau)$, then $[\![p.\varphi]\!]\rho$ is Scott-continuous (i.e. preserves directed unions).*
(3) *If $\varphi \in \mathcal{L}^-(\Sigma;\tau)$, then $[\![p.\varphi]\!]\rho$ is Scott-cocontinuous (i.e. preserves codirected intersections).*

Proof. Item (1) was proven in Lemma 11.4. For items (2) and (3), we prove a more general statement. Let ρ be a valuation of $\Sigma = p_1\colon \sigma_1, \dots, p_n\colon \sigma_n$, and consider the function

$$\begin{array}{rccc} [\![p_1.\cdots.p_n.\varphi]\!]\rho\colon & \mathcal{P}([\![\sigma_1]\!]) \times \cdots \times \mathcal{P}([\![\sigma_n]\!]) & \longrightarrow & \mathcal{P}([\![\tau]\!]) \\ & (S_1, \dots, S_n) & \longmapsto & [\![\varphi]\!]\rho[S_1/p_1, \dots, S_n/p_n] \end{array}$$

We prove the following:
- if $\varphi \in \mathcal{L}^+(\Sigma;\tau)$, then $[\![p_1.\cdots.p_n.\varphi]\!]\rho$ is Scott-continuous;
- if $\varphi \in \mathcal{L}^-(\Sigma;\tau)$, then $[\![p_1.\cdots.p_n.\varphi]\!]\rho$ is Scott-cocontinuous.

We reason by induction on φ.
- The case of $p_i \in \mathcal{L}^{\pm}(\Sigma;\tau)$ is trivial.
- The cases of $[\langle\rangle]$ and $\psi \Vdash\!\!\to \varphi$ are trivial (recall that $\psi \Vdash\!\!\to \varphi$ has no free fixpoint variable).
- The cases of $[\triangle]$ follow from Lemma 11.1 (note that codirected sets are required to be non-empty).
- The cases of True, $\varphi \wedge \psi$, False and $\varphi \vee \psi$, all follow from Lemma 11.7.
- The cases of $(\exists k)\varphi$ and $(\forall k)\psi$, where φ is positive and ψ is negative, follow from items (1) and (3) in Lemma 11.7, respectively.
- In the cases of $(\mu^t q)\varphi$ and $(\nu^t q)\psi$, let $m \coloneqq [\![t]\!]\rho$ and note that

$$[\![(\mu^t q)\varphi]\!]\rho \;=\; ([\![q.\varphi]\!]\rho)^m(\emptyset) \qquad \text{and} \qquad [\![(\nu^t q)\psi]\!]\rho \;=\; ([\![q.\psi]\!]\rho)^m([\![\tau]\!])$$

We can then conclude using Lemma 11.6. ◀

11.2.2 Notes on Example 5.11

We provide some details on Example 5.11.

Streams vs ω-Words. The set of ω-words on a set A is A^ω. The set A^ω can be seen as a subset of the Scott domain $(A_\perp)^\omega$. The induced subspace topology is the usual *product topology* on A^ω (see e.g. [33, Example 8]). This topology is Hausdorff and has a basis of clopens. It is moreover compact if, and only if, the set A is finite (while the Scott topology on $(A_\perp)^\omega$ is always compact).

Boolean Spaces. We claimed that Stone (Boolean) spaces are exactly those spectral spaces for which the compact-saturated sets coincide with the closed sets. Boolean spaces can be defined as those compact Hausdorff spaces which have a basis of clopen sets [11, Definition 1.3.1]. Boolean spaces are of course spectral (see e.g. [11, Theorem 1.3.4]).

In presence of a basis of compact-opens, if compact-saturated sets coincide with closed sets, we get that each element of the basis is in fact clopen. Together with the T_0 separation axiom, this yields that the space is Hausdorff with a basis of clopens. Hence, a space is Boolean provided it is a spectral space in which the compact-saturated sets coincide with the closed sets.

Conversely, in a Boolean space each closed set is an intersection of clopens (since each open is a union of clopens). But in a compact Hausdorff space, clopens coincide with compact-opens. Hence in a Boolean space closed sets coincide with intersections of compact-opens, and thus with compact-saturated sets (Lemma 10.6 and Corollary 10.8, §10).

Well-Filteredness and the Formula in Equation (4)**.** Finally, we made an informal claim corresponding to the following.

▶ **Lemma 11.8.** *Consider a Scott-continuous* $f : [\![\mathsf{Tree\,Bool}]\!] \to [\![\mathsf{Stream\,Nat}]\!]$. *Given* $n \in \mathbb{N}$, *the following are equivalent:*

(i) *f satisfies the formula in Equation* (4) *of Example 3.3.*

(ii) *There is some $m \in \mathbb{N}$ such that f satisfies the formula*

$$\forall\Box[\mathsf{label}]([\mathsf{tt}] \vee [\mathsf{ff}]) \;\wedge\; \exists\bigcirc \bigwedge_{i\leq m} (\exists\bigcirc)^i[\mathsf{label}][\mathsf{tt}] \;\wedge\; \exists\bigcirc \bigwedge_{i\leq m} (\exists\bigcirc)^i[\mathsf{label}][\mathsf{ff}] \quad \Vdash\!\!\mapsto \bigvee_{i\leq m} \bigcirc^i[\mathsf{hd}][\geq \mathsf{n}][\mathsf{tot}]$$

Proof. We show that (i) ⇒ (ii). First, note that for a negative ψ we have

$$\exists\bigcirc\exists\Box\psi \quad \dashv\vdash \quad (\forall\ell_1)(\forall\ell_2)\left([\mathsf{left}](\nu^{\ell_1}p)(\psi \wedge \exists\bigcirc p) \;\vee\; [\mathsf{right}](\nu^{\ell_2}p)(\psi \wedge \exists\bigcirc p)\right)$$

We shall now apply Lemma 5.7. It is actually more convenient to apply it in the form of Lemma 11.14 (used in the proof of the Soundness Theorem 5.13). Hence the formula in Equation (4) implies the following formula

$$\begin{aligned}(\exists\ell_1,\ell_2,\ell_3,\ell_4,k)\Big(\Big(&\forall\Box[\mathsf{label}]([\mathsf{tt}] \vee [\mathsf{ff}])\\ \wedge\; &\left([\mathsf{left}](\nu^{\ell_1}p)([\mathsf{label}][\mathsf{tt}] \wedge \exists\bigcirc p) \;\vee\; [\mathsf{right}](\nu^{\ell_2}p)([\mathsf{label}][\mathsf{tt}] \wedge \exists\bigcirc p)\right)\\ \wedge\; &\left([\mathsf{left}](\nu^{\ell_3}p)([\mathsf{label}][\mathsf{ff}] \wedge \exists\bigcirc p) \;\vee\; [\mathsf{right}](\nu^{\ell_4}p)([\mathsf{label}][\mathsf{ff}] \wedge \exists\bigcirc p)\right)\Big)\\ &\Vdash\!\!\mapsto \quad (\mu^k p)([\mathsf{hd}][\geq \mathsf{n}][\mathsf{tot}] \vee \bigcirc p)\Big)\end{aligned}$$

But now, by monotonicity (Lemma 5.9) we get that if f satisfies the last formula above, then it also satisfies

$$\begin{aligned}(\exists k)\Big(\Big(&\forall\Box[\mathsf{label}]([\mathsf{tt}] \vee [\mathsf{ff}])\\ \wedge\; &\left([\mathsf{left}](\nu^{k}p)([\mathsf{label}][\mathsf{tt}] \wedge \exists\bigcirc p) \;\vee\; [\mathsf{right}](\nu^{k}p)([\mathsf{label}][\mathsf{tt}] \wedge \exists\bigcirc p)\right)\\ \wedge\; &\left([\mathsf{left}](\nu^{k}p)([\mathsf{label}][\mathsf{ff}] \wedge \exists\bigcirc p) \;\vee\; [\mathsf{right}](\nu^{k}p)([\mathsf{label}][\mathsf{ff}] \wedge \exists\bigcirc p)\right)\Big)\\ &\Vdash\!\!\mapsto \quad (\mu^k p)([\mathsf{hd}][\geq \mathsf{n}][\mathsf{tot}] \vee \bigcirc p)\Big)\end{aligned}$$

and thus

$$(\exists k)\Big(\Big(\forall\Box[\mathsf{label}]([\mathsf{tt}] \vee [\mathsf{ff}]) \;\wedge\; \exists\bigcirc(\nu^k p)([\mathsf{label}][\mathsf{tt}] \wedge \exists\bigcirc p) \;\wedge\; \exists\bigcirc(\nu^k p)([\mathsf{label}][\mathsf{ff}] \wedge \exists\bigcirc p)\Big) \Vdash\!\!\mapsto \quad (\mu^k p)([\mathsf{hd}][\geq \mathsf{n}][\mathsf{tot}] \vee \bigcirc p)\Big)$$

Hence the result. ◀

11.2.3 Proof of Proposition 5.12

We recall the statement of Proposition 5.12.

▶ **Proposition 5.12.** *Given $\delta \in \mathcal{L}^\wedge(\tau)$, if $[\![\delta]\!] \neq \emptyset$ then $[\![\delta]\!] = \uparrow d$ for some finite $d \in [\![\tau]\!]$. Conversely, if $d \in [\![\tau]\!]$ is finite, then $\uparrow d = [\![\delta]\!]$ for some $\delta \in \mathcal{L}^\wedge(\tau)$.*

The proof of Proposition 5.12 is split into Lemma 11.9 and Lemma 11.10.

▶ **Lemma 11.9.** *Given $\varphi \in \mathcal{L}^\wedge(\tau)$, if $[\![\varphi]\!] \neq \emptyset$ then $[\![\varphi]\!] = \uparrow d$ for some finite $d \in [\![\tau]\!]$.*

Proof. The proof is by induction on $\varphi \in \mathcal{L}^\wedge(\tau)$.
Case of True. In this case, we have $[\![\varphi]\!] = \uparrow\bot$.
Case of $\varphi \wedge \psi$. First, note that $[\![\varphi]\!], [\![\psi]\!]$ are non-empty since so is their intersection. By induction hypothesis, there are finite $d, e \in [\![\tau]\!]$ such that $[\![\varphi]\!] = \uparrow d$ and $[\![\psi]\!] = \uparrow e$. Since $\uparrow d \cap \uparrow e$ is non-empty, and since $[\![\tau]\!]$ is a Scott domain, we get that $d \vee e$ is defined, finite, and such that $\uparrow(d \vee e) = \uparrow d \cap \uparrow e$. Hence $[\![\varphi \wedge \psi]\!] = \uparrow(d \vee e)$.
Case of $[\langle\rangle]$. Since $[\![[\langle\rangle]]\!] = \uparrow\top$.
Case of $[\triangle]\varphi$ with $\triangle \in \{\pi_1, \pi_2, \mathsf{in}_1, \mathsf{in}_2, \mathsf{fold}\}$. Note that $[\![\varphi]\!]$ is non-empty since so is $[\![[\triangle]\varphi]\!] = [\![[\triangle]]\!]([\![\varphi]\!])$. Hence by induction hypothesis, there is some finite d such that $[\![\varphi]\!] = \uparrow d$. We can then conclude using Lemma 11.2 and Proposition 9.11.
Case of $\psi \Vdash\!\!\rightarrow \varphi$. First, if $[\![\psi]\!] = \emptyset$, then $[\![\psi \Vdash\!\!\rightarrow \varphi]\!] = \uparrow\bot$.
Assume now that $[\![\psi]\!] \neq \emptyset$. In this case, we must also have $[\![\varphi]\!] \neq \emptyset$. Hence by induction hypothesis there are d, e finite such that $\uparrow e = [\![\psi]\!]$ and $\uparrow d = [\![\varphi]\!]$. Then we are done since

$$\begin{array}{rcl} [\![\psi \Vdash\!\!\rightarrow \varphi]\!] & = & \{f \mid \forall x \in [\![\psi]\!],\ f(x) \in [\![\varphi]\!]\} \\ & = & \{f \mid \forall x \geq e,\ f(x) \geq d\} \\ & = & \uparrow(e \searrow d) \end{array}$$

◀

▶ **Lemma 11.10.** *If $d \in [\![\tau]\!]$ is finite, then there is $\varphi \in \mathcal{L}^\wedge(\tau)$ such that $\uparrow d = [\![\varphi]\!]$.*

Proof. We rely on Proposition 9.11 and on the inductive definition of $\mathrm{Fin}([\![\tau]\!])$ in Figure 12. We reason by cases on the rules in Figure 12.
Case of

$$\frac{}{\bot \in \mathrm{Fin}([\![\tau]\!])}$$

Since $\uparrow\bot = [\![\mathsf{True}]\!]$.
Case of

$$\frac{}{\top \in \mathrm{Fin}([\![\mathsf{Unit}]\!])}$$

Since $\uparrow\top = [\![[\langle\rangle]]\!]$.
Case of

$$\frac{d \in \mathrm{Fin}([\![\tau]\!]) \qquad e \in \mathrm{Fin}([\![\sigma]\!])}{(d, e) \in \mathrm{Fin}([\![\tau \times \sigma]\!])}$$

By induction hypothesis, we have $\varphi \in \mathcal{L}^\wedge(\tau)$ and $\psi \in \mathcal{L}^\wedge(\sigma)$ such that $[\![\varphi]\!] = \uparrow d$ and $[\![\psi]\!] = \uparrow e$. Since the order in $[\![\tau \times \sigma]\!]$ is pointwise, we get

$$\begin{array}{rcl} \uparrow(d, e) & = & \uparrow d \times \uparrow e \\ & = & (\uparrow d \times [\![\sigma]\!]) \cap ([\![\tau]\!] \times \uparrow e) \\ & = & [\![[\pi_1]\varphi \wedge [\pi_2]\psi]\!] \end{array}$$

Case of

$$\frac{d \in \mathrm{Fin}(\llbracket\tau\rrbracket_i)}{\llbracket\mathsf{in}_i\rrbracket(d) \in \mathrm{Fin}(\llbracket\tau_1 + \tau_2\rrbracket)}$$

By induction hypothesis, there is $\varphi \in \mathcal{L}^\wedge(\tau_i)$ such that $\llbracket\varphi\rrbracket = {\uparrow}d$. Since $\llbracket\mathsf{in}_i\rrbracket$ is an order embedding (§9.2.3 and §9.2.2),

$$\begin{array}{rcl} {\uparrow}\llbracket\mathsf{in}_i\rrbracket(d) & = & \{\mathsf{in}_i(x) \mid \mathsf{in}_i(x) \geq \mathsf{in}_i(d)\} \\ & = & \{\mathsf{in}_i(x) \mid x \geq d\} \\ & = & \llbracket[\mathsf{in}_i]\varphi\rrbracket \end{array}$$

Case of

$$\frac{d \in \mathrm{Fin}(\llbracket\tau[\mathsf{rec}\,\alpha.\tau/\alpha]\rrbracket)}{\llbracket\mathsf{fold}\rrbracket(d) \in \mathrm{Fin}(\llbracket\mathsf{rec}\,\alpha.\tau\rrbracket)}$$

By induction hypothesis, there is $\varphi \in \mathcal{L}^\wedge(\tau[\mathsf{rec}\,\alpha.\tau/\alpha])$ such that $\llbracket\varphi\rrbracket = {\uparrow}d$. We thus have ${\uparrow}\llbracket\mathsf{fold}\rrbracket(d) = \llbracket[\mathsf{fold}]\varphi\rrbracket$.

Case of

$$\frac{\begin{array}{l}\text{for all } i \in I,\ \ d_i \in \mathrm{Fin}(\llbracket\tau\rrbracket) \ \text{ and } \ e_i \in \mathrm{Fin}(\llbracket\sigma\rrbracket)\ ; \\ \text{for all } J \subseteq I,\ \ \bigvee_{j\in J} d_j \text{ defined in } \llbracket\tau\rrbracket \ \Rightarrow\ \bigvee_{j\in J} e_j \text{ defined in } \llbracket\sigma\rrbracket\end{array}}{\bigvee_{i\in I}(d_i \searrow e_i) \in \mathrm{Fin}(\llbracket\tau \to \sigma\rrbracket)}$$

where I is a finite set.

By induction hypothesis, for each $i \in I$ there are $\varphi_i \in \mathcal{L}^\wedge(\sigma)$ and $\psi_i \in \mathcal{L}^\wedge(\tau)$ such that ${\uparrow}e_i = \llbracket\varphi_i\rrbracket$ and ${\uparrow}d_i = \llbracket\psi_i\rrbracket$. Note that

$$\begin{array}{rcl} {\uparrow}(d_i \searrow e_i) & = & \{f\colon \llbracket\tau\rrbracket \to \llbracket\sigma\rrbracket \mid \forall x \geq d_i,\ f(x) \geq e_i\} \\ & = & \{f\colon \llbracket\tau\rrbracket \to \llbracket\sigma\rrbracket \mid \forall x \in \llbracket\psi_i\rrbracket,\ f(x) \in \llbracket\varphi_i\rrbracket\} \\ & = & \llbracket\psi_i \Vdash\!\!\to \varphi_i\rrbracket \end{array}$$

The result then follows from the fact that

$${\uparrow}\left(\textstyle\bigvee_{i\in I} d_i \searrow e_i\right) \ = \ \textstyle\bigcap_{i\in I} {\uparrow}(d_i \searrow e_i)$$

◀

11.3 Soundness of Deduction

This Appendix 11.3 is devoted to the proof of the Soundness Theorem 5.13. We handle some rules separately.

▶ **Lemma 11.11.** *The following deduction rules are sound.*

$$(\exists\mathsf{M})\frac{}{(\exists k)(\exists\ell)\varphi \vdash (\exists k)\varphi[k/\ell]} \qquad (\forall\mathsf{M})\frac{}{(\forall k)\psi[k/\ell] \vdash (\forall k)(\forall\ell)\psi}$$

Proof. Note that k, ℓ Pos φ, while k, ℓ Neg ψ. By monotonicity (Lemma 5.9), if $x \in \llbracket\varphi\rrbracket\rho[m/k, n/\ell]$ for some $m, n \in \mathbb{N}$, then $x \in \llbracket\varphi\rrbracket\rho[\max(m,n)/k, \max(m,n)/\ell]$. Hence the rule $(\exists\mathsf{M})$.

As for the rule $(\forall\mathsf{M})$, by *anti*monotonicity (Lemma 5.9 again), if $x \notin \llbracket\psi\rrbracket\rho[m/k, n/\ell]$ for some $m, n \in \mathbb{N}$, then $x \notin \llbracket\psi\rrbracket\rho[\max(m,n)/k, \max(m,n)/\ell]$. ◀

▶ **Lemma 11.12.** *The following deduction rules are sound, where $\iota \in \{\mu, \nu\}$.*

$$(\iota/\exists)\frac{k \notin \mathrm{FV}(t)}{(\iota^t p)(\exists k)\varphi \vdash (\exists k)(\iota^t p)\varphi} \qquad (\forall/\iota)\frac{\ell \notin \mathrm{FV}(t)}{(\forall \ell)(\iota^t p)\psi \vdash (\iota^t p)(\forall \ell)\psi}$$

Proof. Note that φ is positive with k Pos φ, while ψ is negative with ℓ Neg ψ. Hence $(\forall/\iota)$ is dual to $(\iota/\exists)$. We only detail $(\iota/\exists)$. The proof relies on the following.

▷ Claim 11.13. Let A be a complete lattice and consider a family of Scott-continuous functions $(f_i \colon A \to A \mid i \in \mathbb{N})$ such that $f_i \leq f_j$ whenever $i \leq j$. Let $f \colon A \to A$ take a to $\bigvee_{i\in\mathbb{N}} f_i(a)$. Then for each $n \in \mathbb{N}$, we have $f^n(a) = \bigvee_{i\in\mathbb{N}} f_i^n(a)$.

Proof. The proof of the Claim is by induction on $n \in \mathbb{N}$. In the base case $n = 0$, we have $f^0(a) = a$ while $\bigvee_{i\in\mathbb{N}} f_i^0(a) = \bigvee_{i\in\mathbb{N}} a = a$.

For the induction step, by induction hypothesis we have $f^{n+1}(a) = f(\bigvee_{i\in\mathbb{N}} f_i^n(a))$. Since the f_i's are in particular monotone, we get $f_i^n \leq f_j^n$ whenever $i \leq j$. It follows that the set $\{f_i^n(a) \mid i \in \mathbb{N}\}$ is directed, so that $f^{n+1}(a) = \bigvee_{i\in\mathbb{N}} f(f_i^n(a))$ by Scott-continuity of f (Lemma 11.7(1)). By definition of f, we thus get $f^{n+1}(a) = \bigvee_{i\in\mathbb{N}} \bigvee_{j\in\mathbb{N}} f_j(f_i^n(a))$. It remains to show that

$$\bigvee_{i\in\mathbb{N}} f_i^{n+1}(a) \quad = \quad \bigvee_{i\in\mathbb{N}} \bigvee_{j\in\mathbb{N}} f_j(f_i^n(a))$$

The inequality $\leq$ is obvious ($f_i^{n+1}(a) = f_i(f_i^n(a))$). For the converse inequality, let $i, j \in \mathbb{N}$ and let $m = \max\{i, j\}$. We have $f_i, f_j \leq f_m$, and thus $f_j(f_i^n(a)) \leq f_m(f_m^n(a))$. Hence $f_j(f_i^n(a)) \leq f_m^{n+1}(a)$, which gives the result. ◁

Returning to the proof of the lemma, let $f_i \coloneqq [\![p.\varphi]\!]\rho[i/k]$ and let $n \coloneqq [\![t]\!]\rho$ (recall that $k \notin \mathrm{FV}(t)$. The f_i's are Scott continuous by Proposition 5.10(2). Moreover, $i \leq j$ implies $f_i \leq f_j$ by Lemma 5.9. Hence the Claim gives the result since $f = [\![p.(\exists k)\varphi]\!]\rho$, so that

$$\begin{array}{rcccl} [\![(\mu^t p)(\exists k)\varphi]\!]\rho & = & ([\![p.(\exists k)\varphi]\!]\rho)^n([\![\mathsf{False}]\!]) & = & f^n([\![\mathsf{False}]\!]) \\ [\![(\nu^t p)(\exists k)\varphi]\!]\rho & = & ([\![p.(\exists k)\varphi]\!]\rho)^n([\![\mathsf{True}]\!]) & = & f^n([\![\mathsf{True}]\!]) \end{array}$$

while

$$\begin{array}{rcl} [\![(\exists k)(\mu^t p)\varphi]\!]\rho & = & \bigcup_{i\in\mathbb{N}} f_i^n([\![\mathsf{False}]\!]) \\ [\![(\exists k)(\nu^t p)\varphi]\!]\rho & = & \bigcup_{i\in\mathbb{N}} f_i^n([\![\mathsf{True}]\!]) \end{array}$$

◀

▶ **Lemma 11.14.** *The following deduction rules are sound.*

$$(\mathsf{WF})\frac{k \notin \mathrm{FV}(\varphi)}{(\forall k)\psi \parallel\!\!\mapsto \varphi \vdash (\exists k)(\psi \parallel\!\!\mapsto \varphi)} \qquad (\mathsf{CC})\frac{\psi \in \mathcal{L}^{\pm}(\sigma) \qquad k \notin \mathrm{FV}(\psi)}{\psi \parallel\!\!\mapsto (\exists k)\varphi \vdash (\exists k)(\psi \parallel\!\!\mapsto \varphi)}$$

Proof. We begin with (WF). Let $f \in [\![(\forall k)\psi \parallel\!\!\mapsto \varphi]\!]\rho$.

Note that $(\forall k)\psi$ is negative. Hence k Neg ψ, and Lemma 5.9 yields that the function $n \in \mathbb{N} \mapsto [\![\psi]\!]\rho[n/k]$ is antimonotone. It then follows from Proposition 5.8 that $\{[\![\psi]\!]\rho[n/k] \mid n \in \mathbb{N}\}$ is a codirected family of compact saturated sets.

Moreover, for each $n \in \mathbb{N}$, $[\![\psi]\!]\rho[n/k]$ is the codirected intersection of all compact opens K such that $[\![\psi]\!]\rho[n/k] \subseteq K$ (see Lemma 10.6 and its proof, §10). Let $\mathcal{Q}$ be the set of all compact opens K such that $[\![\psi]\!]\rho[n/k] \subseteq K$ for some $n \in \mathbb{N}$. Hence $\mathcal{Q}$ is a codirected collection of compact opens with $\bigcap \mathcal{Q} = [\![(\forall k)\psi]\!]\rho$.

On the other hand, φ is positive. Hence $[\![\varphi]\!]\rho$ is open (Proposition 5.8 again), and we get from Lemma 5.7 that $f \in K \Vdash [\![\varphi]\!]\rho$ for some $K \in \mathcal{Q}$. But $K \supseteq [\![\psi]\!]\rho[n/k]$ for some $n \in \mathbb{N}$, so that $f \in [\![\psi \Vdash \varphi]\!]\rho[n/k]$ by contravariance of $\Vdash$ in its first argument (recall that k is not free in φ).

We now turn to (CC). Let $f \in [\![\psi \Vdash (\exists k)\varphi]\!]$. The formula φ is positive. Hence k Pos φ and by Lemma 5.9 and Proposition 5.8 we get that $[\![(\exists k)\varphi]\!]\rho$ is the directed union of the opens $([\![(\exists k)\varphi]\!]\rho[n/k] \mid n \in \mathbb{N})$.

On the other hand, again by Proposition 5.8, we have that $[\![\psi]\!]\rho$ is compact (open). Since f is continuous, we get that that $[\![\psi]\!]\rho$ is a compact set included in the union of the directed family of opens $(f^{-1}([\![(\exists k)\varphi]\!]\rho[n/k]) \mid n \in \mathbb{N})$. Hence $[\![\psi]\!] \subseteq f^{-1}([\![(\exists k)\varphi]\!]\rho[n/k])$ for some $n \in \mathbb{N}$, and $f \in [\![\psi \Vdash \varphi]\!]\rho[n/k]$ (recall that k is not free in φ). ◀

▶ **Lemma 11.15.** *The following rule is sound.*

$$(\Vdash\!\!\to/\vee)\ \frac{\delta \in \mathcal{L}^{\wedge}(\tau)}{\delta \Vdash (\varphi_1 \vee \varphi_2) \vdash (\delta \Vdash \varphi_1) \vee (\delta \Vdash \varphi_2)}$$

Proof. If $[\![\delta]\!] = \emptyset$, then $[\![\delta \Vdash \psi]\!]\rho = [\![\mathsf{True}]\!]$ for all ψ, hence the result. Otherwise, by Proposition 5.12 there is some finite $d \in [\![\tau]\!]$ such that $[\![\delta]\!] = \uparrow d$. Hence, if $f \in [\![\delta \Vdash (\varphi_1 \vee \varphi_2)]\!]\rho$, then in particular $f(d) \in [\![\varphi_1]\!]\rho \cup [\![\varphi_2]\!]\rho$. But since f is monotone and $[\![\varphi_i]\!]\rho$ is saturated (Proposition 5.8), we have $f \in [\![\delta \Vdash \varphi_i]\!]\rho$ whenever $f(d) \in [\![\varphi_i]\!]\rho$. ◀

We now prove the Soundness Theorem 5.13.

▶ **Theorem 5.13** (Soundness of Deduction)**.**
(1) *For all $\delta \in \mathcal{L}^{\wedge}(\tau)$, if $\mathcal{C}(\delta)$ is derivable, then $[\![\delta]\!] \neq \emptyset$.*
(2) *For all $\varphi, \psi \in \mathcal{L}^{s}(\Sigma;\tau)$, if $\psi \vdash \varphi$ is derivable, then $[\![\psi]\!]\rho \subseteq [\![\varphi]\!]\rho$.*

Proof. We reason by mutual induction on $\mathcal{C}(\delta)$ (Figure 5) and $\psi \vdash \varphi$ (Figure 4). We discuss each (sub)figure separately.

Propositional rules in Figure 4a. These rules are all fairly standard, and we omit the proof.

Modal rules in Figure 4b. The rules (I) and the rule for $\Vdash$ were already discussed in §5.3.

The remaining rules all follow from Lemma 11.1.

Quantifier rules in Figure 4c. The rules ($\exists$M) and ($\forall$M) are handled by Lemma 11.11, while the rules ($\iota/\exists$) and ($\forall/\iota$) are handled by Lemma 11.12. The remaining rules are standard.

Rules for bounded iteration in Figure 4d. We discuss the rules

$$\frac{\psi \vdash \varphi}{(\mu^t p)\psi \vdash (\mu^t p)\varphi} \qquad \frac{\psi \vdash \varphi}{(\nu^t p)\psi \vdash (\nu^t p)\varphi}$$

By induction hypothesis we have $[\![\psi]\!]\rho \subseteq [\![\varphi]\!]\rho$. Since moreover the functions $[\![p.\psi]\!]\rho$ and $[\![p.\varphi]\!]\rho$ are monotone (Proposition 5.10(1)), for each (appropriate) S an induction on $n \in \mathbb{N}$ yields

$$([\![p.\psi]\!]\rho)^n(S) \quad \subseteq \quad ([\![p.\varphi]\!]\rho)^n(S)$$

Hence the result.

The other rules all directly follow from the interpretation of bounded iteration in Figure 8.

Rules for the realizability implication in Figure 4e. The rules (WF) and (CC) are handled by Lemma 11.14, while the rule ($\Vdash\!\!\to/\vee$) is dealt-with in Lemma 11.15. All the remaining rules are standard, excepted (C).

$$(\mathsf{C})\ \frac{\mathcal{C}(\delta) \qquad \delta \in \mathcal{L}^{\wedge}}{(\delta \Vdash \mathsf{False}) \vdash \mathsf{False}}$$

By induction hypothesis we have that $[\![\delta]\!] \neq \emptyset$. Hence $[\![\delta \Vdash \mathsf{False}]\!]\rho$ is empty.

Rules for the consistency predicate $\mathcal{C}$ in Figure 5. We only detail a couple of rules, the other one being trivial.

Case of

$$\frac{\mathcal{C}(\varphi) \qquad \mathcal{C}(\psi)}{\mathcal{C}([\pi_1]\varphi \wedge [\pi_2]\psi)}$$

By induction hypothesis we have $[\![\varphi]\!] \neq \emptyset$ and $[\![\psi]\!] \neq \emptyset$. Hence $[\![[\pi_1]\varphi \wedge [\pi_2]\psi]\!] \neq \emptyset$ since

$$\begin{array}{rcl} [\![[\pi_1]\varphi \wedge [\pi_2]\psi]\!] & = & \{z \mid \pi_1(z) \in [\![\varphi]\!]\} \cap \{z \mid \pi_2(z) \in [\![\psi]\!]\} \\ & = & \{(x,y) \mid x \in [\![\varphi]\!] \text{ and } y \in [\![\psi]\!]\} \end{array}$$

Case of

$$\frac{\begin{array}{l} I \text{ finite and } \forall i \in I,\ \mathcal{C}(\psi_i) \text{ and } \mathcal{C}(\varphi_i); \\ \forall J \subseteq I,\ \bigwedge_{j\in J} \psi_j \vdash \mathsf{False} \ \text{ or } \ \mathcal{C}\left(\bigwedge_{j\in J} \varphi_j\right) \end{array}}{\mathcal{C}\left(\bigwedge_{i\in I}(\psi_i \Vdash \varphi_i)\right)}$$

Let τ, σ such that $\varphi_i \in \mathcal{L}^\wedge(\tau)$ and $\psi_i \in \mathcal{L}^\wedge(\sigma)$ for all $i \in I$.
First, by induction hypothesis we have $[\![\psi_i]\!] \neq \emptyset$ and $[\![\varphi_i]\!] \neq \emptyset$ for all $i \in I$. Hence, it follows from Proposition 5.12 that for each $i \in I$, there are finite $d_i \in [\![\tau]\!]$ and $e_i \in [\![\sigma]\!]$ such that $\uparrow d_i = [\![\varphi_i]\!]$ and $\uparrow e_i = [\![\psi_i]\!]$. We thus have $[\![\psi_i \Vdash \varphi_i]\!] = \uparrow(e_i \searrow d_i)$ for each $i \in I$, so that

$$[\![\textstyle\bigwedge_{i\in I}(\psi_i \Vdash \varphi_i)]\!] \quad = \quad \bigcap_{i\in I} \uparrow(e_i \searrow d_i)$$

Assume $[\![\bigwedge_{i\in I}(\psi_i \Vdash \varphi_i)]\!] = \emptyset$. As recalled in §9.2.4 (see also [4, Theorem 1.4.12]), there is some $J \subseteq I$ such that

$$\bigcap_{i\in I} \uparrow e_i \quad \neq \quad \emptyset \qquad \text{and} \qquad \bigcap_{i\in I} \uparrow d_i \quad = \quad \emptyset$$

But the induction hypothesis yields either $\bigcap_{i\in I} \uparrow e_i = \emptyset$ or $\bigcap_{i\in I} \uparrow d_i \neq \emptyset$, a contradiction. ◀

11.4 Completeness for the Neutral Fragment

This Appendix 11.4 is devoted to the proof of Theorem 5.14.

▶ **Theorem 5.14** (Completeness of $\mathcal{L}^\wedge$)**.** *Let $\varphi, \psi \in \mathcal{L}^\wedge(\tau)$.*
(1) *If $[\![\varphi]\!] \neq \emptyset$, then $\mathcal{C}(\varphi)$ is derivable.*
(2) *If $[\![\psi]\!] \subseteq [\![\varphi]\!]$, then $\psi \vdash \varphi$ is derivable.*

The proof uses the following preliminary lemma. The *size* of a formula δ is denoted $|\delta|$.

▶ **Lemma 11.16.** *Let $\delta \in \mathcal{L}^\wedge$ such that neither $\delta \vdash \mathsf{False}$ nor $\mathsf{True} \vdash \delta$ is derivable.*
(1) *If $\delta \in \mathcal{L}^\wedge(\mathsf{Unit})$, then $\delta \dashv\vdash [\langle\rangle]$.*
(2) *If $\delta \in \mathcal{L}^\wedge(\mathsf{rec}\,\alpha.\tau)$, then $\delta \dashv\vdash [\mathsf{fold}]\psi$ with $|[\mathsf{fold}]\psi| \leq |\delta|$.*
(3) *If $\delta \in \mathcal{L}^\wedge(\tau_1 + \tau_2)$, then there is an $i \in \{1,2\}$ such that $\delta \dashv\vdash [\mathsf{in}_i]\psi$ with $|[\mathsf{in}_i]\psi| \leq |\delta|$.*
(4) *If $\delta \in \mathcal{L}^\wedge(\tau_1 \times \tau_2)$, then $\delta \dashv\vdash [\pi_1]\psi_1 \wedge [\pi_2]\psi_2$ with $|[\pi_1]\psi_1 \wedge [\pi_2]\psi_2| \leq |\delta|$.*
(5) *If $\delta \in \mathcal{L}^\wedge(\sigma \to \tau)$, then there is a finite non-empty set J such that*

$$\delta \quad \dashv\vdash \quad \textstyle\bigwedge_{j\in J}(\varphi_j \Vdash \psi_j)$$

with $|\bigwedge_{j\in J}(\varphi_j \Vdash \psi_j)| \leq |\delta|$.

Proof. The proof is a mutual induction on the construction of $\delta \in \mathcal{L}^\wedge$ with the (appropriate) rules in Figure 2. The cases of the rules in Figure 2b and of the rule

$$\frac{\varphi_1 \in \mathcal{L}^\wedge(\tau_1) \qquad \varphi_2 \in \mathcal{L}^\wedge(\tau_2)}{\varphi_1 \Vdash\!\!\!\rightarrow \varphi_2 \in \mathcal{L}^\wedge(\tau_1 \to \tau_2)}$$

are trivial. The only possible rule of Figure 2a is

$$\frac{\varphi_1 \in \mathcal{L}^\wedge \qquad \varphi_2 \in \mathcal{L}^\wedge}{\varphi_1 \wedge \varphi_2 \in \mathcal{L}^\wedge}$$

We handle the different items separately. In each case, if $\mathsf{True} \vdash \varphi_i$, then $\delta \dashv\vdash \varphi_{3-i}$, and the result follows from the induction hypothesis. Also, we can assume $\varphi_i \nvdash \mathsf{False}$ since $\delta \nvdash \mathsf{False}$.

Item (1). By induction hypothesis we have $\varphi_i \dashv\vdash [\langle\rangle]$, and the result follows.

Item (2). By induction hypothesis, there are ψ_i smaller than φ_i such that $\varphi_i \dashv\vdash [\mathsf{fold}]\psi_i$. We have $\delta \dashv\vdash [\mathsf{fold}](\psi_1 \wedge \psi_2)$. Moreover, since $|\varphi_i| \geq |\psi_i| + 1$, we have

$$|\delta| \;\geq\; 1 + |\varphi_1| + |\varphi_2| \;\geq\; 3 + |\psi_1| + |\psi_2| \;\geq\; 2 + |\psi_1 \wedge \psi_2| \;\geq\; 1 + |[\mathsf{fold}](\psi_1 \wedge \psi_2)|$$

Item (3). By induction hypothesis, there are $j_i \in \{1, 2\}$ and ψ_i smaller than φ_i such that $\varphi_i \dashv\vdash [\mathsf{in}_{j_i}]\psi_i$. Since $\delta \nvdash \mathsf{False}$ we must have $j_1 = j_2$, and we can conclude as for item (2).

Item (4). By induction hypothesis, we have $\varphi_i \dashv\vdash [\pi_1]\psi_{i,1} \wedge [\pi_2]\psi_{i,2}$ with $|[\pi_1]\psi_{i,1} \wedge [\pi_2]\psi_{i,2}| \leq |\varphi_i|$. Note that

$$|[\pi_1](\psi_{1,1} \wedge \psi_{2,1}) \wedge [\pi_2](\psi_{1,2} \wedge \psi_{2,2})| \;\leq\; |([\pi_1]\psi_{1,1} \wedge [\pi_2]\psi_{1,2}) \wedge ([\pi_1]\psi_{2,1} \wedge [\pi_2]\psi_{2,2})|$$

Hence we are done with $\delta \dashv\vdash [\pi_1]\psi_1 \wedge [\pi_2]\psi_2$ where ψ_i is $\psi_{1,i} \wedge \psi_{2,i}$.

Item (5). By induction hypothesis. ◀

Theorem 5.14(1) is a consequence of the following.

▶ **Lemma 11.17.** *Let $\delta \in \mathcal{L}^\wedge(\tau)$.*

(1) *If $[\![\delta]\!] \neq \emptyset$, then $\mathcal{C}(\delta)$ is derivable.*

(2) *If $[\![\delta]\!] = \emptyset$, then $\delta \vdash \mathsf{False}$ is derivable.*

Proof. Both statements are proved by a simultaneous induction on the size of $\delta \in \mathcal{L}^\wedge$, using Lemma 11.16.

Note that we may always assume $\mathsf{True} \nvdash \delta$ and $\delta \nvdash \mathsf{False}$. Indeed, if $\mathsf{True} \vdash \delta$ then item (1) is trivial, while item (2) is impossible by the Soundness Theorem 5.13. Similarly, if $\delta \vdash \mathsf{False}$, then item (1) is impossible by the Soundness Theorem 5.13, while item (2) is trivial.

We reason by cases on τ.

Case of Unit. By Lemma 11.16, we have $\delta \dashv\vdash [\langle\rangle]$. Hence $[\![\delta]\!] = \{\top\}$, only item (1) is possible and we get $\mathcal{C}(\delta)$ since $\mathcal{C}([\langle\rangle])$ and $[\langle\rangle] \vdash \delta$.

Case of rec $\alpha.\tau$. By Lemma 11.16, we have $\delta \dashv\vdash [\mathsf{fold}]\psi$ with $|\psi| < |\delta|$.

We first consider item (1). If $[\![\delta]\!] \neq \emptyset$, then $[\![\psi]\!] \neq \emptyset$. Since ψ is smaller than δ, the induction hypothesis yields $\mathcal{C}(\psi)$. Hence $\mathcal{C}([\mathsf{fold}]\psi)$ and we get $\mathcal{C}(\delta)$ since $[\mathsf{fold}]\psi \vdash \delta$.

We now turn to item (2). If $[\![\delta]\!] = \emptyset$ then $[\![\psi]\!] = \emptyset$. Since ψ is smaller than δ, the induction hypothesis yields $\psi \vdash_{\tau[\mathsf{rec}\,\alpha.\tau/\alpha]} \mathsf{False}$, so that $[\mathsf{fold}]\psi \vdash_{\mathsf{rec}\,\alpha.\tau} [\mathsf{fold}]\mathsf{False}$. Then we are done since $[\mathsf{fold}]\mathsf{False} \vdash \mathsf{False}$.

Case of $\tau_1 + \tau_2$. Similar.

Case of $\tau_1 \times \tau_2$. By Lemma 11.16, we have $\delta \dashv\vdash [\pi_1]\psi_1 \wedge [\pi_2]\psi_2$ with $|[\pi_1]\psi_1 \wedge [\pi_2]\psi_2| \leq |\delta|$. We first consider item (1). If $[\![\delta]\!] \neq \emptyset$, then $[\![\psi_1]\!] \neq \emptyset$ and $[\![\psi_2]\!] \neq \emptyset$. Since $|\psi_1|, |\psi_2| < |\delta|$, the induction hypothesis yields $\mathcal{C}(\psi_1)$ and $\mathcal{C}(\psi_2)$. Hence $\mathcal{C}([\pi_1]\psi_1 \wedge [\pi_2]\psi_2)$ and we are done.

We now turn to item (2). If $[\![\delta]\!] = \emptyset$ then we must have (say) $[\![\psi_i]\!] = \emptyset$. Hence $\psi_i \vdash \mathsf{False}$ by induction hypothesis. It follows that $[\pi_i]\psi_i \vdash [\pi_i]\mathsf{False}$, and we are done since $[\pi_i]\mathsf{False} \vdash \mathsf{False}$.

Case of $\sigma \to \tau$. This is the most important and most difficult case.

By Lemma 11.16, there is a finite set J such that

$$\delta \quad \dashv\vdash \quad \textstyle\bigwedge_{j\in J} (\varphi_j \Vdash\!\!\mapsto \psi_j)$$

with $|\bigwedge_{j\in J} (\varphi_j \Vdash\!\!\mapsto \psi_j)| \leq |\delta|$.

Assume first that for some $j \in J$, we have $[\![\psi_j]\!] = \emptyset$ with $[\![\varphi_j]\!] \neq \emptyset$. Then $[\![\varphi_j \Vdash\!\!\mapsto \psi_j]\!] = \emptyset$ and $[\![\delta]\!] = \emptyset$. Hence we must be in the case of item (2). Moreover, by induction hypothesis we have $\mathcal{C}(\varphi_j)$ and $\psi_j \vdash \mathsf{False}$, so that we can derive $\delta \vdash \mathsf{False}$ using the rule (C) in Figure 4e.

Assume now that we have $[\![\psi_j]\!] \neq \emptyset$ for all $j \in J$ such that $[\![\varphi_j]\!] \neq \emptyset$.

Given $j \in J$ such that $[\![\varphi_j]\!] = \emptyset$, by induction hypothesis we have $\varphi_j \vdash \mathsf{False}$, and since $\mathsf{True} \vdash (\mathsf{False} \Vdash\!\!\mapsto \psi_j)$, we get $\mathsf{True} \vdash (\varphi_j \Vdash\!\!\mapsto \psi_j)$. Hence $\bigwedge_{i\neq j}(\varphi_i \Vdash\!\!\mapsto \psi_i) \dashv\vdash \delta$.

We can therefore reduce the case of $\bigwedge_{j\in J}(\varphi_j \Vdash\!\!\mapsto \psi_j)$ where $[\![\psi_j]\!] \neq \emptyset$ and $[\![\varphi_j]\!] \neq \emptyset$ for all $j \in J$. In, particular, the induction hypothesis yields $\mathcal{C}(\psi_j)$ and $\mathcal{C}(\varphi_j)$ for all $j \in J$.

Regarding item (1), if $[\![\delta]\!] \neq \emptyset$, then for all $I \subseteq J$ we have either $\left[\!\!\left[\bigwedge_{i\in I} \varphi_i\right]\!\!\right] = \emptyset$ or $\left[\!\!\left[\bigwedge_{i\in I} \psi_i\right]\!\!\right] \neq \emptyset$, so that either $\bigwedge_{i\in I} \varphi_i \vdash \mathsf{False}$ or $\mathcal{C}(\bigwedge_{i\in I} \psi_i)$ by induction hypothesis. We can thus obtain $\mathcal{C}(\delta)$ by using the last rule in Figure 5.

Concerning item (2), if $[\![\delta]\!] = \emptyset$, then there is some $I \subseteq J$ such that $\left[\!\!\left[\bigwedge_{i\in I} \varphi_i\right]\!\!\right] \neq \emptyset$ while $\left[\!\!\left[\bigwedge_{i\in I} \psi_i\right]\!\!\right] = \emptyset$. Hence $\mathcal{C}(\bigwedge_{i\in I} \varphi_i)$ and $\bigwedge_{i\in I} \psi_i \vdash \mathsf{False}$ by induction hypothesis. We have

$$\delta \quad \vdash \quad \textstyle\bigwedge_{j\in J} \left((\bigwedge_{i\in I} \varphi_i) \Vdash\!\!\mapsto \psi_j\right)$$

and thus

$$\delta \quad \vdash \quad (\textstyle\bigwedge_{i\in I} \varphi_i) \Vdash\!\!\mapsto \bigwedge_{i\in I} \psi_i$$

and we obtain $\delta \vdash \mathsf{False}$ using the rule (C) in Figure 4e. ◀

We now turn to Theorem 5.14(2). We shall use the following preliminary result.

▶ **Lemma 11.18.** *Given $\delta \in \mathcal{L}^{\wedge}(\tau)$, if $[\![\delta]\!] = [\![\mathsf{True}]\!]$, then $\mathsf{True} \vdash \delta$ is derivable.*

Proof. By the Soundness Theorem 5.13, if $\delta \vdash \mathsf{False}$ then $[\![\delta]\!] = \emptyset$. We can thus assume $\delta \nvdash \mathsf{False}$. As consequence, we are done if we show that $[\![\delta]\!] \neq [\![\mathsf{True}]\!]$ whenever $\delta \nvdash \mathsf{False}$ and $\mathsf{True} \nvdash \delta$. The proof is by induction in the size of δ, using Lemma 11.16. We reason by cases on τ.

Case of Unit. By Lemma 11.16, we have $\delta \dashv\vdash [\langle\rangle]$. Hence we are done since $[\![[\langle\rangle]]\!] = \{\top\}$ while $[\![\mathsf{Unit}]\!] = \{\bot, \top\}$.

Case of $\tau_1 + \tau_2$. By Lemma 11.16, we have $\delta \dashv\vdash [\mathsf{in}_i]\psi$ for some $i \in \{1, 2\}$. Hence $[\![\delta]\!] \neq [\![\mathsf{True}]\!]$.

Case of rec $\alpha.\tau$. By Lemma 11.16, we have $\delta \dashv\vdash [\mathsf{fold}]\psi$ with $|\psi| < |\delta|$.

If $\psi \vdash \mathsf{False}$ then $[\mathsf{fold}]\psi \vdash [\mathsf{fold}]\mathsf{False}$, so that $\delta \vdash \mathsf{False}$ since $[\mathsf{fold}]\mathsf{False} \vdash \mathsf{False}$. Similarly, if $\mathsf{True} \vdash \psi$ then $\mathsf{True} \vdash \delta$ since $\mathsf{True} \vdash [\mathsf{fold}]\mathsf{True}$.

Hence, we have $\mathsf{True} \nvdash \psi$ and $\psi \nvdash \mathsf{False}$. The induction hypothesis yields $[\![\psi]\!] \neq [\![\mathsf{True}]\!]$, from which we get $[\![\delta]\!] \neq [\![\mathsf{True}]\!]$ since $[\![\mathsf{unfold}]\!]$ is an iso.

Case of $\tau_1 \times \tau_2$. By Lemma 11.16, we have $\delta \dashv\vdash [\pi_1]\psi_1 \wedge [\pi_2]\psi_2$ with $|[\pi_1]\psi_1 \wedge [\pi_2]\psi_2| \leq |\delta|$.
Assume toward a contradiction that $\psi_i \vdash \mathsf{False}$. Then $\delta \vdash \mathsf{False}$ since $[\pi_i]\mathsf{False} \vdash \mathsf{False}$. Hence we have $\psi_1 \nvdash \mathsf{False}$ and $\psi_2 \nvdash \mathsf{False}$. Moreover, since $\mathsf{True} \vdash [\pi_i]\mathsf{True}$ and $\mathsf{True} \nvdash \delta$, must have either $\mathsf{True} \nvdash \psi_1$ or $\mathsf{True} \nvdash \psi_2$.
Assume (say) $\mathsf{True} \nvdash [\pi_i]\psi_i$. Then by induction hypothesis we get $[\![\psi_i]\!] \neq [\![\mathsf{True}]\!]$, which implies $[\![\delta]\!] \neq [\![\mathsf{True}]\!]$.

Case of $\sigma \to \tau$. By Lemma 11.16, there is a finite set J such that

$$\delta \quad \dashv\vdash \quad \bigwedge_{j\in J} (\varphi_j \Vdash \psi_j)$$

with $|\bigwedge_{j\in J} (\varphi_j \Vdash \psi_j)| \leq |\delta|$.
Let $i \in J$ such that $\mathsf{True} \vdash \psi_i$. Hence $\mathsf{True} \vdash \varphi_i \Vdash \psi_i$ and $\delta \dashv\vdash \bigwedge_{j\neq i}(\varphi_j \Vdash \psi_j)$. Since $\mathsf{True} \nvdash \delta$, we can thus reduce to the case where $\mathsf{True} \nvdash \psi_j$ for all $j \in J$.
Similarly, let $i \in J$ such that $\varphi_i \vdash \mathsf{False}$. Hence $\mathsf{True} \vdash \varphi_i \Vdash \psi_i$ and $\delta \dashv\vdash \bigwedge_{j\neq i}(\varphi_j \Vdash \psi_j)$. Since $\mathsf{True} \nvdash \delta$, we can thus reduce to the case where $\varphi_i \nvdash \mathsf{False}$ for all $j \in J$. In particular, by Lemma 11.17 we have $[\![\varphi_j]\!] \neq \emptyset$ and $\mathcal{C}(\varphi_j)$ for all $j \in J$.
Moreover, $\psi_j \vdash \mathsf{False}$ would imply $\delta \vdash \mathsf{False}$ using the rule (C) in Figure 4e. We thus have $\psi_j \nvdash \mathsf{False}$ for all $j \in J$.
Assume toward a contradiction that $[\![\delta]\!] = [\![\mathsf{True}]\!]$. Hence in particular $(\bot \searrow \bot) \in [\![\delta]\!]$, so that $(\bot \searrow \bot) \in [\![\varphi_j \Vdash \psi_j]\!]$ for all $j \in J$. For each $j \in J$, since $[\![\varphi_j]\!] \neq \emptyset$ we have $\bot \in [\![\psi_j]\!]$. But this implies $[\![\psi_j]\!] = [\![\mathsf{True}]\!]$, which by induction hypothesis contradicts that $\mathsf{True} \nvdash \psi_j$ and $\psi_j \nvdash \mathsf{False}$. ◀

▶ **Lemma 11.19.** *For all $\varphi, \psi \in \mathcal{L}^{\wedge}(\tau)$, if $[\![\psi]\!] \subseteq [\![\varphi]\!]$, then $\psi \vdash \varphi$ is derivable.*

Proof. The proof is by induction on the sum of the sizes of ψ and φ.
First, note that the result is trivial if either $\mathsf{True} \vdash \varphi$ or $\psi \vdash \mathsf{False}$. Moreover, if $\varphi \vdash \mathsf{False}$ then $[\![\varphi]\!] = \emptyset$ by the Soundness Theorem 5.13, and we also have $[\![\psi]\!] = \emptyset$, so that $\psi \vdash \mathsf{False}$ by Lemma 11.17. Furthermore, if $\mathsf{True} \vdash \psi$, then $[\![\psi]\!] = [\![\mathsf{True}]\!]$, so that $[\![\varphi]\!] = [\![\mathsf{True}]\!]$ as well, and Lemma 11.18 implies $\mathsf{True} \vdash \varphi$.
Altogether, we can thus assume $\mathsf{True} \nvdash \varphi$, $\varphi \nvdash \mathsf{False}$, $\mathsf{True} \nvdash \psi$ and $\psi \nvdash \mathsf{False}$. Hence Lemma 11.16 applies to both φ and ψ.
Moreover, note that if $\varphi = \varphi_1 \wedge \varphi_2$, then $[\![\psi]\!] \subseteq [\![\varphi]\!]$ implies $[\![\psi]\!] \subseteq [\![\varphi_i]\!]$ for all $i = 1, 2$, and we can obtain $\psi \vdash \varphi$ from the induction hypotheses. We can thus assume that φ is not a conjunction.
We now reason by cases on τ.

Case of Unit. By Lemma 11.16 we have $\psi \dashv\vdash [\langle\rangle]$ and $\varphi \dashv\vdash [\langle\rangle]$. Hence $\psi \vdash \varphi$.

Case of $\mathsf{rec}\,\alpha.\tau$. By Lemma 11.16 we have $\psi \dashv\vdash [\mathsf{fold}]\psi'$ and $\varphi \dashv\vdash [\mathsf{fold}]\varphi'$ with $|\psi'| < |\psi|$ and $|\varphi'| < |\varphi|$. Since moreover $[\![\psi']\!] \subseteq [\![\varphi']\!]$, we can conclude by induction hypothesis.

Case of $\tau_1 + \tau_2$. By Lemma 11.16 we have $\psi \dashv\vdash [\mathsf{in}_i]\psi'$ and $\varphi \dashv\vdash [\mathsf{in}_j]\varphi'$ with $|\psi'| < |\psi|$ and $|\varphi'| < |\varphi|$. Since $[\![\psi]\!] \subseteq [\![\varphi]\!]$, we must have $i = j$, and the result follows using the induction hypothesis.

Case of $\tau_1 \times \tau_2$. By Lemma 11.16 we have $\psi \dashv\vdash [\pi_1]\psi'_1 \wedge [\pi_2]\psi'_2$ and $\varphi \dashv\vdash [\pi_i]\varphi'$ with $|\psi'_1|, |\psi'_2| < |\psi|$ and $|\varphi'| < |\varphi|$.
Since $[\![\psi]\!] \subseteq [\![\varphi]\!]$, we must have $[\![\psi'_i]\!] \subseteq [\![\varphi']\!]$, so that $\psi'_i \vdash \varphi'$ by induction hypothesis, and thus $[\pi_i]\psi'_i \vdash [\pi_i]\varphi'$. We then obtain $\psi \vdash \varphi$, as expected.

Case of $\sigma \to \tau$. This is again the most important and most difficult case. The proof is an adaptation to our setting of the proof of [4, Proposition 10.5.2].
First, by Lemma 11.16 the formula φ must be of the form $\varphi'' \Vdash \varphi'$. If $[\![\varphi'']\!] = \emptyset$, then by Lemma 11.17 we obtain $\varphi'' \vdash \mathsf{False}$, so that $\mathsf{True} \vdash (\varphi'' \Vdash \varphi')$. Hence $\psi \vdash \varphi$ in this

case. We can thus assume $[\![\varphi'']\!] \neq \emptyset$. Since $\varphi \not\vdash \mathsf{False}$, by Lemma 11.17 we get $[\![\varphi]\!] \neq \emptyset$, which implies that $[\![\varphi']\!] \neq \emptyset$ as well. Hence, by Proposition 5.12 there are finite d'', d' such that $[\![\varphi'']\!] = {\uparrow}d''$ and $[\![\varphi']\!] = {\uparrow}d'$.

On the other hand, Lemma 11.16 yields that ψ is of the form $\bigwedge_{i\in I}(\psi_i'' \Vdash \psi_i')$ for some finite set I, with $|\bigwedge_{i\in I}(\psi_i'' \Vdash \psi_i')| \leq |\psi|$. Furthermore, since $\mathsf{True} \not\vdash \psi$, reasoning as in the proof of Lemma 11.18, we can reduce to the case where $[\![\psi_i'']\!] \neq \emptyset$ for all $i \in I$. Lemma 11.17 then yields $\mathcal{C}(\psi_i'')$ for all $i \in I$. Since moreover $\psi \not\vdash \mathsf{False}$, Lemma 11.17 (again) yields $[\![\psi_i']\!] \neq \emptyset$ for all $i \in I$. Again by Proposition 5.12, for each $i \in I$ there are finite e_i'', e_i' such that $[\![\psi_i'']\!] = {\uparrow}e_i''$ and $[\![\psi_i']\!] = {\uparrow}e_i'$. Moreover, $\bigvee_{i\in I}(e_i'' \searrow e_i')$ exists since $[\![\psi]\!] \neq \emptyset$ by Lemma 11.17 (as $\psi \not\vdash \mathsf{False}$).

Hence $[\![\psi]\!] \subseteq [\![\varphi]\!]$ means

$$\uparrow \bigvee_{i\in I} (e_i'' \searrow e_i') \quad \subseteq \quad \uparrow(d'' \searrow d')$$

which implies

$$d'' \searrow d' \quad \leq \quad \bigvee_{i\in I} (e_i'' \searrow e_i')$$

We now evaluate the two functions above at d''. Since

$$\left(\bigvee_{i\in I} e_i'' \searrow e_i'\right)(x) \quad = \quad \bigvee\{e_i' \mid x \geq e_i''\}$$

we get

$$d' \quad \leq \quad \bigvee_{d''\geq e_i''} e_i$$

that is

$$\uparrow \bigvee_{\uparrow d''\subseteq \uparrow e_i''} e_i' \quad \subseteq \quad \uparrow d'$$

In other words,

$$[\![\bigwedge_{[\![\varphi'']\!]\subseteq[\![\psi_i'']\!]} \psi_i']\!] \quad \subseteq \quad [\![\varphi']\!]$$

and by induction hypothesis

$$\bigwedge_{\varphi''\vdash\psi_i''} \psi_i' \quad \vdash \quad \varphi'$$

Hence we are done since

$$\psi \quad \vdash \quad \bigwedge_{i\in I} \left(\left(\bigwedge_{\varphi''\vdash\psi_i''} \psi_i''\right) \Vdash \psi_i'\right)$$

and thus

$$\psi \quad \vdash \quad \left(\bigwedge_{\varphi''\vdash\psi_i''} \psi_i''\right) \Vdash \bigwedge_{\varphi''\vdash\psi_i''} \psi_i'$$

while $\varphi'' \vdash \bigwedge_{\varphi''\vdash\psi_i''} \psi_i''$. ◀

We note here that the (closed) neutral fragment is complete. We do not need this result.

▶ **Corollary 11.20.** *Let $\varphi, \psi \in \mathcal{L}^{\pm}(\tau)$ be* closed. *If $[\![\psi]\!] \subseteq [\![\varphi]\!]$, then $\psi \vdash \varphi$ is derivable.*

Proof. By Lemma 3.8 and Lemma 3.6 (together with Soundness Theorem 5.13), we can assume $\varphi, \psi \in \mathcal{L}^{\vee\wedge}(\tau)$. If ψ is a disjunction, say $\psi_1 \vee \psi_2$, then $[\![\psi]\!] \subseteq [\![\varphi]\!]$ implies $[\![\psi_1]\!] \subseteq [\![\varphi]\!]$ and $[\![\psi_2]\!] \subseteq [\![\varphi]\!]$. We can thus reduce to the case where $\psi \in \mathcal{L}^{\wedge}(\tau)$. If $[\![\psi]\!] = \emptyset$, then Lemma 11.17 yields that $\psi \vdash \mathsf{False}$, and we are done.

Otherwise, by Proposition 5.12 there is some finite $d \in [\![\tau]\!]$ such that $[\![\psi]\!] = {\uparrow}d$. Hence $[\![\psi]\!] \subseteq [\![\varphi]\!]$ implies $d \in [\![\varphi]\!]$. If φ is a disjunction, say $\varphi_1 \vee \varphi_2$, then either $d \in [\![\varphi_1]\!]$ and $[\![\psi]\!] \subseteq [\![\varphi_1]\!]$, or $d \in [\![\varphi_2]\!]$ and $[\![\psi]\!] \subseteq [\![\varphi_2]\!]$. We can thus reduce to the case were $\varphi \in \mathcal{L}^{\wedge}(\tau)$, and conclude with Lemma 11.19. ◀

12 Proofs of §5.4 (Semantics of the Refinement Type System)

We begin with the (long) proof of Theorem 5.18.

▶ **Theorem 5.18.** *If $\mathcal{E}^\wedge \vdash M : T^\wedge$ is sound, then it is derivable.*

Proof. Note that Lemma 4.3 restricts to types $V^\wedge$, in the sense that given $V^\wedge$, there is some $\varphi \in \mathcal{L}^\wedge(|V^\wedge|)$ such that $V^\wedge \simeq \{|V^\wedge| \mid \varphi\}$.

In the following, we write $\mathcal{E} \vdash M : T$ for $\mathcal{E}^\wedge \vdash M : T^\wedge$.

Let $\varphi \in \mathcal{L}^\wedge(|T|)$ such that $T \simeq \{|T| \mid \varphi\}$. Assume $\mathcal{E} = x_1 : U_1, \dots, x_n : U_n$ and let $\psi_i \in \mathcal{L}^\wedge(|U_i|)$ such that $U_i \simeq \{|U_i| \mid \psi_i\}$ for each $i = 1, \dots, n$.

Note that if $[\![U_i]\!] = \emptyset$ for some $i \in \{1, \dots, n\}$, since $[\![|U_i|]\!]$ is not empty (as it is a Scott domain), we must have $[\![\psi]\!]_i = \emptyset$ and thus $\psi_i \vdash \mathsf{False}$ by completeness for $\mathcal{L}^\wedge$ (Theorem 5.14). We can then conclude with the rule for False in Figure 7b (§4). We can thus reduce the case of a typing context such that $[\![U_i]\!] \neq \emptyset$ (or equivalently $\psi_i \nvdash \mathsf{False}$) for each $i = 1, \dots, n$. We call such typing contexts *consistent*.

Note also that if $\mathsf{True} \vdash \varphi$, then we can derive $\mathcal{E} \vdash M : T$ from $|\mathcal{E}| \vdash M : |T|$ by subtyping. We can thus assume $\mathsf{True} \nvdash \varphi$. Moreover, since $\mathcal{E}$ is consistent we must have $[\![\varphi]\!] \neq \emptyset$, that is, $\varphi \nvdash \mathsf{False}$. We can thus always apply Lemma 11.16 to φ.

We now reason by induction on the typing derivation of $|\mathcal{E}| \vdash M : |T|$.

Case of

$$\frac{(x : |T|) \in |\mathcal{E}|}{|\mathcal{E}| \vdash x : |T|}$$

We have $(x : U) \in \mathcal{E}$ for some type U with $|U| = |T|$. Let $\varphi, \psi \in \mathcal{L}^\wedge(|T|)$ such that $T \simeq \{|T| \mid \varphi\}$ and $U \simeq \{|T| \mid \psi\}$. Since $\mathcal{E} \vdash M : T$ is sound, we have $[\![\psi]\!] \subseteq [\![\varphi]\!]$. Hence $\psi \vdash \varphi$ by completeness for $\mathcal{L}^\wedge$ (Theorem 5.14). We then conclude by subtyping.

Case of

$$\frac{}{|\mathcal{E}| \vdash \langle\rangle : |T|}$$

We have $|T| = \mathsf{Unit}$ and $\varphi \dashv\vdash [\langle\rangle]$. We conclude with the rule

$$\frac{}{\mathcal{E} \vdash \langle\rangle : \{\mathsf{Unit} \mid [\langle\rangle]\}}$$

Case of

$$\frac{|\mathcal{E}| \vdash N_1 : \tau_1 \qquad |\mathcal{E}| \vdash N_2 : \tau_2}{|\mathcal{E}| \vdash \langle N_1, N_2\rangle : \tau_1 \times \tau_2}$$

where $|T| = \tau_1 \times \tau_2$ and $M = \langle N_1, N_2\rangle$. We have $\varphi \dashv\vdash [\pi_1]\varphi_1 \wedge [\pi_2]\varphi_2$. The judgements $\mathcal{E} \vdash N_1 : \{\tau_1 \mid \varphi_1\}$ and $\mathcal{E} \vdash N_2 : \{\tau_2 \mid \varphi_2\}$ are both sound since so is $\mathcal{E} \vdash \langle N_1, N_2\rangle : \{\tau_1 \times \tau_2 \mid [\pi_1]\varphi_1 \wedge [\pi_2]\varphi_2\}$. We can thus conclude with the induction hypothesis and

$$\dfrac{\dfrac{\dfrac{\vdots}{\mathcal{E} \vdash N_1 : \{\tau_1 \mid \varphi_1\}}}{\mathcal{E} \vdash \langle N_1, N_2\rangle : \{\tau_1 \times \tau_2 \mid [\pi_1]\varphi_1\}} \quad \dfrac{\dfrac{\vdots}{\mathcal{E} \vdash N_2 : \{\tau_2 \mid \varphi_2\}}}{\mathcal{E} \vdash \langle N_1, N_2\rangle : \{\tau_1 \times \tau_2 \mid [\pi_2]\varphi_2\}}}{\mathcal{E} \vdash \langle N_1, N_2\rangle : \{\tau_1 \times \tau_2 \mid [\pi_1]\varphi_1 \wedge [\pi_2]\varphi_2\}}$$

Case of

$$\frac{|\mathcal{E}| \vdash N : \tau \times \sigma}{|\mathcal{E}| \vdash \pi_i(N) : |T|}$$

where $i = 1, 2$ and $M = \pi_1(N)$. Assume w.l.o.g. $i = 1$ (so that $|T| = \tau$). Note that $\mathcal{E} \vdash N\colon \{\tau \times \sigma \mid [\pi_1]\varphi\}$ is sound since so is $\mathcal{E} \vdash \pi_1(N)\colon \{\tau \mid \varphi\}$. We then conclude with the induction hypothesis and the rule

$$\frac{\mathcal{E} \vdash N\colon \{\tau \times \sigma \mid [\pi_1]\varphi\}}{\mathcal{E} \vdash \pi_1(N)\colon \{\tau \mid \varphi\}}$$

Case of

$$\frac{|\mathcal{E}| \vdash N\colon \tau[\mathsf{rec}\,\alpha.\tau/\alpha]}{|\mathcal{E}| \vdash \mathsf{fold}(N)\colon \mathsf{rec}\,\alpha.\tau}$$

where $|T| = \mathsf{rec}\,\alpha.\tau$ and $M = \mathsf{fold}(M)$. We have $\varphi \dashv\vdash [\mathsf{fold}]\psi$. Moreover, $\mathcal{E} \vdash N\colon \{\tau[\mathsf{rec}\,\alpha.\tau/\alpha] \mid \psi\}$ is sound since so is $\mathcal{E} \vdash \mathsf{fold}(N)\colon \{\mathsf{rec}\,\alpha.\tau \mid [\mathsf{fold}]\psi\}$. We can thus conclude using the induction hypothesis and the rule

$$\frac{\mathcal{E} \vdash N\colon \{\tau[\mathsf{rec}\,\alpha.\tau/\alpha] \mid \psi\}}{\mathcal{E} \vdash \mathsf{fold}(N)\colon \{\mathsf{rec}\,\alpha.\tau \mid [\mathsf{fold}]\psi\}}$$

Case of

$$\frac{|\mathcal{E}| \vdash N\colon \mathsf{rec}\,\alpha.\tau}{|\mathcal{E}| \vdash \mathsf{unfold}(N)\colon \tau[\mathsf{rec}\,\alpha.\tau/\alpha]}$$

where $|T| = \tau[\mathsf{rec}\,\alpha.\tau/\alpha]$ and $M = \mathsf{unfold}(N)$. The judgement $\mathcal{E} \vdash N\colon \{\mathsf{rec}\,\alpha.\tau \mid [\mathsf{fold}]\varphi\}$ is sound since so is $\mathcal{E} \vdash \mathsf{unfold}(N)\colon \{\tau[\mathsf{rec}\,\alpha.\tau/\alpha] \mid \varphi\}$ and we conclude using the induction hypothesis and the rule

$$\frac{\mathcal{E} \vdash N\colon \{\mathsf{rec}\,\alpha.\tau \mid [\mathsf{fold}]\varphi\}}{\mathcal{E} \vdash \mathsf{unfold}(N)\colon \{\tau[\mathsf{rec}\,\alpha.\tau/\alpha] \mid \varphi\}}$$

Case of

$$\frac{|\mathcal{E}|, x\colon \sigma \vdash N\colon \tau}{|\mathcal{E}| \vdash \lambda x.N\colon \sigma \to \tau}$$

where $|T| = \sigma \to \tau$, and where $M = \lambda x.N$.
We have $\varphi \dashv\vdash \bigwedge_{i\in I}(\varphi_i'' \Vdash \varphi_i')$ for some finite set I. Let $i \in I$. The judgement $\mathcal{E} \vdash \lambda x.N\colon \{\sigma \to \tau \mid \varphi_i'' \Vdash \varphi_i'\}$ is sound, and so is $\mathcal{E}, x\colon \{\sigma \mid \varphi_i''\} \vdash N\colon \{\tau \mid \varphi_i'\}$. Using the induction hypothesis, we derive $\mathcal{E} \vdash \lambda x.N\colon \{\sigma \to \tau \mid \varphi_i'' \Vdash \varphi_i'\}$. We can then obtain $\mathcal{E} \vdash \lambda x.N\colon \{\sigma \to \tau \mid \varphi\}$.

Case of

$$\frac{|\mathcal{E}| \vdash N\colon \sigma \to \tau \qquad |\mathcal{E}| \vdash V\colon \sigma}{|\mathcal{E}| \vdash NV\colon \tau}$$

where $|T| = \tau$ and where $M = NV$. For each $i \in \{1, \dots, n\}$, since $[\![\psi_i]\!] \neq \emptyset$, by Proposition 5.12 there is a finite $e_i \in [\![|U_i|]\!]$ such that $[\![\psi_i]\!] = {\uparrow}e_i$. Similarly, since $[\![\varphi]\!] \neq \emptyset$, again by Proposition 5.12 there is a finite $d \in [\![\tau]\!]$ such that $[\![\varphi]\!] = {\uparrow}d$.
We have $[\![M]\!](\vec{e}) \in [\![\varphi]\!]$. Note that $[\![M]\!](\vec{e}) = [\![N]\!](\vec{e})\,([\![V]\!](\vec{e}))$. Since $[\![\sigma]\!]$ is a Scott domain, it is algebraic, and $[\![V]\!](\vec{e})$ is the directed l.u.b. of the finite $e \leq [\![V]\!](\vec{e})$. Since $[\![N]\!](\vec{e})$ is Scott-continuous, we thus get that $[\![M]\!](\vec{e})$ is the l.u.b. of the directed set

$$\{[\![N]\!](\vec{e})(e) \mid e \text{ finite and } \leq [\![V]\!](\vec{e})\}$$

Since $d \leq [\![M]\!](\vec{e})$ and since d is finite, it follows that we have $d \leq [\![N]\!](\vec{e})(e)$ for some finite $e \leq [\![V]\!](\vec{e})$. By Proposition 5.12, there is a formula $\psi \in \mathcal{L}^{\wedge}(\sigma)$ such that $[\![\psi]\!] = {\uparrow}e$.

Since $d \leq [\![N]\!](\vec{e})(e)$, we have $(e \searrow d) \leq [\![N]\!](\vec{e})$, so that $[\![N]\!](\vec{e}) \in [\![\psi \Vdash \varphi]\!]$. Since $[\![N]\!]$ is monotone, it follows that $\mathcal{E} \vdash N\colon \{\sigma \to \tau \mid \psi \Vdash \varphi\}$ is sound. Hence, this judgement is derivable by induction hypothesis.
Similarly, since $e \leq [\![V]\!](\vec{e})$, we obtain that the judgement $\mathcal{E} \vdash V\colon \{\sigma \mid \psi\}$ is sound and thus derivable.
We can then easily derive $\mathcal{E} \vdash M\colon \{\tau \mid \varphi\}$.

Case of

$$\frac{|\mathcal{E}|, x\colon |T| \vdash N\colon |T|}{|\mathcal{E}| \vdash \mathsf{fix}\, x.N\colon |T|}$$

where $M = \mathsf{fix}\, x.N$. Similarly as above, for each $i \in \{1, \dots, n\}$ there is a finite $e_i \in [\![\psi_i]\!]$ such that $[\![U_i]\!] = \uparrow e_i$. There is also a finite $d \in [\![|T|]\!]$ such that $[\![\varphi]\!] = \uparrow d$.
Let $f\colon [\![|T|]\!] \to [\![|T|]\!]$ be the Scott-continuous function which takes $a \in [\![|T|]\!]$ to $[\![N]\!](\vec{e}, a)$. We have

$$[\![\mathsf{fix}\, x.N]\!](\vec{e}) \quad = \quad \bigvee_{k \in \mathbb{N}} f^k(\bot)$$

Since $d \leq [\![\mathsf{fix}\, x.N]\!](\vec{e})$ with d finite, there is some $k \in \mathbb{N}$ such that $d \leq f^k(\bot)$. Write d_k for d. Since $[\![|T|]\!]$ is algebraic, by induction, for each $j = k-1, \dots, 0$ there is some finite d_j such that $d_{j+1} \leq f(d_j)$ and $d_j \leq f^j(\bot)$. In particular, $d_0 = \bot$. For each $j = 0, \dots, k$, let φ_j such that $[\![\varphi_j]\!] = \uparrow d_j$. Note that $\varphi_k = \varphi$. Moreover, since $d_0 = \bot$, we can take $\varphi_0 = \mathsf{True}$.
Again reasoning similarly as above, we obtain that $\mathcal{E}, x\colon \{|T| \mid \varphi_j\} \vdash N\colon \{|T| \mid \varphi_{j+1}\}$ is sound and thus derivable for each $j = 0, \dots, k-1$. Moreover, $\mathcal{E} \vdash \mathsf{fix}\, x.N\colon \{|T| \mid \varphi_0\}$ is derivable. We can then derive $\mathcal{E} \vdash \mathsf{fix}\, x.N\colon \{|T| \mid \varphi\}$ by iterated applications of the rule

$$\frac{\mathcal{E} \vdash \mathsf{fix}\, x.N\colon \{\tau \mid \psi\} \qquad \mathcal{E}, x : \{\tau \mid \psi\} \vdash N\colon \{\tau \mid \psi'\}}{\mathcal{E} \vdash \mathsf{fix}\, x.N\colon \{\tau \mid \psi'\}}$$

Case of

$$\frac{|\mathcal{E}| \vdash N\colon \tau_i}{|\mathcal{E}| \vdash \mathsf{in}_i(N)\colon \tau_1 + \tau_2}$$

where $|T| = \tau_1 + \tau_2$ and $M = \mathsf{in}_i(N)$. We have $\varphi \dashv\vdash [\mathsf{in}_j]\psi$, and since $\mathcal{E} \vdash M\colon T$ is sound, we must have $i = j$. But then $\mathcal{E} \vdash N\colon \{\tau_i \mid \psi\}$ is also sound, and we can conclude by induction hypothesis and the rule

$$\frac{\mathcal{E} \vdash N\colon \{\tau_i \mid \psi\}}{\mathcal{E} \vdash \mathsf{in}_i(N)\colon \{\tau_1 + \tau_2 \mid [\mathsf{in}_i]\psi\}}$$

Case of

$$\frac{|\mathcal{E}| \vdash N\colon \tau_1 + \tau_2 \qquad |\mathcal{E}|, x_1\colon \tau_1 \vdash N_1\colon \sigma \qquad |\mathcal{E}|, x_2\colon \tau_2 \vdash N_2\colon \sigma}{|\mathcal{E}| \vdash \mathsf{case}\ N\ [x_1 \mapsto N_1 \mid x_2 \mapsto N_2]\colon \sigma}$$

where $|T| = \sigma$ and where M is the term $(\mathsf{case}\ N\ [x_1 \mapsto N_1 \mid x_2 \mapsto N_2])$.
Similarly as in the cases of $\mathsf{fix}\, x.N$ and NV above, for each $i \in \{1, \dots, n\}$ there is a finite $e_i \in [\![|U_i|]\!]$ such that $[\![\psi_i]\!] = \uparrow e_i$. Also, there is a finite $d \in [\![\sigma]\!]$ such that $[\![\varphi]\!] = \uparrow d$.
Since $[\![\varphi]\!] \neq [\![\mathsf{True}]\!]$, we must have $\bot \notin [\![\varphi]\!]$, so that $[\![M]\!](\vec{e}) \neq \bot$ and thus $[\![N]\!](\vec{e}) \neq \bot$. Hence $[\![N]\!](\vec{e}) = [\![\mathsf{in}_i]\!](z)$ for some $z \in [\![\tau_i]\!]$. Since $[\![M]\!](\vec{e}) = [\![N_i]\!](\vec{e}, z) \in \uparrow d$, by algebraicity there is a finite $e \leq z$ such that $d \leq [\![N_i]\!](\vec{e}, e)$. With $\psi \in \mathcal{L}^\wedge(\tau_i)$ such that $[\![\psi]\!] = \uparrow e$, we get that the judgements $\mathcal{E} \vdash N\colon \{\tau_1 + \tau_2 \mid [\mathsf{in}_i]\psi\}$ and $\mathcal{E}, x\colon \{\tau_i \mid \psi\} \vdash$

$N_i\colon \{\sigma \mid \varphi\}$ are sound. We can then conclude using the induction hypothesis and the rule

$$\frac{\mathcal{E} \vdash N\colon \{\tau_1 + \tau_2 \mid [\mathsf{in}_i]\psi\} \qquad \mathcal{E}, x_i\colon \{\tau_i \mid \psi\} \vdash N_i\colon T \qquad |\mathcal{E}|, x_{3-i}\colon \tau_{3-i} \vdash N_{3-i}\colon |T|}{\mathcal{E} \vdash \mathsf{case}\ N\ [x_1 \mapsto N_1 \mid x_2 \mapsto N_2]\colon T}$$

◀

We now prove our main result, namely the Positive Completeness Theorem 5.17.

▶ **Theorem 5.17** (Positive Completeness)**.** *If $\mathcal{E}^- \vdash M\colon T^+$ is sound, then it is derivable.*

Proof. Write $\mathcal{E} \vdash M\colon T$ for $\mathcal{E}^- \vdash M\colon T^+$. Let $\mathcal{E} = x_1\colon U_1, \dots, x_n\colon U_n$. By Lemma 4.3 and Proposition 3.9, for each $i \in \{1, \dots, n\}$ there is a $\psi_i \in \mathcal{L}^{\pm}(|U_i|)$ such that $U_i \simeq \{|U_i| \mid (\forall \ell_i)\psi_i\}$. We similarly get a $\varphi \in \mathcal{L}^{\pm}(|T|)$ such that $T \simeq \{|T| \mid (\exists k)\varphi\}$.

Let $\mathcal{Q} = \{S_m \mid m \in \mathbb{N}\}$, where for each $m \in \mathbb{N}$,

$$S_m \quad = \quad [\![\psi_1[m/\ell_1]]\!] \times \cdots \times [\![\psi_n[m/\ell_n]]\!]$$

▷ Claim 12.1. $\mathcal{Q}$ is a codirected family of compact saturated sets. Moreover

$$\bigcap \mathcal{Q} \quad = \quad [\![U_1]\!] \times \cdots \times [\![U_n]\!]$$

Proof. It is clear that each $S_m \in \mathcal{Q}$ is compact-saturated as a (finite) product of compact-saturated sets (see e.g. [13, Theorem 4.5.12]). Moreover, $\mathcal{Q}$ is codirected since (it is non-empty and) $m \leq p$ implies $S_p \subseteq S_m$ by antimonotonicity (Lemma 5.9).

Finally, the equality follows from the fact that

$$\begin{array}{lll} (u_1, \dots, u_n) \in \bigcap \mathcal{Q} & \text{iff} & \forall m \in \mathbb{N},\ (u_1, \dots, u_n) \in S_m \\ & \text{iff} & \forall m \in \mathbb{N},\ \forall i \in \{1, \dots, n\},\ u_i \in [\![\psi_i[m/\ell_i]]\!] \\ & \text{iff} & \forall i \in \{1, \dots, n\},\ \forall m \in \mathbb{N},\ u_i \in [\![\psi_i[m/\ell_i]]\!] \\ & \text{iff} & \forall i \in \{1, \dots, n\},\ u_i \in [\![U_i]\!] \end{array}$$

◁

Since $\mathcal{E} \vdash M\colon T$ is sound, we have $\bigcap \mathcal{Q} \subseteq [\![M]\!]^{-1}([\![T]\!])$, where $[\![T]\!]$ is Scott-open and M is Scott-continuous. By well-filteredness (Proposition 5.4) and the Claim, there is some $S_m \in \mathcal{Q}$ such that $S_m \subseteq [\![M]\!]^{-1}([\![T]\!])$. Hence there is some $m \in \mathbb{N}$ such that the following is sound:

$$x_1\colon \{|U_1| \mid \psi_1[m/\ell_1]\},\ \dots,\ x_n\colon \{|U_n| \mid \psi_n[m/\ell_n]\} \quad \vdash \quad M\colon T \tag{12}$$

Now, by Lemmas 3.8 and 3.6, each $\psi_i[m/k] \in \mathcal{L}^{\pm}(|U_i|)$ is equivalent to a disjunction $\bigvee_{j \in J_i} \delta_{i,j}$, where $\delta_{i,j} \in \mathcal{L}^{\wedge}(|U_i|)$. For each tuple $(j_1, \dots, j_n) \in J_1 \times \cdots \times J_n$, the judgement

$$x_1\colon \{|U_1| \mid \delta_{1,j_1}\},\ \dots,\ x_n\colon \{|U_n| \mid \delta_{n,j_n}\} \quad \vdash \quad M\colon T \tag{13}$$

is sound. If $\delta_{i,j_i} \vdash \mathsf{False}$, then we can derive (13) using the rule for False in Figure 7b (§4).

Assume $\delta_{i,j_i} \not\vdash \mathsf{False}$ for all $i = 1, \dots, n$. Hence by Proposition 5.12, for each $i = 1, \dots, n$ there is some finite $d_i \in [\![|U_i|]\!]$ such that $[\![\delta_i]\!] = {\uparrow} d_i$. The soundness of (13) then yields $[\![M]\!](\vec{d}) \in [\![T]\!]$. Hence there is $p \in \mathbb{N}$ such that $[\![M]\!](\vec{d}) \in [\![\varphi[p/k]]\!]$. By Lemmas 3.8 and 3.6 again, the formula $\varphi[p/k]$ is equivalent to a disjunction of $\gamma \in \mathcal{L}^{\wedge}(|T|)$. Since $[\![M]\!](\vec{d}) \in [\![\varphi[p/k]]\!]$, we have $[\![M]\!](\vec{d}) \in [\![\gamma]\!]$ for at least one of these γ's. It follows that the judgement

$$x_1\colon \{|U_1| \mid \delta_{1,j_1}\},\ \dots,\ x_n\colon \{|U_n| \mid \delta_{n,j_n}\} \quad \vdash \quad M\colon \{|T| \mid \gamma\}$$

is sound, and thus derivable by Theorem 5.18. We then derive (13) by subtyping, using that $\gamma \vdash \varphi[p/k]$.

Since we have (13) for each tuple $(j_1, \ldots, j_n) \in J_1 \times \cdots \times J_n$, we then obtain (12) by multiple applications of the rule for $\vee$ in Figure 7b (§4). Then we are done by subtyping since $(\forall \ell_i)\psi_i \vdash \psi_i[m/\ell_i]$ for each $i = 1, \ldots, n$. ◀

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